Probability

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Probability

RANDOM EXPERIMENTS

In our day-to-day life, we perform many activities which have a fixed results no matter the number of times they are repeated. For example, given the triangle, without knowing the three angles, we can definitely say that the sum of measure of angle is 180°.

We also perform many experimental activities, where the results may not be same, when they are repeated under identical conditions. For example, when a coin is tossed it may turn up a head or a tail, but we are not sure which one of these results will actually be obtained. Such experiments are called random experiments.

An experiment is called **random experiment** if it satisfies the following two conditions:

(i) It has more than one possible outcome.

(ii) It is not possible to predict the outcome in advance.

Example :

(i) Tossing a coin is a random experiment.

(ii) Throwing a dice is a random experiment.

(iii) Drawing a card from a well shuffled deck of 52 playing cards is also a random experiment.

OUTCOMES AND SAMPLE SPACE :

A possible result of a random experiment is called its outcome.

Consider the experiment of rolling a die. The outcomes of this experiment are 1, 2, 3, 4, 5 or 6, if we are interested in the number of dots on the upper face of the die.

The set of outcomes {1, 2, 3, 4, 5, 6} is called the sample space of the experiment.

Definition

Thus, the set of all possible outcomes of a random experiment is called the sample space associated with the experiment. Sample space is denoted by the symbol S.

Each element of the sample space is called a sample point. In other words, each outcome of the random experiment is also called sample point.

Q. **Two coins (a one rupee coin and a two rupee coin) are tossed once. Find a sample space.**

Sol. Clearly the coins are distinguishable in the sense that we can speak of the first coin and the second coin Since either coin can turn up Head (H) or Tail (T), the possible outcomes may be
Head on both coins = (H. H) = HH
Head on first coin and tail on second coin = HT
Tail on first coin and Head on the other coin = (T, H) = TH
Tail on both coins = (T, T) = TT
Thus, the sample space is S = {HH, HT, TH, TT}

Q. **A coin is tossed twice. If head appears in the second throw then a dice is thrown. Write down the sample space of the experiment.**

Sol. When a coin is tossed two times then possible outcomes are {(TT),(HT),(TH),(HH)} if head appears in the second throw then dice is thrown.

$\therefore$ All possible outcomes of the experiment are

S = {(TT), (HT), (TH1), (TH2), (TH3), (TH4), (TH5), (TH6), (HH1), (HH2), (HH3), (HH4), (HH5), (HH6)}

EVENT :

Consider the experiment of tossing a coin two times. An associated sample space is

S = {HH, HT, TH, TT}

Now suppose that we are interested in those outcomes which correspond to the occurrence of exactly one head. We find that HT and TH are the only elements of S corresponding to the occurrence of this happening (event). These two elements form the set E={HT,TH}.

We know that the set E is a subset of the sample space S. Similarly, we find the following correspondence between events and subsets of S.

Definition

Any subset E of a sample space S is called an event.

Point to remember!!!

The maximum number of events which can be associated with an experiment is 2^n, where n is the number of elements in the sample space.
i.e., ${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots + {}^nC_n = 2^n$

Description of events	Corresponding subset of 'S'
Number of tail is exactly 2	$A = \{TT\}$
Number of tails is atleast one	$B = \{HT, TH, TT\}$
Number of heads is atmost one	$c = \{HT, TH, TT\}$
Second toss is not head	$D = \{HT, TT\}$
Number of tails is atmost two	$S = \{HH, HT, TH, TT\}$
Number of tails is more than two	ϕ

Q. **In throwing a pair of dice write down two possible events.**
E_1 = sum of the numbers appearing on both the dice is 7.
E_2 = The sum of the numbers appearing on both the dice is divisible by 3.

Sol. $E_1 = \{(6,1),(5,2),(4,3),(3,4),(2,5),(1,6)\}$
$E_2 = \{(2,1),(1,2),(5,1),(4,2),(3,3),(2,4),(1,5),(6,3)(5,4)(4,5),(3,6),(6,6)\}$

Occurrence of an event :

Consider the experiment of throwing a die. Let E denotes the event "a number less than 4 appears". If actually '1' had appeared on the die then we say that event E has occurred. As a matter of fact if outcomes are 2 or 3, we say that even E has occurred.

Impossible and sure Events :

The empty set ϕ and the sample space S describe events. In fact, ϕ is called **impossible event** and S, i.e., the whole sample space is called the sure event.

To understand these let us consider the experiment of rolling a die. The associated sample space is

$S = \{1, 2, 3, 4, 5, 6\}$

Definition

Thus, the event E of a sample space S is said to have occurred if the outcome ω of the experiment is such that $\omega \in E$. If the outcome ω is such that $\omega \notin E$, we say that the event E has not occurred.

Let E be the event "the number appears on the die is multiple of 7".

Clearly no outcome satisfies the condition given in the event, i.e., no element of the sample space ensure the occurrence of the event E. Thus, we say that the empty set only correspond to the event E. In other words we can say that it impossible to have a multiple of 7 on the upper face of the die. Thus, the event $E = \phi$ is an impossible event.

Now let us use another event F "the number turns up is odd or even".

Clearly $F = \{1,2,3,4,5,6\} = S$, i.e., all outcomes of the experiment ensure the occurrence of the even F. Thus, the event F = S is a sure event.

Simple Event :

If an event E has only one sample point of a sample space, it is called a simple (or elementary) event.

In a sample space containing n distinct elements, there are exactly n simple events.

$$S = \{HH, HT, TH, TT\}$$

There are four simple events corresponding of this sample space, These are

$$E_1 = \{HH\}, E_2 = \{HT\}, E_3 = \{TH\} \text{ and } E_4 = \{TT\}.$$

Compound Event :

If an event has more than one sample point, it is called a compound event.

For example, in the experiment of " tossing a coin thrice" the events

E : 'Exactly one head appeared'

F : 'Atleast one head appeared'

G : 'Atmost one head appeared' etc.

are all compound events. The subsets associated with these events are

$$E = \{HTT, THT, TTH\}$$

$$F = \{HTT, THT, TTH, HHT, HTH, THH, HHH\}$$

$$G = \{TTT, THT, HTT, TTH\}$$

Know the facts



Each of the above subsets contain more than one sample point, hence they are all compound events.

ALGEBRA OF EVENTS :

In the Chapter on Sets, we have studied about different ways of combining two or more sets, viz, union, intersection, difference, complement of a set etc. Like-wise we can combine two or more events by using the analogous set notations.

Let A, B, C be events associated with an experiment whose sample space is S.

Complementary Event :

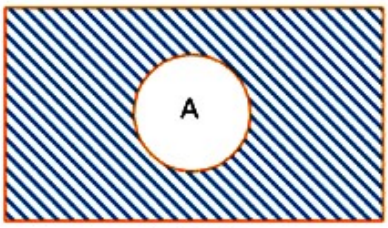
For every event A, there corresponds another event A' or $\bar{A}$ called the complementary event to A. It is also called the event 'not A'.

For example, take the experiment 'of tossing three coins'. An associated sample space is

$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

Let $A = \{HTH, HHT, THH\}$ be the event 'only one tail appears'.

Clearly for the outcome HTT, the event A has not occurred. But we may say that the event 'not A' has occurred. Thus, with every outcome which is not in A, we say that 'not A' occurs.



A' or $\bar{A}$

Thus the complementary event 'not A' to the event A is

$A' = \{HHH, HTT, THT, TTH, TTT\}$

Or $A' = \{\omega : \omega \in S \text{ and } \omega \notin A\} = S - A$

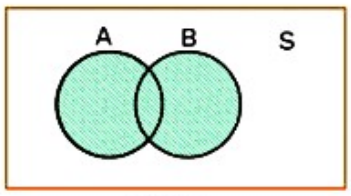
The Event 'A or B' :

Recall that union of two sets A and B denoted by $A \cup B$ contains all those elements which are either in A or in B or in both.

When the sets A and B are two events associated with a sample space, then '$A \cup B$ is the either' A or B or both'. This event '$A \cup B$' is also called 'A or B'.

Therefore Event

'A or B' = $A \cup B = \{w : w \in A \text{ or } w \in B\}$



$A \cup B$

The Event 'A and B' :

We know that intersection of two sets $A \cap B$ is the set of those elements which are common to both A and B. i.e., which belong to both 'A and B'.

If A and B are two events, then the set $A \cap B$ denotes the events ' A and B'.

Thus, $A \cap B = \{\omega : \omega \in A \text{ and } \omega \in B\}$

For example, in the experiment of 'throwing a die twice' Let A be the event 'score on the first throw is six' and B is 'sum of scores is atleast 11' then

$$A = \{(6,1),(6,2),(6,3),(6,4)(6,5)(6,6)\}$$

And $B = \{(5,6),(6,5),(6,6)\}$

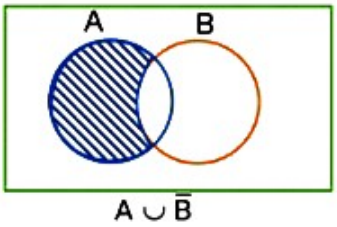
So $A \cap B = \{(6,5),(6,6)\}$

Note that the set $A \cap B = \{(6,5),(6,6)\}$ may represent the event 'the score on the first throw is six and the sum of the scores is atleast 11.'

The Event 'A but not B' :

We know that A – B is the set of all those elements which are in A but not in B. Therefore, the set A – B may denote the event 'A but not B'. We know that

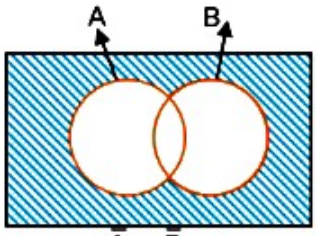
$A - B = A \cap B'$



$A \cup \bar{B}$

The Event 'neither A nor B' :

The set of elements which neither in set A not in set B. i.e. $S - (A \cup B)$ and which is denoted by $\bar{A} \cap \bar{B}$.



$\bar{A} \cap \bar{B}$

Q. **Consider the experiment of rolling a die. Let A be the event 'getting a prime number', B be the event 'getting an odd number'. Write the sets representing the events**

(i) A or B **(ii) A and B** **(ii) A but not B** **(iv) 'not A'**

Sol. Here $S = \{1,2,3,4,5,6\}, A = \{2,3,5\}$

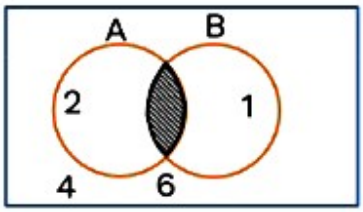
Obviously

(i) 'A or B' $= A \cup B = \{1,2,3,5\}$

(ii) 'A and B' $= A \cap B = \{3,5\}$

(iii) 'A but not B' $= A - B = \{2\}$

(iv) 'not A' $= A' = \{1,4,6\}$



THREE MOST IMPORTANT EVENTS :

(1) Equally Likely Events :

Events are said to be equally likely when no particular event is given preference to occur in relation to the other event.

Examples :

(i) The outcomes as result of throwing a die are equally likely, as no particular face is more likely to occur as compared to other faces. That is why we normally write as fair die or unbiased die.

(ii) The outcomes as result of drawing a card from a well shuffled pack of 52 playing cards are equally likely to occur. Each card is as likely to be withdrawn as any other card.

(iii) However, getting of a total of 7 is not as equally likely as getting of a total of 12 when a pair of dice are rolled once. It is also to be noted that it is 6 times more likely to get a total of 7 than to get a total of 12 in a single throw with the pair of dice.

Points to remember!!!

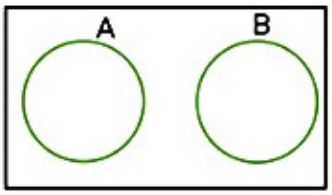
(i) $\overline{A \cup B} = \bar{A} \cap \bar{B}$

(ii) $\overline{A \cap B} = \bar{A} \cup \bar{B}$

} **De Morgan's Law**

(2) Mutually Exclusive / Disjoint / Incompatible Events:

Two events A and B are said to be mutually exclusive events if their simultaneous occurrence is impossible, i.e. both the events cannot occur together.



$A \cap B = \phi$

Examples:

(i) In throwing a fair die, two events A and B are such that
A : getting an odd number
B : getting an even number
then A & B are mutually exclusive events.

(ii) In drawing a card from a well shuffled pack of 52 playing cards, two events A and B are such that
A : getting an ace
B : getting a red card
then A and B are not mutually exclusive events.

(3) Exhaustive Event :

If $E_1, E_2, ..., E_n$ are n events associated with an experiment whose sample space is S and if

$$E_1 \cup E_2 \cup E_3 \cup ... \cup E_n = \bigcup_{i=1}^{n} E_i = S$$

then $E_1, E_2, ..., E_n$ are called exhaustive events. In other words, events $E_1, E_2, ..., E_n$ are said to be exhaustive if atleast one of them necessarily occurs whenever the experiment is performed.

Further, if $E_1 \cap E_j = \phi$ for $i \neq j$ i.e., events E_i and E_j are pairwise disjoint and $\bigcup_{i=1}^{n} E_i = S$, then event $E_1, E_2, ..., E_n$ are called mutually exclusive and exhaustive events.

Example :

Consider the experiment of throwing a die. We have
S = {1,2,3,4,5,6}. Let us define the following events
A : 'a number less than 4 appears'.
B: 'a number greater than 2 but less than 5 appears'
and C: 'a number greater than 4 appears'.
Then A = {1, 2, 3}, B = {3, 4} and C = {5, 6}. We observe that $A \cup B \cup c = \{1,2,3\} \cup \{3,4\} \cup \{5,6\} = S$.

Such events A, B and C are called exhaustive events.

CLASSICAL (A PRIORI) DEFINITION OF PROBABILITY:

If an experiment results in total of (m + n)outcomes which are equally likely and mutually exclusive with one another and if 'm' outcomes are favourable to an event 'A' while 'n' are unfavourable, then the probability of occurrence of the event 'A', denoted by P(A), is defined by

$$\frac{m}{m+n} = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

i.e. $P(A) = \frac{m}{m+n}$.

Note that $P(\bar{A})$ or $P(A')$ or $P(A^C)$, i.e. probability of non-occurrence of

$$A = \frac{n}{m+n} = 1 - P(A)$$

In the above we shall denote the number of outcomes favourable to the event A by n(A) and the total number of outcomes in the sample space S by n(S).

$$\therefore P(A) = \frac{n(A)}{n(S)}.$$

If	$P(A) = 0$	Event is impossible
	$P(A) = 1$	Event is sure
	$P(A) \neq 1$ and $P(A) \neq 0$	

Note :

(i) More is the probability of an event, more are chances of its happening.

(ii) $P(\phi) = 0$ & $P(S) = 1$ i.e. nothing outside sample space can occur.

Throwing A pair Dice

Sample Space :

1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

Q. **If two dice are thrown, find the probability that :**
(i) sum equals 2

Sol. n(s) = 36

Number of favourable outcomes = 1 · (1, 1)

$\therefore$ Probability = p = $\frac{1}{36}$

Q. **(ii) sum equals 3**

Sol. Number of favourable outcomes

= 2 $((1, 2), (2, 1))$

$\Rightarrow p = \frac{2}{36} = \frac{1}{18}$

Q. **(iii) sum equals 4**

Sol. Favourable outcomes : $(1,3),(2,2),(3,1)$

$\Rightarrow p = \frac{3}{36} = \frac{1}{12}$

Q. **(iv) sum equals 12**

Sol. Favourable outcomes : (6, 6)

$\Rightarrow p = \frac{1}{36}$

Q. **(v) sum equals 6**

Sol. Favourable outcomes : $(1,5),(2,4)(3,3),(4,2),(5,1)$

$$\Rightarrow p = \frac{5}{36}$$

Q. **(vi) sum is a prime number**

Sol. Favourable outcomes : (1, 1), (1, 2), (1, 4),(1, 6),(2, 1),(2, 3), (2, 5),(3, 2), (3, 4), (4, 1), (4, 3),(5, 2), (5, 6),(6, 1),(6, 5)

$$\Rightarrow p = \frac{15}{36} = \frac{5}{12}$$

Q. **(vii) we get a Doublet**

Sol. Favourable outcomes : (1, 1), (2, 2), (3, 3),(4, 4),(5, 5),(6, 6)

$$\Rightarrow p = \frac{6}{36} = \frac{1}{6}$$

Q. **(viii) Two numbers appearing on dice are different.**

Sol. Two different number means not getting a doublet.

$$\Rightarrow p = 1 - \frac{1}{6} = \frac{5}{6}$$

A coin is Tossed four times

Sample space:

HHHH, HHHT, HHTH, HTHH, THHH, HHTT, HTHT, HTTH, TTHH, THTH, THHT, HTTT, THTT, TTHT, TTTH, TTTT (Total 16 outcomes)

Q. **A coin is tossed 4 times. Find probability of getting**
(i) Exactly 0 head

Sol. Exactly 0 head = TTTT

$$\Rightarrow p = \frac{1}{16}$$

Q. **(ii) Exactly 1 head**

Sol. Exactly 1 head ≡ HTTT, THTT, TTHT, TTTH

$\Rightarrow p = \frac{4}{16} = \frac{1}{4}$

Q. **(iii) Exactly 2 head**

Sol. Let us fill results in 4 positions:

_ _ _ _

For 2 heads, number of cases $= {}^4C_2 = 6$

$\Rightarrow p = \frac{6}{16} = \frac{3}{8}$

Q. **(iv) Exactly 3 head**

Sol. Just like previous case, for 3 heads, number of cases $= {}^4C_3 = 4$.

$\Rightarrow p = \frac{4}{16} = \frac{1}{4}$

Q. **(v) Exactly 4 head**

Sol. Probability of exactly 4 head $= \frac{1}{16}$ (obvious).

Q. **(vi) 2 consecutive head**

Sol. 2 consecutive head : HHTT,HHTH,THHT,TTHH,HTHH.

$\Rightarrow$ There are 5 such cases.

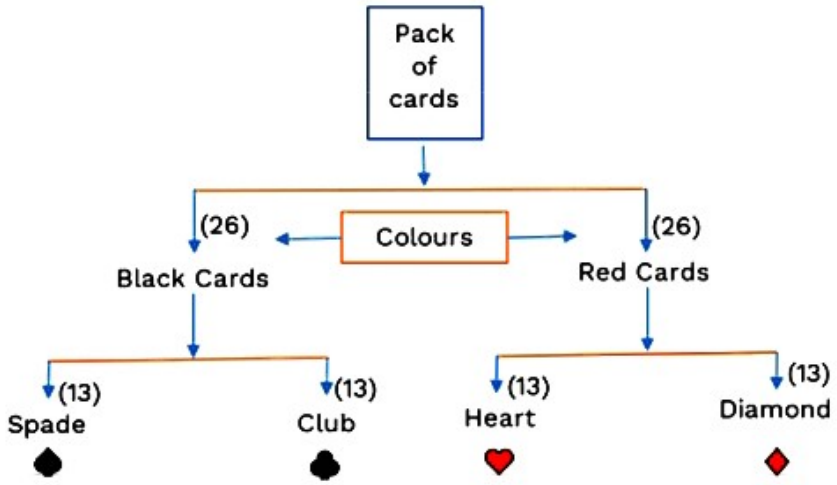
$\Rightarrow p = \frac{5}{16}$

Designation of cards :

Colours : There are two colours. Red & Black

Suits : There are four(4) suits (types)

Each suit contains 13 cards



Recognition of Cards :

	K King	Q Queen	J Jack	A Ace
♥	1	1	1	1
♦	1	1	1	1
♠	1	1	1	1
♣	1	1	1	1
	4	4	4	4

(i) Face Cards:

Face cards contain 12 cards all of K, Q and J having designed a figure of a person i.e., face cards = 4 + 4 + 4 = 12,

(ii) Honours cards:

It contains all face cards and also a card marked A.

i.e. Honours cards = (4 + 4 + 4) + 4 = 16 cards.

(iii) Knave cards:

(10, J, Q) = 4 + 4 + 4 = 12 cards

Comparative study of Equally likely, Mutually, Exclusive and Exhaustive events.

	Experiment	Events	E/L	M/E	Exhaustive
1	Throwing of a dice	A: throwing an odd face {1, 3, 5} B: throwing a composite face {4, 6}	No	Yes	No
2	A ball is drawn from an urn containing 2W, 3R and 4G balls	E_1 : getting a W ball E_2 :getting a R ball E_3 : getting a G ball	No	Yes	Yes
3	Throwing a pair of dice	A : throwing a doublet {11, 22,33,44,55,66} B : throwing a total of 10 or more {46, 64, 55, 56, 65,66}	Yes	No	No
4	From a well shuffled pack of cards a card is drawn	E_1 : getting a heart E_2 : getting a spade E_3 : getting a diamond E_4 : getting a club	Yes	Yes	Yes
5	From a well shuffled pack of cards a card is drawn	A= getting a heart B= getting a face card	No	No	No

Q. **An old man while dialling a seven digit telephone number, after having dialled the five digits, suddenly forgets the last two. But he remembered that the last two digits were different . On this assumption he randomly dials the last two digits. What is the probability that the correct telephone number is dialled?**

Sol. Note that total number of ways in which the last two digit (different) can be dialled is $10 \times 9 = 90$. Out of these 90 Equally likely / Mutually exclusive and exhaustive outcomes only one of them favours happening of the event'' correct telephone is dialed''. Hence $P(E) = \frac{1}{90}$.

What the probability would have been if he did not even remember the last two digits were different:

Here $n(S) = 10 \times 10 = 100$ Hence $P(E) = \frac{1}{100}$.

Q. **4 Apples and 3 Oranges are randomly placed in a line. Find the chances that the extreme fruits are both oranges.**

Sol. $n(S)=\frac{7!}{4!3!};n(A)=\frac{5!}{4!}\Rightarrow P=\frac{5!}{4!}.\frac{4!3!}{7!}=\frac{1}{7}$

Note that whether fruits of the same species are different or alike that probability of the required event remains the same.

Q. **Two natural numbers are randomly selected from the set of first 20 natural numbers. Find the probability that (A) their sum is odd (B) sum is even (C) selected pair is twin prime.**

Sol. $S=\{1,2,3,...,19,20\};n(S)={}^{20}C_2$

$n(A)={}^{10}C_1.{}^{10}C_1=100\Rightarrow P(A)=\frac{100}{190}=\frac{10}{19}$ (sum odd $\Rightarrow$ one odd and one even)

$n(B)={}^{10}C_2+{}^{10}C_2=2.{}^{10}C_2=90\Rightarrow P(B)=\frac{90}{190}=\frac{9}{19}$

(sum even $\Rightarrow$ both odd or both even)

$n(C)=\{(3,5),(5,7),(11,13),(17,19)\}\Rightarrow P(C)=\frac{4}{190}=\frac{2}{95}$

Q. **What is the chance that fourth power of an integer chosen randomly ends in the digit six?**

Sol. Any integer randomly selected can end in 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. These are Equally likely / Mutually exclusive and Exhaustive cases. Out of these 10 case only four cases, when the integer ends in 2, 4, 6 and 8 favours happening of the required event. Hence

$$P(\text{required event})=\frac{4}{10}=40\%$$

It will be incorrect to think this problem as:

4th power of an integer can end in 0, 1, 5 and 6. Hence the probability $=\frac{1}{4}$ which is wrong. Note that four events are Mutually Exclusive and exhaustive but not equally likely hence the definition of probability can not be based on them. Infact 4th power of an integer.

ending in '0' is favoured by only 1 case {0}
ending in '1' is favoured by only 4 cases {1, 3, 7, 9}
ending in '5' is favoured by only 1 cases {5}
ending in '6' is favoured by only 4 cases {2, 4, 6, 8}

$\Rightarrow P(0)=\frac{1}{10};P(1)=\frac{4}{10};P(5)=\frac{1}{10};P(6)=\frac{4}{10}$

Q. **Pair of dice has been rolled/thrown /cast once. Find the probability that atleast one of the dice shows up the face one.**

Sol. There are four Mutually Exclusive and Exhaustive cases

E_1 : 1st dice only shows up the face one.

E_2 : 2nd dice only shows up the face one.

E_3 : both dice shows up the face one.

E_4 : None of the dice shows up the face one.

Out of these, first 3 cases favours happening of the required event. Hence

$$P(\text{required even}) = 1 - P(E_4) = 1 - \frac{5 \times 5}{36} = \frac{11}{36}$$

Note that E_1, E_2, E_3, E_4 are not equally likely.

Q. **A leap year is selected at random. Find the probability that it has**
(A) 53 Sundays
(B) 53 Sundays and Mondays
(C) 53 Sundays or 53 Mondays

Sol. Leap year means which is divisible by 4 if it is not a century year. If it is a century year it must be divisible by 400 as well. A leap year has 366 days out of which 364 days are consumed for 52 weeks i.e. 52 times S, M, T, W, Th, F and Sat. Remaining 2 days of the leap year can be SM, MT, TW, WTh, ThF, F Sat and Sat Sun.

$$\Rightarrow P(A) = \frac{2}{7}; P(B) = \frac{1}{7}; P(C) = \frac{3}{7}$$

Q. **A card is drawn randomly from a well shuffled pack of 52 cards. Find the probability that the drawn card is "neither a heart nor a face card".**

Sol. Note that there are 22 cards which either H or face cards (All K, Q and J) hence

$$P(\text{either a H or Face card}) = \frac{22}{52} = \frac{11}{26}$$

$$\therefore \quad P(\text{neither a H nor FC}) = 1 - \frac{11}{26} = \frac{15}{26}$$

It is to be noted that

$$P(\text{not A or } \bar{A} \text{ or } A^c) = 1 - P(A)$$

Note that A and A^c makes an event a sure event and probability of a event is one.

Q. **6 married couples are enjoying a Birthday party.**
(i) 4 are chosen at random. Find the probability that there is one married couple.

Sol. Total number of possible outcomes
$= {}^{12}C_4$ (Selecting 4 out of 12).

For exactly one married couple to be chosen, number of ways

$$= \underbrace{{}^{6}C_1}_{\text{Selecting 1 out of 6 couple}} \times \underbrace{{}^{5}C_2 \times^2 C_1 \times^2 C_1}_{\text{Selecting 2 couple out of rest of 5 couples and selecting 1 person out of each}}$$

$$\therefore p = \frac{{}^{6}C_1 \times^5 C_2 \times^2 C_1 \times^2 C_1}{{}^{12}C_4}$$

$$= \frac{6 \times 10 \times 2 \times 2}{495}$$

$$= \frac{16}{33}$$

Q. **(ii) 4 are chosen at random. Find the probability that there are 2 married couples.**

Sol. Number of ways of selecting 2 couples $= {}^{6}C_2$

$$\Rightarrow \frac{{}^{6}C_2}{{}^{12}C_4} = \frac{15}{495} = \frac{1}{33}$$

Q. **(iii) 3 are chosen at random. Find the probability that there is one married couple.**

Sol. Total number of ways $= {}^{12}C_3$

Number of ways of selecting one couple and one other person

$$= {}^{6}C_1 \times \underbrace{{}^{10}C_1}_{\text{Selecting one out of remaining 10}}$$

$$\Rightarrow p = \frac{{}^{6}C_1 \times^{10} C_1}{{}^{12}C_3} = \frac{60}{220} = \frac{3}{11}$$

Q. **Consider the path in 2-D, where roads are parallel to coordinate axes at each integral x and y. Sweety has to reach (4,5) from (0,0). At point (2,2), there is a wolf. Find the probability that Sweety reaches safely.**

Sol. Total number of way $= {}^{4+5}C_4 = {}^9C_4$

To reach safely, sweety has to avoid (2,2).

Number of ways of going through (2,2) to (4,5) is $\underbrace{\left({}^{2+2}C_2\right)}_{\text{number of ways to go from (0,0) to (2,2)}} \quad \underbrace{\left({}^{2+3}C_3\right)}_{\text{number of ways to go from (2,2) to (4,5)}}$

So, number of ways of reaching (4,5) safely $= {}^9C_4 - \left({}^4C_2\right)\left({}^5C_2\right)$

$$\Rightarrow \text{Probability} = \frac{{}^9C_4 - \left({}^4C_2\right)\left({}^5C_2\right)}{{}^9C_4}$$

$$= \frac{126 - 60}{126} = \frac{66}{126}$$

$$= \frac{11}{21}$$

Q. **2 numbers are randomly selected from the set of first six natural number. Find the Probability that the selected pair is coprime.**

Sol. Total number of ways $= {}^6C_2 = 15$

Possible coprime pairs are (1, 2),(1, 3),(1, 4),(1, 5),(1, 6),(2, 3),(2, 5),(3, 4)(3, 5), (4, 5), (5, 6) i.e total 11 pairs.

$\therefore$ probability$= \frac{11}{15}$

Q. **Let S = {1, 2, 3,...,20}. A subset B of S is said to be "nice", if sum of the elements is 203. Then the probability that a randomly chosen subset of S is 'nice' is :**

Sol. Number of subsets of $S = 2^{20}$

Sum of elements in $S = 1 + 2 + ... + 20 = \frac{20(21)}{2}$

$$= 210$$

So, we must remove the elements with total sum 7 to get 203.

Such cases are $S - \{7\}, S - \{6,1\}, S - \{5,2\}, S - \{4,3\}, S - \{4,2,1\}$.

$\therefore$ Number of favourable cases = 5

$$\Rightarrow \text{probability} = \frac{5}{2^{20}}$$

Q. **A has 3 shares in a lottery in which there are 3 prizes and 6 blanks. B has 1 share in a lottery in which there is 1 prize and 2 blanks. Then the ratio of A's chance of success to B's chance of sucess is**

(A) 1 : 1 **(B) 1 : 9** **(C) 6 : 13** **(D) 16 : 7**

Sol. **(D)**

A and B will succeed as long as they have atleast 1 prize respectively.

So, for A, total number of outcomes $= {}^{9}C_3 = 84$

Number of ways of losing $= {}^{6}C_3 = 20$

($\because$ choosing 3 out of 6 blanks).

$\therefore$ number of favourable outcomes for A = 84 − 20 = 64

$\Rightarrow$ Probability of A winning $= \frac{64}{84} = \frac{16}{21}$

Similarly, probability of 'B' winning $= \frac{1}{3}$

Ratio of A's success to B's success $= \frac{\frac{16}{21}}{\frac{1}{3}}$

$= \frac{16}{7} = 16 : 7$

Q. **If 2 of the 64 squares are chosen at random in a chessboard, find the probability that they have one side in common.**

Sol. Total possible outcomes $= {}^{64}C_2$

For 2 squares to have a common side, they must be consecutive either vertically or horizontally.

Number of ways of choosing 2 consecutive square in a row = 7

$\Rightarrow$ For 8 rows, number of ways = 8 × 7 = 56

Similarly, number of ways of choosing 2 consecutive square vertically in all 8 columns = 8 × 7 = 56.

$\therefore$ total number of favourable outcomes = 2 × 56

$\Rightarrow$ probability $= \frac{2 \times 56}{{}^{64}C_2} = \frac{2 \times 56}{64 \times 63} \times 2 = \frac{1}{18}$

Q. **Out of 11 consecutive natural number if three numbers are selected at random (without repetition), then the probability that they are in A. P with positive common difference is**

(A) $\frac{10}{99}$ **(B)** $\frac{5}{33}$ **(C)** $\frac{15}{101}$ **(D)** $\frac{5}{101}$

Sol. **(B)**

Let 3 numbers selected be a, b, c

$\Rightarrow b = \frac{a+c}{2}$ = integer (to be in A. P)

$\Rightarrow a + c$ is even $\Rightarrow$ Both a and c are even or both are odd numbers.

So, to select 3 numbers in A. P, just select 1st and 3rd number such that either both are even or both are odd.

The middle number will only have 1 way to be selected.

$\Rightarrow$ Number of such cases = $\underset{(odd)}{{}^{6}C_{2}} + \underset{(even)}{{}^{5}C_{2}}$ (if first number is odd)

Or

$= \underset{(odd)}{{}^{5}C_{2}} + \underset{(even)}{{}^{6}C_{2}}$ (if first number is even)

$= 25$

Total number of possible outcomes ${}^{11}C_{3} = 165$

$\Rightarrow$ Probability $= \frac{25}{165} = \frac{5}{33}$.

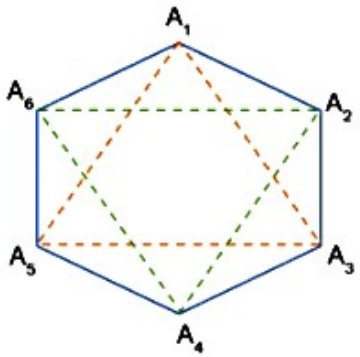
Q. **3 vertices of a regular hexagon are selected randomly. Find the probability that an equilateral triangle is formed.**

Sol.

From above figure, it is clear that number of favourable outcomes = 2.

Number of total outcomes $= {}^{6}C_{3} = 20$

$\Rightarrow$ Probability $= \frac{2}{20} = \frac{1}{10}$



ODDS IN FAVOUR AND ODDS AGAINST OF AN EVENT:

If an experiment has (m + n) as a total number of outcomes which are equally likely, mutually exclusive and exhaustive, and if 'm' outcomes are in favour of an event 'A' and n outcomes are not in favour of that event A, means n outcomes are in against of event A, then we can say-

Odds in favour of event A

$$= \frac{m}{n} = \frac{\text{No. of outcomes which are in favour of event A}}{\text{No. of outcomes which are not in favour of even A}}$$

Odds in against of event A

$$= \frac{n}{m} = \frac{\text{No. of outcomes which are not in favour of event A}}{\text{No. of outcomes which are in favour of even A}}$$

Note : If $P(A) = \frac{a}{b}$ then

(i) odd in favour of event A = a : b – a.

(ii) odds against of event A = b – a : a.

Q. **5 different marbles are placed in 5 different boxes randomly. Find the odds in favour that exactly two boxes remain empty. Given each box can hold any number of marbles.**

Sol. $n(S) = 5^5$; For computing favourable outcomes.

2 boxes which are to remain empty, can be selected in ${}^5C_2$ ways and 5 marbles can be placed in the remaining 3 boxes in group of 221 or 311 in

$$3! \times \left[\frac{5!}{2!2!2!} + \frac{5!}{3!2!}\right] = 150 \text{ ways}$$

$$\therefore P(E) = {}^5C_2 \times \frac{150}{5^5} = \frac{12}{25}$$

Hence, odds is favour of event E = 12 : 13

Q. **A bag contains 5 red and 4 white balls. 3 balls are drawn randomly. Find the odds against these being all red.**

Sol. Total number of outcomes $= {}^{5+4}C_3 = {}^9C_3 = 84$

Number of ways of selecting all red balls $= {}^5C_3 = 10$

$\Rightarrow$ Probability of drawing 3 red balls $= \frac{10}{84} = \frac{5}{42}$

$\Rightarrow$ Odds against three red $= \dfrac{1-p}{p}$

$$= \frac{1-\frac{5}{42}}{\frac{5}{42}} = \frac{37}{5} = 37:5$$

Q. **The chance of one event happening is the square of the chance of a 2nd event, but odds against the first are the cubes of odds against the 2nd. Find the chances of each. (Assume that both events are neither sure nor impossible).**

Sol. Let probability of 1st and 2nd event be p^2 and p respectively.

Odds against the 1st event $= \dfrac{1-p^2}{p^2}$

Odds against the 2nd event $= \dfrac{1-p}{p}$

$$\Rightarrow \frac{1-p^2}{p^2} = \left(\frac{1-p}{p}\right)^3$$

$$\Rightarrow 1+p = \frac{(1-p)^2}{p}$$

$$\Rightarrow p+p^2 = 1+p^2-2p$$

$$\Rightarrow 3p = 1 \Rightarrow p = \frac{1}{3}$$

$\Rightarrow$ chance of 1st event $= \dfrac{1}{9}$

$\Rightarrow$ chance of 2nd event $= \dfrac{1}{3}$

DEPENDENT AND INDEPENDENT EVENTS

(i) Two people, one holding a normal dice and the other a coin, throw once, then getting a 6 on normal dice and getting a head on the coin are the example of events which are independent.

(ii) From an urn containing 2R, 3G and 4W balls, a ball is drawn and its colour is noted, the ball is replaced in the urn and another ball is drawn. Getting a red and a red ball on both the occasion are the example of events which are independent.

Definition

Independent events-Events A and B are said to be independent if occurrences or non-occurrence of one does not affect the probability of occurrence or non-occurrence of the other.

(iii) Similar example can be given in playing cards ' getting an ace' and 'an ace' in two successive drawn from a well shuffled pack of 52 cards when the first drawn is replaced in the pack before the second is drawn. If it is not replaced the events become dependent or contingent.

ADDITION THEOREM ON PROBABILITY :

If A and B are two events associated with an experiment then $P(A \cup B)$ is called the sum of the probabilities of all the sample points in $A \cup B$ or probability of occurrence of atleast one of the events from A and B and the expression for $P(A \cup B)$ is called the addition theorem on probability.

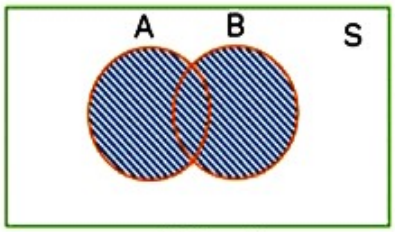
From the Venn diagram it is clear that

P(Occurrence of atleast one of the events from A and B)
P(A or B or both)
or
P(A + B)
$\Rightarrow P(A \cup B)$

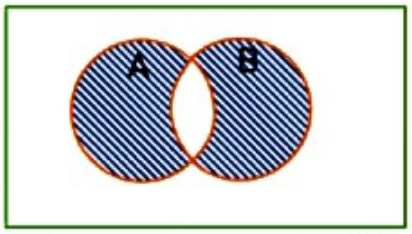
$$= P(A) + P(B) - P(A \cap B)$$
$$= P(A) + P(\bar{A} \cap B)$$
$$= P(B) + (A \cap \bar{B})$$
$$= P(A \cap \bar{B}) + P(A \cap B) + P(\bar{A} \cap B)$$
$$= 1 - P(\bar{A} \cap \bar{B})$$
$$= 1 - P(\overline{A \cup B})$$

P(occurrence of exactly one of the events)
or
P(A or B but not both)

$$\begin{cases} P(A \cap \bar{B}) + P(\bar{A} \cap B) \\ P(A) + P(B) - 2P(A \cap B) \end{cases}$$



$P(A \cup B)$



P(A or B but not both)

Points to remember!!!

(i) If A and B are mutually exclusive events then-

$$P(A \cup B) = P(A) + P(B)$$
$$\{\because P(A \cap B) = 0\}$$

(ii) If A and B are exhaustive events then $P(A \cup B) = 1$

(iii) $P(A \cup B) = 1 - P(\overline{A \cup B})$

Q. **Two students Anil and Ashima appeared in an examination. The probability that Anil will qualify the examination is 0.05 and that Ashima will qualify the examination is 0.10. The probability that both will qualify the examination is 0.02. Find the probability that**

(a) Both Anil and Ashima will not qualify the examination

(b) Atleast one of them will not qualify the examination and

(c) Only one of them will qualify the examination.

Sol. Let E and F denote the events that Anil and Ashima will qualify the examination, respectively. Given that $P(E) = 0.05$, $P(F) = 0.10$ and $P(E \cap F) = 0.02$

Then

(a) The event ' both Anil and Ashima will not qualify the examination' may be expressed as $E' \cap F'$. Since, E' is not E, i.e., Anil will not qualify the examination and F' is ' not F, i.e., Ashima will not qualify the examination.

Also $E' \cap F' = (E \cup F)'$ (By Demorgan's Law)

Now $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

Or $P(E \cup F) = 0.05 + 0.10 - 0.02 = 0.13$

Therefore $P(E' \cap F') = P(E \cup F)' = 1 - P(E \cup F) = 1 - 0.13 = 0.87$

(b) P(atleast one of them will not qualify)

= 1 – P(both of them will qualify)

= 1 – 0.02 = 0.98

(c) The event only one of them will qualify the examination is same as the event either (Anil will qualify, and Ashima will not qualify) or (Anil will not qualify and Ashima will qualify) i.e., $E \cap F'$ or $E' \cap F$, where $E \cap F'$ and $E' \cap F$ are mutually exclusive.

Therefore, P(only one of them will qualify) $= P(E \cap F' \text{ or } E' \cap F)$

$= P(E \cap F') + P(E' \cap F) = P(E) - P(E \cap F) + P(F) - P(E \cap F)$

$= 0.05 - 0.02 + 0.10 - 0.02 = 0.11$

Q. **A and B are any two events such that $P(A) = 0.3, P(B) = 0.1$ and $P(A \cap B) = 0.16$.**

Find the probability that exactly one of the events happens.

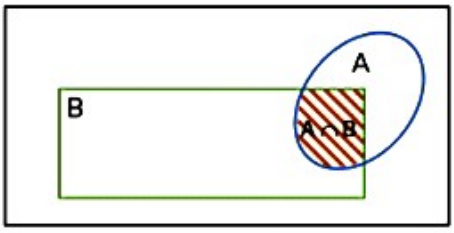
Sol. Exactly one of the events happens $= P(A \cap B')$ or $P(A' \cap B)$

$P(A \cap B') + P(A' \cap B) = P(A) + P(B) - 2P(A \cap B)$

$= 0.3 + 0.1 - 2 \times 0.16 = 0.08$

CONDITIONAL PROBABILITY

Let A and B be any events associated with a random experiment. The probability of occurrence of event A when the even B has already occurred is called as P(A/B). The conditional probability P(A/B) is meaningful only when $P(B) \neq 0$, i.e., when B is not an impossible event.



By definition,

$P\left(\frac{A}{B}\right)$ = Probability of occurrence of event A when the event B has already occurred

$$= \frac{\text{Number of cases favourable to B which are also favourable to A}}{\text{Number of cases favourable to B}}$$

$$\therefore P\left(\frac{A}{B}\right) = \frac{\text{Number of cases favourable to } A \cap B}{\text{Number of cases favourable to B}}$$

$$\text{Also, } P\left(\frac{A}{B}\right) = \frac{\dfrac{\text{Number of cases favourable to } A \cap B}{\text{Number of cases in the sample space}}}{\dfrac{\text{Number of cases favourable to B}}{\text{Number ofases in the sample space}}}$$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}, \text{ provided } P(B) \neq 0.$$

Similarly, we have

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}, \text{ provided } P(A) \neq 0.$$

Q. **Roll a fair die twice. Let A be the event that the sum of the two rolls equals six, and let B be the event that the same number comes up twice. What is P(A/B)?**
(A) 1/6 **(B) 5/36** **(C) 1/5** **(D) None**

Sol. **(A)**

$A = \{(1,5),(2,4),(3,3),(4,2),(5,1)\}$, $B = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)} = \frac{1}{6}$$

Q. **In a class, 30% of the students failed in Physics, 25% failed in Mathematics and 15% failed in both Physics and Mathematics. If a student is selected at random failed in Mathematics, find the probability that he failed in Physics also.**

Sol. Let A be the event " failed in Physics" and B be the event "failed in Mathematics". We want to find $P\left(\frac{A}{B}\right)$. It is given that

$P(A) = \frac{30}{100}$ and $P(B) = \frac{25}{100}$

Also $P(A \cap B) = \frac{15}{100}$

Therefore $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{15}{100}}{\frac{25}{100}} = \frac{15}{25} = \frac{3}{5}$

Q. **Let A and B be two event such that** $P(A) = 0.3, P(B) = 0.6$ and $P\left(\frac{B}{A}\right) = 0.5$ **Then** $P\left(\frac{\bar{A}}{\bar{B}}\right)$ **equals**

(A) $\frac{3}{4}$ **(B)** $\frac{5}{8}$ **(C)** $\frac{9}{40}$ **(D)** $\frac{1}{4}$

Sol. **(B)**

$P(A \cap B) = P(A)P\left(\frac{B}{A}\right) = (0.3)(0.5) = 0.15$

Now $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.6 - 0.15 = 0.75$

Also $P\left(\frac{\bar{A}}{\bar{B}}\right) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{1 - P(A \cup B)}{1 - 0.6} = \frac{0.25}{0.4} = \frac{250}{400} = \frac{5}{8}$

MULTIPLICATION THEOREM ON PROBABILITY

Let A and B be two events associated with a sample space S. Clearly, the set $A \cap B$ denotes the event that both A and B have occurred. In other words, $A \cap B$ denotes the simultaneous occurrence of the events A and B. The event $A \cap B$ is also written as AB.

We known that the conditional probability of event A given that B has occurred is denoted by $P(A \mid B)$ and is given by

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

From this results, we can write $P(A \cap B) = P(B).P(A \mid B)$ ...(i)

Also, we know that

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}, P(A) \neq 0$$

Thus, $P(A \cap B) = P(A).P(B \mid A)$ ...(ii)

Combining (i) and (ii), we find that

$P(A \cap B) = P(A)P(B \mid A) = P(B)P(A \mid B)$ provided $P(A) \neq 0$ and $P(B) \neq 0$

The above results is known as the **Multiplication rule of probability**.

Theorem-I

Let A and B events associated with a random experiment. If A and B are independent, then show that the events (i) $\bar{A}, B$ (ii) $A, \bar{B}$ (iii) $\bar{A}, \bar{B}$ are also independent.

Proof:

The events A and B are independent.

$\therefore P(A \cap B) = P(A)P(B)$ ...(i)

(i) $(A \cap B) \cap (\bar{A} \cap B) = (A \cap \bar{A}) \cap (B \cap B) = \phi \cap B = \phi$

and $(A \cap B) \cup (\bar{A} \cap B) = (A \cup \bar{A}) \cap B = S \cap B = B$

$\therefore$ The events $A \cap B$ and $\bar{A} \cap B$ are mutually exclusive and their union is B.

$\therefore$ By addition theorem, we have

$P(B) = P(A \cap B) + P(\bar{A} \cap B)$ ...(i)

Points to remember!!!

If A & B independent events then

$p\left(\frac{A}{B}\right) = P(A)$ and $P\left(\frac{B}{A}\right) = P(B)$ and

in this case multiplication theorem $P(A \cap B) = P(A).P(B)$.

$\Rightarrow P(\bar{A} \cap B) = P(B) - P(A \cap B) = P(B) - P(A)P(B)$

$= (1 - P(A))P(B) = P(\bar{A})P(B)$ (Using(i))

$\therefore P(\bar{A} \cap B) = P(\bar{A})P(B)$ i.e., $\bar{A}$ and B are independent.

(ii) $(A \cap B) \cap (A \cap \bar{B}) = (A \cap A) \cap (B \cap \bar{B}) = A \cap \phi = \phi$

and $(A \cap B) \cup (A \cap \bar{B}) = A \cap (B \cup \bar{B}) = A \cap S = A$

$\therefore$ The events $A \cap B$ and $A \cap \bar{B}$ are mutually exclusive and their union is A.

$\therefore$ By addition theorem, we have $P(A) = P(A \cap B) + P(A \cap \bar{B})$ ...(i)

$\Rightarrow P(A \cap \bar{B}) = P(A)P(\bar{B})$ i.e., A and $\bar{B}$ are independent.

(iii) $(\bar{A} \cap B) \cap (\bar{A} \cap \bar{B}) = (\bar{A} \cap \bar{A}) \cap (B \cap \bar{B}) = \bar{A} \cap \phi = \phi$

and $(\bar{A} \cap B) \cup (\bar{A} \cap \bar{B}) = \bar{A} \cap (B \cup \bar{B}) = \bar{A} \cap S = \bar{A}$

$\therefore$ The events $\bar{A} \cap B$ and $\bar{A} \cap \bar{B}$ are mutually exclusive and their union is $\bar{A}$.

$\therefore$ By addition theorem, we have

$P(\bar{A} \cap \bar{B}) = P(\bar{A}) - P(\bar{A} \cap B) = P(\bar{A}) - P(\bar{A})P(B)$

$= P(\bar{A})(1 - P(B)) = P(\bar{A})P(\bar{B})$ (Using(i))

$\therefore P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$ i.e., $\bar{A}$ and $\bar{B}$ are independent.

Q. **A die is thrown. If E is the event ' the number appearing is a multiple of 3' and F be the event 'the number appearing is even' then find whether E and F are independent?**

Sol. We know that the sample space is $S = \{1,2,3,4,5,6\}$

Now $E = \{3,6\}, F = \{2,4,6\}$ and $E \cap F = \{6\}$

Then $P(E) = \frac{2}{6} = \frac{1}{3}, P(F) = \frac{3}{6} = \frac{1}{2}$ and $P(E \cap F) = \frac{1}{6}$

Clearly $P(E \cap F) = P(E).P(F)$

Hence E and F are independent events.

Q. **Three coins are tossed simultaneously. Consider the event E' three head or three tails', F' at least two heads' and G' at most two head'. Of the pairs (E, F), (E, G) and (F, G), which are independent? Which are dependent?**

Sol. The sample space of the experiment is given by

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Clearly $E = \{HHH, TTT\}, F = \{HHH, HHT, HTH, THH\}$

and $G = \{HHT, HTH, THH, HTT, , THT, TTH, TTT\}$

Also $E \cap F = \{HHH\}, E \cap G = \{TTT\}, F \cap G = \{HHT, HTH, THH\}$

Therefore $P(E) = \frac{2}{8} = \frac{1}{4}, P(F) = \frac{4}{8} = \frac{1}{2}, P(G) = \frac{7}{8}$

and $P(E \cap F) = \frac{1}{8}, P(E \cap G) = \frac{1}{8}, P(F \cap G) = \frac{3}{8}$

Also $P(E).P(F) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}, P(E).P(G) = \frac{1}{4} \times \frac{7}{8} = \frac{7}{32}$

and $P(F).P(G) = \frac{1}{2} \times \frac{7}{8} = \frac{7}{16}$

Thus $P(E \cap F) = P(E).P(F)$

$P(E \cap G) \neq P(E).P(G)$ and $P(F \cap G) \neq P(F).P(G)$

Hence, the events (E and F) are independent, and the event (E and G) and (F and G) are dependent.

Q. **A pair of fair dice is thrown. Find the probability that either of the dice shows 2 if the sum is 6.**

Sol. The sample space of the experiment "throwing a pair of fair dice" consists of $36 (= 6 \times 6)$ ordered pair (a, b), where a and b can be any integer from 1 to 6. Let A be the event "2 appears on either of the dice" and B be the event "sum is 6". We want to find $P\left(\frac{A}{B}\right)$. Note that

$A = [(2,b) \mid 1 \leq b \leq 6] \cup [(a,2) \mid 1 \leq a \leq 6]$ and $B = [(1,5), (2,4), (3,3), (4,2), (5,1)]$

Also, $A \cap B = [(2,4), (4,2)]$

Therefore $P(B) = \frac{5}{36}$ and $P(A \cap B) = \frac{2}{36}$

So $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{36}}{\frac{5}{36}} = \frac{2}{5}$

Q. **A jar contains 10 white balls and 6 blue balls, all are of equal size. Two balls are drawn without replacement. Find the probability that the second ball is white if it is known that the first is white.**

Sol. Let E_1 be the event " the first ball drawn is white" and E_2 be the event" the second ball drawn is white again. Then

$$P(E_1) = \frac{10}{16}$$

since 10 out of 10 + 6 balls are white. But, after one ball is chosen, there remain 9 white balls and 6 blue balls. Therefore, the required probability is

$$P\left(\frac{E_2}{E_1}\right) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{\frac{10}{16} \cdot \frac{9}{15}}{\frac{10}{16}} = \frac{9}{15} = \frac{3}{5}$$

Q. **There are four machines and it is known that exactly two of them are faulty. They are tested one by one, in a random order till both the faulty machines are identified. The probability that only two tests are needed is**

(A) $\frac{1}{3}$ **(B)** $\frac{1}{6}$ **(C)** $\frac{1}{2}$ **(D)** $\frac{1}{4}$

Sol. **(A)**

The procedure ends in first two tests if either both are faulty or both are good. Therefore, the probability is

$$= P(G \cap G) + P(F \cap F) = P(G).P\left(\frac{G}{G}\right) + P(F).P\left(\frac{F}{F}\right) = \frac{2}{4}.\frac{1}{3} + \frac{2}{4}.\frac{1}{3} = \frac{1}{3}$$

Q. **If a dice is thrown, find the probability of occurrence of a composite number when even number has already occurred.**

Sol. Let A be event when composite number occurs

Let B be event when even number occurs.

$$P(A) = \frac{2}{6} = \frac{1}{3}$$

$$P(B) = \frac{3}{6} = \frac{1}{2}$$

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}$$

$$\Rightarrow P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

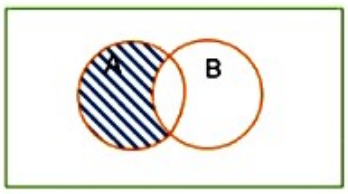
Q. **For any two events A and B, $P(A \cup B) = \frac{7}{8}$, $P(A \cap B) = \frac{1}{4}$ and $P(A^c) = \frac{5}{8}$. Find $P(A), P(B), P(A \cap B^c)$.**

Sol. $P(A) = 1 - P(A^c) = 1 - \frac{5}{8} = \frac{3}{8}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow \frac{7}{8} = \frac{3}{8} + P(B) - \frac{1}{4}$

$\Rightarrow P(B) = \frac{3}{4}$



From above figure, we can see that,

$$P(A \cap B') = P(A - B) = P(A) - P(A \cap B)$$

$$= \frac{3}{8} - \frac{1}{4} = \frac{1}{8}$$

Q. **If A and B are two events with $P(A) = \frac{3}{8}, P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$. Find**

(i) $P(A \cup B)$

Sol. $P(A \cup B) = \frac{3}{8} + \frac{1}{2} - \frac{1}{4} = \frac{5}{8}$

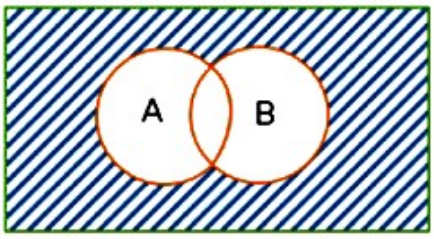
Q. **(ii) $P(A^c)$ and $P(B^c)$**

Sol. $P(A^C) = 1 - \frac{3}{8} = \frac{5}{8}$

$P(B^C) = 1 - \frac{1}{2} = \frac{1}{2}$

Q. **(iii) $P(A^c \cap B^c)$**

Sol. From figure,



$$P(A^c \cap B^c) = 1 - P(A \cup B)$$

$$= 1 - \frac{5}{8} = \frac{3}{8}$$

Q. **(iv) $P(A^c \cup B^c)$**

Sol. $$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c \cap B^c)$$

$$= \frac{5}{8} + \frac{1}{2} - \frac{3}{8}$$

$$= \frac{3}{4}$$

Q. **(v) $P(A \cap B^c)$**

Sol. $$P(A \cap B^c) = P(A) - P(A \cap B)$$

$$= \frac{3}{8} - \frac{1}{4} = \frac{1}{8}$$

Q. **(vi) $P(B \cap A^c)$**

Sol. $$P(B \cap A^c) = P(B) - P(A \cap B)$$

$$= \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

Q. **$P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$. Find**

(i) $P\left(\frac{A}{B}\right)$

Sol. $$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{1}{3}} = \frac{3}{4}$$

Q. (ii) $P\left(\frac{B}{A}\right)$

Sol. $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$

Q. (iii) $P(A \cup B)$

Sol. $P(A \cup B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$

Q. (iv) $P\left(\frac{A^c}{B^c}\right)$

Sol. $P\left(\frac{A^c}{B^c}\right) = \frac{P\left(A^c \cap B^c\right)}{P\left(B^c\right)} = \frac{P\left((A \cup B)^c\right)}{P\left(B^c\right)}$

$$= \frac{1 - P(A \cup B)}{1 - P(B)}$$

$$= \frac{1 - \frac{7}{12}}{1 - \frac{1}{3}} = \frac{\frac{5}{12}}{\frac{2}{3}} = \frac{5}{8}$$

Q. (v) $P\left(\frac{B^c}{A^c}\right)$

Sol. $P\left(\frac{B^c}{A^c}\right) = \frac{P\left(B^c \cap A^c\right)}{P\left(A^c\right)} = \frac{P\left((B \cup A)^c\right)}{1 - P(A)}$

$$= \frac{1 - \frac{7}{12}}{1 - \frac{1}{2}}$$

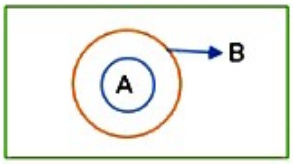
$$= \frac{\frac{5}{12}}{\frac{1}{2}}$$

$$= \frac{5}{6}$$

Q. **Find $P\left(\frac{B}{A}\right)$ if**

(i) A is subset of B

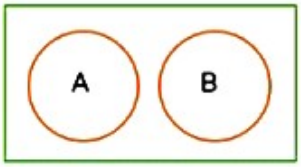
Sol. $\Rightarrow P(A \cap B) = P(A)$



$$\Rightarrow P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = 1$$

Q. **(ii) A and B are disjoint**

Sol. $\Rightarrow P(A \cap B) = 0$



$$\Rightarrow P\left(\frac{B}{A}\right) = \frac{0}{P(A)} = 0$$

Q. **Assume that each born child is equally likely to be a boy or a girl. If two families have two children each, then the conditional probability that all children are girls given atleast two are girls is**

(A) $\frac{1}{11}$ **(B)** $\frac{1}{17}$ **(C)** $\frac{1}{10}$ **(D)** $\frac{1}{12}$

Sol. Probability of boy = Probability of a girl $= \frac{1}{2}$



$\Rightarrow$ Total 4 places.

$$\text{Required conditional probability} = \frac{P(\text{All girls})}{P(\text{All girls}) + P(3\text{girls } 1\text{ boy}) + P(2\text{girls } 2\text{boys})}$$

$$= \frac{\left(\frac{1}{2}\right)^4}{\left(\frac{1}{2}\right)^4 + {}^4C_3\left(\frac{1}{2}\right)^4 + {}^4C_2\left(\frac{1}{2}\right)^4}$$

$$= \frac{1}{1+4+6} = \frac{1}{11}$$

Q. **A lot contains 12 items of which 4 are defective. 3 items are drawn one after the other. Find the probability that all 3 are non-defective.**

Sol. 4 items are defective and 8 items are non-defective.
Let probability of drawing non-defective on 1st ,2nd , 3rd draw be A_1, A_2 & A_3 respectively.

$$\Rightarrow P(A_1 \cap A_2 \cap A_3) = P(A_1)P\left(\frac{A_2}{A_1}\right)P\left(\frac{A_3}{A_1 \cap A_2}\right)$$

1st draw 2nd draw 3rd draw

$$P(A_1) = \frac{8}{12} \quad P\left(\frac{A_2}{A_1}\right) = \frac{7}{11} \quad P\left(\frac{A_3}{A_1 \cap A_2}\right) = \frac{6}{10}$$

$$\Rightarrow P(A_1 \cap A_2 \cap A_3) = \frac{8}{12} \times \frac{7}{11} \times \frac{6}{10} = \frac{14}{55}$$

Q. **The probability of student getting an "A" in the first quiz is 0.2. Given that a student got an "A" in the first quiz, the probability that she will get an "A" also in the 2nd quiz is 0.7. What is the probability of a student getting "A"s in both the 1st and 2nd quizzes?**
(A) 0.7 **(B) 0.2** **(C) 0.14** **(D) 0.35**

Sol. Let the event of student getting "A" in 1st and 2nd quiz be A_1 and A_2 respectively.
$\Rightarrow$ Probability of a student getting "A" in both quizzes $= P(A_1 \cap A_2)$

$$= P(A_1).P\left(\frac{A_2}{A_1}\right)$$

$P(A_1) = 0.2$ and $P\left(\frac{A_2}{A_1}\right) = 0.7$ (Given)

$\Rightarrow$ Required probability = (0.2)(0.7)=0.14

Q. **A fair die is rolled twice. Let A be the event that the sum of two rolls equals 5 and let B the event that an even number comes up in the first roll. State whether events A and B are independent or not.**

Sol. Favourable outcomes of A : $(1,4);(2,3);(3,2);(4,1)$

$$\Rightarrow P(A) = \frac{4}{36} = \frac{1}{9}$$

$$P(B) = \frac{1}{2}$$

$$P(A \cap B) = \frac{2}{36} = \frac{1}{18}\left((2,3) \ \& \ (4,1)\right)$$

$$= \left(\frac{1}{2}\right).\left(\frac{1}{9}\right) = P(B).P(A)$$

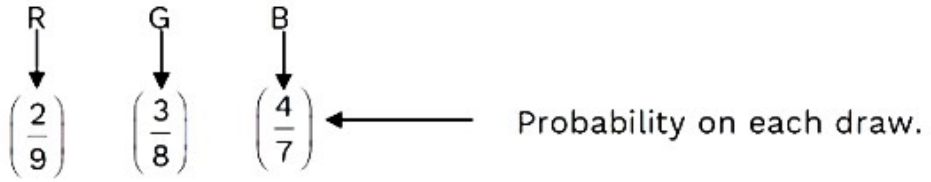
$\Rightarrow$ A and B are independent.

Q. **Probability that an anti-air craft gun can hit an enemy plane at the 1st , 2nd , 3rd shot are 0.6, 0.7 and 0.1 respectively. Find the probability that the gun hits the plane?**

Sol. Required Probability=1-P(gun does not hit the plane in any shot)
Since all the shorts are independent of each other, probability of not getting shot = (0.4)(0.3)(0.9)=0.108
$\Rightarrow$ Required probability = 1−0.108=0.892

Q. **A bag contains 2 red and 3 green and 4 black balls. 3 balls are randomly drawn. Find the probability that they are found of different colours**
(i) When not replaced

Sol. Let the order of balls drawn be R G B
Where R is red, G is green and B is black.



$\Rightarrow$ Probability of this order $= \left(\frac{2}{9}\right)\left(\frac{3}{8}\right)\left(\frac{4}{7}\right) = \frac{1}{21}$

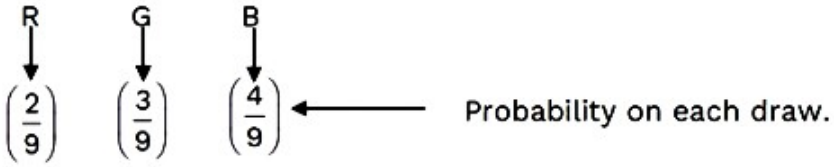
There is total 3! number of orders.

$\Rightarrow$ Total probability $= 3!\left(\frac{1}{21}\right)$

$$= \frac{6}{21} = \frac{2}{7}.$$

Q. **(ii) When replaced**

Sol. If balls are replaced,



(Total number of balls available remains constant)

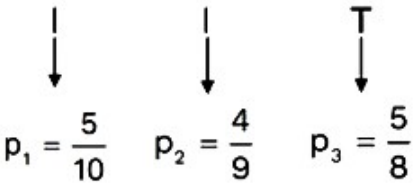
$\Rightarrow$ probability of this order $= \left(\frac{2}{9}\right)\left(\frac{3}{9}\right)\left(\frac{4}{9}\right)$

$\Rightarrow$ Again, since there are 6 types of order,

Total probability $= 6 \times \frac{8}{243} = \frac{16}{81}$.

Q. **A purse contains 10 tickets, five printed with I and five printed with T. 3 tickets are drawn without replacement and arranged in the same order in which they are drawn on the table. Find the probability that IIT is formed.**

Sol.



Where p_1 is probability of getting I on 1st draw, p_2 is probability of getting I after an I was drawn in 1st draw and p_3 is probability of getting T in 3rd draw after I in 1st and 2nd draw.

$\Rightarrow$ Required probability $= p_1p_2p_3$

$$= \frac{5}{10} \times \frac{4}{9} \times \frac{5}{8} = \frac{5}{36}$$

Q. **A problem in mathematics is given to A and B who solve it independently. If probability of A solving it is $\frac{1}{2}$ and probability of B solving it is $\frac{2}{3}$, find the probability that the problem is solved.**

Sol. Problem will be solved if atleast one of A and B solve the problem.
Let P(A) be probability that A solves the problem.
Let P(B) be probability that B solves the problem.

$$\Rightarrow \text{Required Probability} = P(A \cup B)$$
$$= 1 - P(\bar{A} \cap \bar{B})$$
$$= 1 - P(\bar{A})(\bar{B}) \quad \{\text{independent events}\}$$
$$= 1 - \left(\frac{1}{2}\right)\left(\frac{1}{3}\right)$$
$$= \frac{5}{6}$$

Q. **A person takes 3 tests in succession. The probability of his passing the first test is p, that of his passing each successive test is p or $\frac{p}{2}$ according as he passes or fails in the preceding one. He gets selected provided he passes atleast two tests. Determine the probability that the person is selected.**

Sol. Let S and F be the events where person passes or fails the test respectively.
$\Rightarrow$ favourable outcomes for a person to be selected are : SSS, SSF, SFS, FSS

$$P(SSS) = p^3$$
$$P(SSF) = p^2(1-p)$$
$$P(SFS) = p(1-p)\left(\frac{p}{2}\right) = \frac{p^2(1-p)}{2}$$
$$P(FSS) = (1-p)\left(\frac{p}{2}\right)p = \frac{p^2(1-p)}{2}$$

$$\text{Total probability} = p^3 + p^2(1-p) + \left(\frac{p^2(1-p)}{2}\right)2$$
$$= p^3 + 2p^2(1-p) = p^2(2-p)$$

Q. **An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is**
(A) 2, 4 or 8 **(B) 3, 6 or 9** **(C) 4 or 8** **(D) 5 or 10**

Sol. **(D)**

Let B consists of n outcomes and $A \cap B$ consists of m outcomes.

If A and B are independent, then

$$P(A \cap B) = P(A)P(B)$$

$$\Rightarrow m\left(\frac{1}{10}\right) = \left(\frac{4}{10}\right)\left(\frac{n}{10}\right)$$

$\left(\text{probability of each outcome is } \frac{1}{10}\right)$

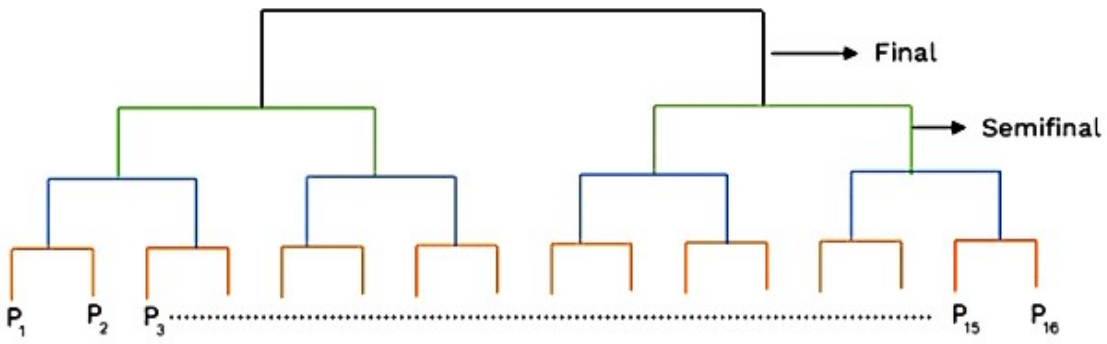
$$\Rightarrow m = \frac{2n}{5}$$

$\Rightarrow$ n is multiple of 5

$\Rightarrow$ n can be 5 or 10.

Q. **16 players take part in tennis tournament. The order of matches is chosen at random. There is always a player better, than other one. The better wins. Find the probability that**
(i) best 4 reaches to semi-final.
(ii) P_1 and P_2 plays finals.
(iii) Sixth best reaches to finals.

Sol.



Total number of ways, players can be divided into 4 groups of 4 players each is

$$\frac{16!}{(4!)^4\,4!}$$

(i) For best 4 to reach semifinal, each of then must be assigned into different groups.

$$\text{Number of ways} = \left(\frac{12!}{(3!)^4\,4!}\right)4!$$

(Assigning 3 players each to best 4 players)

$$\Rightarrow \text{Probability} = \frac{\frac{12!}{(3!)^4}}{\frac{16!}{(4!)^4\,4!}}$$

$$= \frac{\left(4^4\right)4!}{13\times14\times15\times16}$$

$$= \frac{4\times4\times4\times3\times2}{13\times14\times15}$$

$$= \frac{64}{455}$$

(ii) For P_1 and P_2 to be in finals, they must not be in the same group of 8.

$$\text{Number of ways so that they are in different group} = \left(\frac{14!}{(7!)^2\,2!}\right)2!$$

$$\text{Total number of ways of dividing players into 2 groups of 8} = \frac{16!}{(8!)^2\,2!}$$

$$\Rightarrow \text{Probability} = \frac{\frac{14!}{(7!)^2}}{\frac{16!}{(8!)^2\,(2!)}} = \frac{8\times8\times2}{15\times16} = \frac{8}{15}$$

(iii) For P_6 to reach finals, P_1 to P_5 players must be in other group.

Number of ways of selecting 3 players other than P_6 in group with P_1, P_2, P_3, P_4 and $P_5 = {}^{10}C_3$

$$\text{Total number of ways} = \frac{16!}{(8!)^2\,(2!)}$$

$$\Rightarrow \text{Probability} = \frac{\frac{10!}{7!\,3!}}{\frac{16!}{(8!)^2\,(2!)}}$$

$$= \frac{(10!)(8!)^2\,(2!)}{16!\,7!\,3!}$$

$$= \frac{(8!)\,8}{(11\times 12\times 13\times 14\times 15\times 16)\,3} = \frac{8}{429}$$

Q. **16 players $S_1, S_2, \ldots, S_{16}$ play in a tournament. They are divided into 8 pairs at random. From each pair, a winner is decided on the basis of game played between 2 players of the pair. Assume that all players have equal strength. Find probability that exactly one of players S_1 and S_2 is among the winners.**

Sol. **Case-I:** S_1 and S_2 are a pair

In this a case, exactly one of them will be winner.

$\Rightarrow$ Probability of case I $= \frac{1}{15}$

Case-II: S_1 and S_2 are not a pair.

Probability that both of them loose $= \frac{1}{2}\times\frac{1}{2}$

$$= \frac{1}{4}$$

Similarly, probability that both of them wins $= \frac{1}{4}$

$\Rightarrow$ Probability that exactly one of them wins $= 1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}$

Also, probability of case II $= \frac{14}{15}$

$\Rightarrow$ Total probability $= P_{\text{CaseI}} + P_{\text{CaseII}}$

$$= \frac{1}{15}\times 1 + \frac{14}{15}\times\frac{1}{2} = \frac{8}{15}$$

Q. **Probability that a teacher takes a surprise test is $\frac{1}{3}$. If a student remains absent for two days, then find the probability that he misses.**

(i) exactly one test

(ii) atleast one test

(iii) atmost one test

Sol. Let the event that teacher takes the test be T.

$$\Rightarrow P(T) = \frac{1}{3} \text{ \& } P(\overline{T}) = \frac{2}{3}$$

(i) To miss exactly one test, cases possible for it are: $(T\overline{T})$ and $(\overline{T}T)$.

$$\Rightarrow P(T\overline{T}) + P(\overline{T}T) = \frac{1}{3}\times\frac{2}{3} + \frac{2}{3}\times\frac{1}{3} = \frac{4}{9}$$

(ii) Probability that person misses atleast one test

= 1-probability that person misses no test

$$= 1 - P(\overline{T}\,\overline{T}) = 1 - \frac{2}{3}\times\frac{2}{3} = \frac{5}{9}$$

(iii) Probability that person mises at most 1 test = 1 – probability of missing both test

$$= 1 - P(TT)$$

$$= 1 - \left(\frac{1}{3}\right)\left(\frac{1}{3}\right)$$

$$= \frac{8}{9}$$

Q. **An urn contains 1 Red, 2 Green and 3 Black balls. 3 People A, B & C in order draw one ball from the urn and put it back after noting its colour. They continue doing it indefinitely until one who draws a red ball first wins the game. Compute their respective chances of winning the games.**

Sol. Let event of drawing a red ball be called S(success) and drawing green or black ball be called F (failure).

$$\Rightarrow P(S) = \frac{1}{1+2+3} = \frac{1}{6}$$

$$\Rightarrow P(F) = \frac{2+3}{1+2+3} = \frac{5}{6}$$

$\Rightarrow$ Cases where A wins the game will be:

S or (FFF)S or (FFF)(FFF)S or (FFF)(FFF)(FFF)S and so on.

Thus $P(A \text{ wins}) = \frac{1}{6} + \left(\frac{5}{6}\right)^3\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^6\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^9\left(\frac{1}{6}\right) + \ldots$

$$= \frac{\frac{1}{6}}{1-\left(\frac{5}{6}\right)^3} = \frac{36}{91}$$

$\Rightarrow$ Similarly, cases where B wins are:
FS or (FFF) FS or (FFF) (FFF) FS or ... so on.

$$\Rightarrow P(B \text{ wins}) = \frac{5}{6}\times\frac{1}{6} + \left(\frac{5}{6}\right)^4\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^7\left(\frac{1}{6}\right) + \ldots$$

$$= \frac{\frac{5}{36}}{1-\left(\frac{5}{6}\right)^3} = \frac{30}{91}$$

$$\Rightarrow P(C \text{ wins}) = 1 - P(A \text{ wins}) - P(B \text{ wins})$$

$$= 1 - \frac{36}{91} - \frac{30}{91} = \frac{25}{91}$$

Previous Year's Question

In a game, two players A & B take turns in throwing a pair of fair dice starting with player A and total of scores on the two dice, in each throw is noted. A wins the game if he throws a total of 6 before B throws a total of 7 else B wins the game. The game stops as soon as either of them wins. Probability of A winning the game is: **[JEE Mains 2020]**

(A) $\frac{5}{31}$ (B) $\frac{31}{61}$ (C) $\frac{30}{61}$ (D) $\frac{5}{6}$

Sol. P (getting a sum of six) $= \frac{5}{36}$

P (getting a sum of seven) $= \frac{6}{36} = \frac{1}{6}$

A wins when : A gets total of six on 1st turn

Or

Both A and B fails on their first turn and then A gets total of six

Or

⋮

So on

Let event when A gets the sum of six be called R and when B gets sum of seven be called Q.

$\Rightarrow$ Required Probability $= P(R) + P(\bar{R}\bar{Q}R) + P(\bar{R}\bar{Q}\bar{R}\bar{Q}Q) + \ldots$

$$= \frac{5}{36} + \left(\frac{31}{36}\right)\left(\frac{30}{36}\right)\left(\frac{5}{36}\right) + \left(\frac{31}{36}\times\frac{30}{36}\right)^2\left(\frac{5}{36}\right) + \ldots$$

$$= \frac{\frac{5}{36}}{1-\frac{31}{36}\times\frac{30}{36}} = \frac{\frac{5}{36}}{\frac{216-155}{216}}$$

$$= \frac{30}{61}$$

Q. **A pair of dice is rolled until a total of 5 or 7 is obtained. Find the probability that total of 5 comes before a total of 7.**

Sol. $P(5) = \frac{4}{36} = \frac{1}{9}$ $\quad((1,4),(2,3),(3,2),(4,1))$

$P(7) = \frac{6}{36} = \frac{1}{6}$

Let A be event of getting a sum of 5 and B be event that total sum does not equal to 5 or 7.

$\Rightarrow P(A) = \frac{1}{9}$ and $P(B) = 1 - \frac{4}{36} - \frac{6}{36}$

$$= \frac{26}{36} = \frac{13}{18}$$

Required Probability $= P(A) + P(BA) + P(BBA) + P(BBBA) + \ldots$

$$= \frac{1}{9} + \frac{13}{18}\times\frac{1}{9} + \left(\frac{13}{18}\right)^2\times\frac{1}{9} + \left(\frac{13}{18}\right)^3\times\frac{1}{9} + \ldots$$

$$= \frac{\frac{1}{9}}{1-\frac{13}{18}} = \frac{2}{5}$$

Q. **Two persons A and B one by one in order draw one ball each from a purse containing 5 white (W) and 1 Red (R) balls and retain it. The person who gets a red ball wins the game.**

E: event that A wins the game.

F: event that B wins the game.

Prove that E and F are equiprobable.

Sol. $P(A \text{ wins}) = P(R) + P(WWR) + P(WWWWR)$

$$= \frac{1}{6} + \left(\frac{5}{6} \times \frac{4}{5} \times \frac{1}{4}\right) + \left(\frac{5}{6} \times \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3} \times \frac{1}{2}\right) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

$\Rightarrow P(B \text{ wins}) = 1 - P(A \text{ wins})$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

Q. **All face cards from a pack of 52 cards are removed. From the remaining 40 cards, 4 are drawn . Find the probability**
(i) that all the 4 cards are of the same denomination
(ii) that there are two cards of one denomination, two cards of other denomination and all four are of different suits
(iii) that they are of different suit & different denominations.

Sol. **(i)** Now there are 10 types of denomination available with 4 cards each.

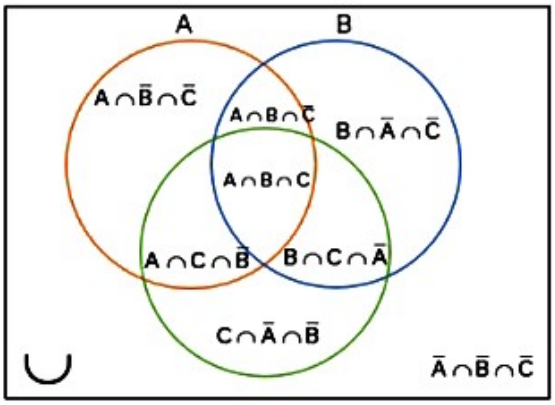
$$\Rightarrow \text{Required probability} = \frac{{}^{10}C_1}{{}^{40}C_4} = \frac{10 \times 4!}{37 \times 38 \times 39 \times 40} = \frac{1}{9139}$$

(ii) Required Probability $= \dfrac{{}^{10}C_2 \times \left(\dfrac{4!}{2!2!2!} \times 2!\right)}{{}^{40}C_4} = \dfrac{{}^{10}C_2 \times \dfrac{4!}{2!2!}}{{}^{40}C_4}$

(iii) Number of favourable ways $\underbrace{{}^{10}C_1}_{\text{heart}} \times \underbrace{{}^{9}C_1}_{\text{spade}} \times \underbrace{{}^{8}C_1}_{\text{club}} \times \underbrace{{}^{7}C_1}_{\text{Diamond}}$.

$$\Rightarrow \text{Probability} = \frac{10 \times 9 \times 8 \times 7}{{}^{40}C_4}$$

THREE EVENTS ASSOCIATED WITH AN EXPERIMENTAL PERFORMANCE:

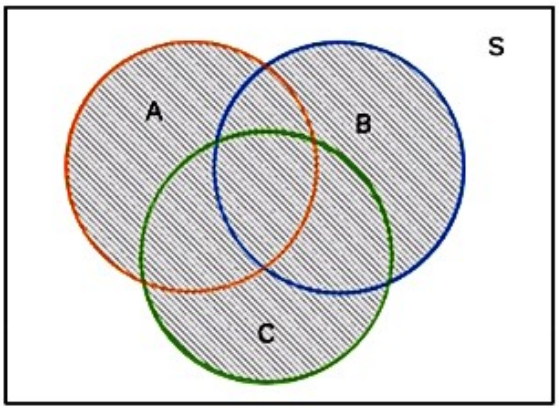


The addition theorem can be extended when three events are associated with the experiment.

If A, B and C are three events then

$P(A \cup B \cup C)$ denotes the sum of probabilities of all the sample points in $(A \cup B \cup C)$ or probability of occurrence of atleast one of the events.

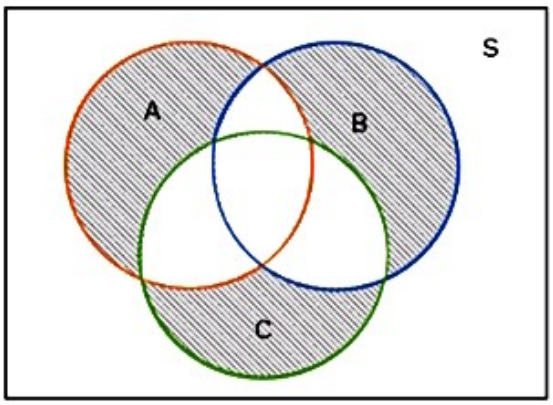
(i) $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$



$(A \cup B \cup C)$

(ii) P(occurrence of exactly one of the event) $= P(E-1)$

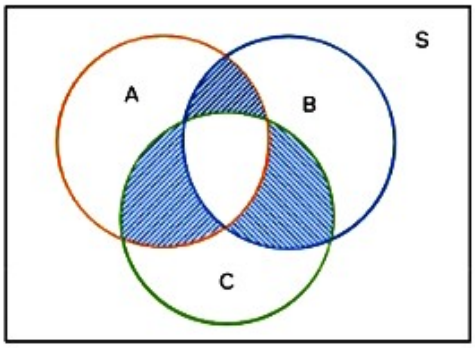
$= P(A) + P(B) + P(C) - 2\left[P(A \cap B) + P(B \cap C) + P(C \cap A)\right] + 3P(A \cap B \cap C)$



$(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)$

(iii) P(occurrence of exactly two of the events) $= P(E-2)$

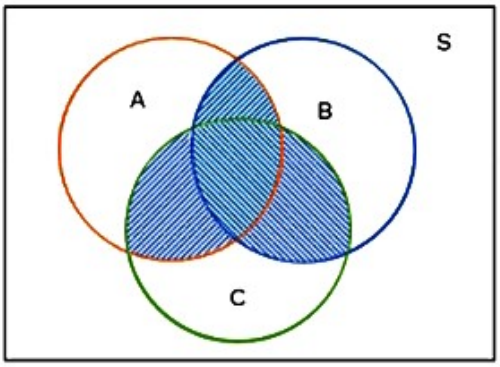
$= P(A \cap B) + P(B \cap C) + P(C \cap A) - 3P(A \cap B \cap C)$



$(A \cap B \cap \bar{C}) \cup (\bar{A} \cap B \cap C) \cup (A \cap \bar{B} \cap C)$

(iv) P(occurrence of atleast two of the events)

$= P(A \cap B) + P(B \cap C) + P(C \cap A) - 2P(A \cap B \cap C)$



$(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C) \cup (A \cap B \cap C)$

Note:

(a) If A, B, C are pair wise independent $\not\Rightarrow$ they are independent. Infact for 3 events A, B and C to be independent they must be

$$P(A \cap B) = P(A) \cdot P(B); P(B \cap C) = P(B) \cdot P(C); P(C \cap A) = P(C) \cdot P(A)$$
$$\text{and } P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

(b) Similarly for n independent events, the number of conditions would be

$${}^{n}C_2 + {}^{n}C_3 + \ldots + {}^{n}C_n = 2^n - n - 1$$

Q. **A, B and C are three newspapers from a city. 25% of the population reads A, 20% reads B, 15% reads C, 16% reads both A and B, 10% reads both B and C, 8% reads both A and C and 4% reads all the three. Find the percentage of the population who read atleast one of A, B and C.**

Sol. We are given that

$$P(A)=\frac{25}{100}, P(B)=\frac{20}{100}, P(C)=\frac{15}{100}$$

$$P(A\cap B)=\frac{16}{100}, P(B\cap C)=\frac{10}{100}, P(C\cap A)=\frac{8}{100} \text{ and } P(A\cap B\cap C)=\frac{4}{100}$$

We have to find $P(A\cup B\cup C)$. We can use the formula

$$P(A\cup B\cup C)=P(A)+P(B)+P(C)-P(A\cap B)-P(B\cap C)-P(C\cap A)+P(A\cap B\cap C)$$

$$=\frac{1}{100}(25+20+15-16-10-8+4)=\frac{30}{100}$$

Thus 30% of the people read atleast one of the newspapers.

Q. **Let A, B and C be three events such that p = P (exactly one of A or B) = P(exactly one of B or C)=P(exactly one of C or A) and P(A, B, C simultaneously) = p^2 where $0<p<\frac{1}{2}$. Then P(at least one of A, B or C) is equal to**

(A) $\frac{3p+2p^2}{2}$ **(B)** $\frac{2p+3p^2}{2}$ **(C)** $\frac{2p+3p^2}{4}$ **(D)** $\frac{3p+2p^2}{4}$

Sol. Exactly one of A or B means A or B but not both

So $P(A)+P(B)-2P(A\cap B)=p$ ...(i)

Similarly, P (exactly one of B or C)

$P(B)+P(B\cap C)-2P(B\cap C)=p$ ...(ii)

and $P(C)+P(A)-P(C\cap A)=p$ ...(iii)

Adding equation (i)-(iii), we have

$$2\left[P(A)+P(B)+P(C)-P(A\cap B)-P(B\cap C)-P(C\cap A)\right]=3p \quad \text{...(iv)}$$

Now P(atleast one A, B or C) is given by

$$P(A\cup B\cup C)=P(A)+P(B)+P(C)-P(A\cap B)-P(B\cap C)-P(C\cap A)+P(A\cap B\cap C)$$

$$=\frac{3p}{2}+p^2 \qquad \left[\text{Equation(iv) and } P(A\cap B\cap C)=p^2\right]$$

$$=\frac{3p+2p^2}{2}$$

Q. **A fair coin is tossed two times.**
Event A: Head occurring on 1st toss.
Event B: Head occurring on 2nd toss.
Event C: Head occurring exactly on one toss.
Show that A, B, C are not independent but they are pair-wise independent.

Sol. Sample space : {HH, HT, TH, TT}

$\Rightarrow P(A) = \frac{1}{2}$

$\Rightarrow P(B) = \left(\frac{1}{2}\right)$

$\Rightarrow P(C) = \frac{1}{2}$

$\Rightarrow P(A \cap B) = \frac{1}{4} = P(A)P(B)$

$\Rightarrow$ A & B independent.

Similarly, $P(B \cap C) = \frac{1}{4} = P(B)P(C)$ and $P(C \cap A) = \frac{1}{4} = P(A)P(C)$

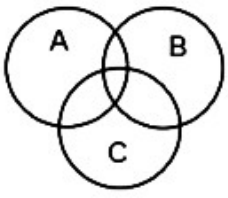
$\Rightarrow$ A, B and C are pairwise independent events.

But $P(A \cap B \cap C) = 0 \neq P(A)P(B)P(C)$.

$\Rightarrow$ A, B and C are not independent.

Q. **100 management students who read atleast one of the 3 business magazines are surveyed to study the readership pattern. It is found that 80 read Business India, 50 read Business world and 30 read Business Today. Five students read all the three magazines. A student was selected randomly. Find the probability that he reads exactly 2 magazines.**

Sol. A : Students reading Business India
B : Students reading Business World.
C : Students Business Today.



$A \cup B \cup C = A + B + C - (A \cap B + B \cap C + C \cap A) + A \cap B \cap C$

$\Rightarrow 100 = 80 + 50 + 30 - \sum(A \cap B) + 5$

$\Rightarrow \sum(A \cap B) = 65$

P(Student reading exactly two magazines)

$= P(A \cap B) + P(B \cap C) + P(C \cap A) - 3P(A \cap B \cap C)$

$= \frac{65 - 3 \times 5}{100} = \frac{1}{2}$

Q. **There are 3 clubs A, B and C in a town with 40, 50 and 60 members. 10 people are members of all three clubs. 70 people are members of exactly one club. A member is randomly selected. Find the probability that he had membership of exactly 2 clubs.**

(A) $\frac{7}{15}$ **(B)** $\frac{1}{6}$ **(C)** $\frac{3}{21}$ **(D)** $\frac{5}{21}$

Sol. E_1 = people with membership in only 1 club.

E_2 = people with membership in 2 clubs.

E_3 = people with membership in 3 clubs.

$E_1 = 70$ $E_3 = 10$

$A \cup B \cup C = E_1 + E_2 + E_3$

$\Rightarrow A \cup B \cup C = 80 + E_2$ ...(i)

$A \cup B \cup C = A + B + C - \sum (A \cap B) + A \cap B \cap C$

$\Rightarrow A \cup B \cup C = 40 + 50 + 60 - \sum (A \cap B) + 10$

$\Rightarrow A \cup B \cup C = 160 - \sum (A \cap B)$ ...(ii)

Also, $E_2 = \sum (A \cap B) - 3(A \cap B \cap C)$ ...(iii)

$= \sum (A \cap B) - 30$

From (i) , (ii) and (iii)

$160 - \sum (A \cap B) = 80 + \sum (A \cap B) - 30$

$\Rightarrow 160 - \sum (A \cap B) = 50 + \sum (A \cap B)$

$\Rightarrow \sum (A \cap B) = 55$

$\Rightarrow E_2 = 25$

$\Rightarrow A \cup B \cup C = 160 - 55 = 105$

$\Rightarrow P(E_2) = \frac{n(E_2)}{n(S)} = \frac{25}{105} = \frac{5}{21}$

Q. **3 persons A, B and C independently fire at a target. Suppose $P(A) = \frac{1}{6}; P(B) = \frac{1}{4}$ and $P(C) = \frac{1}{3}$ denote their probability of hitting the target.**

(i) Find the probability that atleast one of them hits the target.

(ii) Find the probability that exactly one of them hits the target.

(iii) If the target is hit only once, find the probability that it was man A.

Sol. **(i)** P(Atleast one of them hits the target)

=1–P(none of them hit the target)

$$= 1 - P\left(\bar{A} \cap \bar{B} \cap \bar{C}\right)$$

$$= 1 - \left(\frac{5}{6} \times \frac{3}{4} \times \frac{2}{3}\right) \qquad (\because \text{independent events})$$

$$= 1 - \frac{5}{12} = \frac{7}{12}$$

(ii) P(Exactly one of them hits the targets)

$$= P\left(A \cap \bar{B} \cap \bar{C}\right) + P\left(\bar{A} \cap B \cap \bar{C}\right) + P\left(\bar{A} \cap \bar{B} \cap C\right)$$

$$= \frac{1}{6} \times \frac{3}{4} \times \frac{2}{3} + \frac{5}{6} \times \frac{1}{4} \times \frac{2}{3} + \frac{5}{6} \times \frac{3}{4} \times \frac{1}{3}$$

$$= \frac{1}{12} + \frac{5}{36} + \frac{5}{24} = \frac{31}{72}$$

(iii) Let E_1 be event that exactly one hit the target

E_2 : A hits the target not others.

$$\text{Required probability} = P\left(\frac{E_2}{E_1}\right)$$

$$= \frac{P(E_2 \cap E_1)}{P(E_1)}$$

$$= \frac{P\left(A \cap \bar{B} \cap \bar{C}\right)}{P(E_1)}$$

$$= \frac{\frac{1}{12}}{\frac{31}{72}} \qquad (\because \text{(ii) part})$$

$$= \frac{6}{31}$$

BINIOMIAL PROBABILITY:

Let an experiment has n-independent trials, and each of the trial has two possible outcomes :

(i) success

(ii) failure

If probability of getting a success is P(S) = p and probability of getting a failure is P(F) = q such that p + q = 1, then, P(r successes) $= {}^{n}C_{r}p^{r}q^{n-r}$

Proof:

Consider the compound event where r successes are in succession and (n – r) failures are in succession.

$$P\left(\underbrace{SSS...S}_{r}\underbrace{FFF...F}_{(n-r)}\right) = \underbrace{P(S).P(S)...P(S)}_{r \text{ times}}\underbrace{P(F).P(F)...P(F)}_{(n-r)\text{times}} = p^r.q^{n-r}$$

But these r successes and (n−r) failures can be arranged in $\frac{n!}{r!(n-r)!} = {}^nC_r$ ways and in each arrangement the probability will be $p^r.q^{n-r}$

Hence total probability = $P(r) = {}^n C_r p^r q^{n-r}$ ...(1)

Recurrence relation

$$P(r+1) = {}^n C_{r+1} p^{r+1}.q^{n-r-1}$$

$$\therefore \frac{P(r+1)}{P(r)} = \frac{{}^nC_{r+1}}{{}^nC_r}\frac{p}{q} = \frac{n-r}{r+1}\frac{p}{1-p}$$

$$\therefore P(r+1) = \frac{n-r}{r+1}.\frac{p}{1-p}P(r) \quad ...(2)$$

Equation (2) is used for finding the probability of P(1); P(2); P(3); ... etc. once P(0) is determined.

Q. **A pair of dice is thrown 6 times, getting a doublet is considered a success. Compute the probability of**

(i) no success

(ii) exactly one success

(iii) at least one success

(iv) at most one success

Sol. Total possible outcomes = 36

In which six doublets then

$$P = \frac{6}{36} = \frac{1}{6}; q = 1 - \frac{1}{6} = \frac{5}{6}$$

(i) No success for r = 0

$$\therefore P(0) = {}^6 C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^6 = \left(\frac{5}{6}\right)^6$$

(ii) Exactly one success for r = 1

$$\therefore P(1) = {}^6 C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^5 = \left(\frac{5}{6}\right)^5$$

(iii) For at least one success for r = 1, 2, 3, 4, 5, 6.

$$\therefore \sum_{r=1}^{6} {}^6C_r p^r q^{6-r} = {}^6C_1\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^5 + {}^6C_2\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^4 + {}^6C_3\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^3 + {}^6C_4\left(\frac{1}{6}\right)^4\left(\frac{5}{6}\right)^2 + {}^6C_5\left(\frac{1}{6}\right)^5\left(\frac{5}{6}\right)^1 + {}^6C_5\left(\frac{1}{6}\right)^6\left(\frac{5}{6}\right)^0$$

(iv) For at most one success for r =0, 1

$$\sum_{r=0}^{1} {}^6C_r\left(\frac{1}{6}\right)^6\left(\frac{5}{6}\right)^{6-r} = {}^6C_0\left(\frac{1}{6}\right)^0\left(\frac{5}{6}\right)^6 + {}^6C_1\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^5$$

Q. **In a hurdle race a man has to clear 9 hurdles. Probability that he clears a hurdle is 2/3 and the probability that he knocks down the hurdle is 1/3. Find the probability that he knocks down fewer than 2 hurdles.**

Sol. For probability that he knocks down fewer than two hurdles for r = 0, 1

Where $p = \frac{1}{3}, \quad q = \frac{2}{3}$

$$\therefore \sum_{r=0}^{1} {}^9C_r\left(\frac{1}{3}\right)^r\left(\frac{2}{3}\right)^{9-r} = {}^9C_0\left(\frac{1}{3}\right)^0\left(\frac{2}{3}\right)^9 + {}^9C_1\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^8$$

Q. **A drunkard takes a step forward or backward. The probability that he takes a step forward is 0.4. Find the probability that at the end of 11 steps he is one step away from the starting point.**

Sol. At the end of 11 steps, he is one step away from the starting point by two ways

(i) Man has taken 6 steps forward and 5 steps backward

(ii) Man has taken 6 steps backward and 5 steps forward

here p=probability of forward steps $= \frac{2}{5}$

q = probability of backward steps $= \frac{3}{5}$

$$\therefore \text{Probability} = {}^{11}C_6\left(\frac{2}{5}\right)^6\left(\frac{3}{5}\right)^5 + {}^{11}C_5\left(\frac{2}{5}\right)^5\left(\frac{3}{5}\right)^6$$

Q. **100 identical coins each falling head wise with probability p (0 < p < 1) are tossed once. If the probability of 50 coins showing up head is equal to the probability of showing heads on 51 coins, find the value of p.**

Sol. $P(H) = p \Rightarrow P(T) = 1 - p$

Using Binomial Probability distribution,

$P(n = 100, r = 50) = P(n = 100, r = 51)$

$\Rightarrow {}^{100}C_{50} p^{50} (1-p)^{50} = {}^{100}C_{51} p^{51} (1-p)^{49}$

$\Rightarrow \frac{100!}{50!50!}(1-p) = \frac{100!}{51!49!} \times p$

$\Rightarrow \frac{1-p}{50} = \frac{p}{51}$

$\Rightarrow 51 = 101p$

$\Rightarrow p = \frac{51}{101}$

Q. **A rifleman is firing at a distant target with 10% chance of hitting. Number of rounds he must fire so that he has more than 50% chance of hitting is?**

Sol. Let number of rounds fired = n.

Probability of getting atleast 1 hit = 1 – probability of getting 0 hit.

$\Rightarrow 1 - {}^{n}C_0 (0.1)^0 (0.9)^n \geq 0.5$

$\Rightarrow 0.5 \geq (0.9)^n$

$\Rightarrow n \leq 7$

$\therefore$ Rifleman must fire atleast 7 times.

Q. **A coin is likely to land heads twice than tails. In a sequence of 5 independent trials, find the probability that the 3rd head occur on 5th toss.**

Sol. $P(\text{Head}) = P(H) = p = \frac{2}{3}$

$P(\text{Tail}) = P(T) = \frac{1}{3}$

Since 3rd head is to occur on 5th toss, there will be two heads occurring in first four tosses.

$\Rightarrow$ Required Probability = P(2 Head in 4 tosses)× P(Head on 5th toss)

$= \left({}^{4}C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2 \right) \times \frac{2}{3}$

$= 6 \times \frac{4}{9} \times \frac{1}{9} \times \frac{2}{3} = \frac{16}{81}$

Q. **A fair coin is flipped 'n' times. Let E be the event "a Head is obtained on 1st flip". Let F_k be the event "exactly K heads are obtained". For which of the following pairs (n, k), are E and F_k independent?**
(A) (12, 4) **(B) (20, 10)** **(C) (40, 10)** **(D) (100, 51)**

Sol. $P(E) = \frac{1}{2}$

$$P(F_k) = {}^nC_k \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k} = {}^nC_k \left(\frac{1}{2}\right)^n$$

$$P(E \cap F_k) = \frac{1}{2} \times \left({}^{n-1}C_{k-1}\right)\left(\frac{1}{2}\right)^{n-1}$$

(Head on first toss and K-1 head in rest of tosses)
For E and F_k to be independent

$$P(E \cap F_k) = P(E)P(F_k)$$

$$\Rightarrow {}^{n-1}C_{k-1}\left(\frac{1}{2}\right)^n = \frac{1}{2} \times {}^nC_k \left(\frac{1}{2}\right)^n$$

$$\Rightarrow \frac{(n-1)!}{(k-1)!(n-k)!} = \frac{1}{2} \times \frac{n!}{k!(n-k)!}$$

$$\Rightarrow 1 = \frac{1}{2} \times \frac{n}{k}$$

$$\Rightarrow n = 2k$$

$\Rightarrow$ 'B' is the correct option.

Q. **A fair coin is tossed 10 times. Find the probability that head never occurs consecutively.**

Sol. $P(H) = P(T) = \frac{1}{2}$ {H : Head, T : Tail}

Consecutive Head can never occur only in following cases:

Case-I : 0 Head

$$\text{Probability} = p_0 = {}^{10}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{10} = \left(\frac{1}{2}\right)^{10}$$

Case-II : 1 Head

$$p_1 = {}^{10}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^9 = 10\left(\frac{1}{2}\right)^{10}$$

Case-III : 2 Heads

Heads must not be consecutive.

$$\Rightarrow p_2 = \left({}^{10}C_2 - \underbrace{9}_{\text{Number of Cases with consecutive heads}} \right)\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^8 = 36\left(\frac{1}{2}\right)^{10}$$

Case-IV : 3 heads

Using gap Method of permutation and combination,

$$\Rightarrow p_3 = {}^{10-3+1}C_3\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^7 = {}^{8}C_3\left(\frac{1}{2}\right)^{10} = 56\left(\frac{1}{2}\right)^{10}$$

Case-V : 4 Head

Similarly, $p_4 = {}^{10-4+1}C_4\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^6 = {}^{7}C_4\left(\frac{1}{2}\right)^{10} = 35\left(\frac{1}{2}\right)^{10}$

Case-VI : 5 Head

$$\Rightarrow p_5 = {}^{10-5+1}C_5\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^5 = 6\left(\frac{1}{2}\right)^{10}$$

Total probability $= p_1 + p_2 + p_3 + p_4 + p_5 + p_0$

$$= (1 + 10 + 36 + 56 + 35 + 6)\left(\frac{1}{2}\right)^{10}$$

$$= (144)\left(\frac{1}{2}\right)^{10}$$

$$= \frac{9}{2^6} = \frac{9}{64}$$

TOTAL PROBABILITY THEOREM :

Let $E_1, E_2, \ldots E_n$ be n mutually exclusive and exhaustive events, with non-zero probabilities, of a random experiment. If A be any arbitrary event of the sample of the above random experiment with $P(A) > 0$, then

$$P(A) = P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + \ldots + P(E_n)P\left(\frac{A}{E_n}\right)$$

Proof:

Let S be the sample space of the random experiment.

Since $E_1, E_2, \ldots, E_n$ are exhaustive, we have $S = E_1 \cup E_2 \cup \ldots \cup E_n$.

Now $A = S \cap A = (E_1 \cup E_2 \cup \ldots \cup E_n) \cap A$

$$\Rightarrow A = (E_1 \cap A) \cup (E_2 \cap A) \cup \ldots \cup (E_n \cap A) \qquad \ldots(i)$$

Since $E_1, E_2, \ldots, E_n$ are mutually exclusive, we have $E_i \cap E_j = \phi$ for $i \neq j$

Now $(E_i \cap A) \cap (E_j \cap A) = (E_i \cap E_j) \cap A = \phi \cap A = \phi$

$\therefore E_1 \cap A, E_2 \cap A, \ldots, E_n \cap A$ are also mutually exclusive.

By using addition theorem, (i) implies

$$P(A) = P(E_1 \cap A) + P(E_2 \cap A) + \ldots + P(E_n \cap A)$$

$$\Rightarrow P(A) = P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + \ldots + P(E_n)P\left(\frac{A}{E_n}\right).$$

Remarks:

In practical problems, it is found convenient to write as follows:

$P(A) = P(E_1A \text{ or } E_2A \text{ or} \ldots E_nA) = P(E_1A) + P(E_2A) + \ldots + P(E_nA)$

Q. **A box contains three coins; one coins is fair, one coin is two-headed, and one coin is weighted so that the probability of head appearing is 1/3. A coin is selected and tossed. Find the probability that (i) head appears (ii) tail appears.**

Sol. Let E_1, E_2 and E_3 be the events of selecting at random first coin, second coin and third coin respectively.

$\therefore P(E_1) = \frac{1}{3}, P(E_2) = \frac{1}{3}$ and $P(E_3) = \frac{1}{3}$

Let H and T be events of getting head and tail respectively.

$\therefore P\left(\frac{H}{E_1}\right) = \frac{1}{2}, \quad P\left(\frac{T}{E_1}\right) = \frac{1}{2}$ ($\because$ First coin is fair)

$P\left(\frac{H}{E_2}\right) = 1, \quad P\left(\frac{T}{E_2}\right) = 0$ ($\because$ Second coin is two-headed)

(i) P(getting head) $= P(H) = P(E_1H \text{ or } E_2H \text{ or } E_3H)$

$$= P(E_1H) + P(E_2H) + P(E_3H)$$

$$= P(E_1)P\left(\frac{H}{E_1}\right) + P(E_2)P\left(\frac{H}{E_2}\right) + P(E_3)P\left(\frac{H}{E_3}\right)$$

$$= \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times 1 + \frac{1}{3} \times \frac{1}{3} = \frac{11}{18}$$

(ii) P(getting tail) = P(T)

$$= P(E_1T \text{ or } E_3T) = P(E_1T) + P(E_3T) = P(E_1)P\left(\frac{T}{E_1}\right) + P(E_3)P\left(\frac{T}{E_3}\right)$$

$$= \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times \frac{2}{3} = \frac{7}{18}$$

Q. **There are two bags. The first bag contains 5 white and 3 black balls and the second bag contains 3 white and 5 black balls. Two balls are drawn at random from the first bag and are put into the second bag, without noticing their colours. Then two balls are drawn from the second bag. Find the probability that the balls drawn are white and black.**

Sol.



Bag-I Bag-II

Let E_1, E_2 and E_3 be the events of transferring 2 white, 1 white and 1 black, 2 black balls respectively from the first bag to the second bag.

$$\therefore P(E_1) = \frac{{}^5C_2}{{}^8C_2} = \frac{10}{28} = \frac{5}{14}$$

$$P(E_2) = \frac{({}^5C_1) \times ({}^3C_1)}{{}^8C_2} = \frac{5 \times 3}{28} = \frac{15}{28}$$

$$P(E_3) = \frac{{}^3C_2}{{}^8C_2} = \frac{3}{28}$$

Let A be the event of drawing one white and one black ball from the second bag.

$$P(A) = P(E_1A \text{ or } E_2A \text{ or } E_3A)$$

$$= P(E_1A) + P(E_2A) + P(E_3A)$$

$$= P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + P(E_3)\left(\frac{A}{E_3}\right)$$

$$= \frac{5}{14} \times \frac{{}^5C_1 \times {}^5C_1}{{}^{10}C_2} + \frac{15}{28} \times \frac{{}^4C_1 \times {}^6C_1}{{}^{10}C_2} + \frac{3}{28} \times \frac{{}^3C_1 \times {}^7C_1}{{}^{10}C_2}$$

$$= \frac{5}{14} \times \frac{5}{9} + \frac{15}{28} \times \frac{8}{15} + \frac{3}{28} \times \frac{7}{15} = \frac{673}{1260}$$

Q. **Two machines A and B produce respectively 60% and 40% of the total numbers of items of a factory. The percentages of defective output of these machines are respectively 2% and 5%. If an item is selected at random, what is the probability that the item is (i) defective (ii) non-defective?**

Sol. Let E_1, E_2 be the events of drawing an item produced by machine A and machine B respectively. Let A be the event of selecting a defecting item.

$\therefore \bar{A}$ represent the event of selecting a non-defective item.

We have $P(E_1) = 60\%; \quad P(E_2) = 40\%$

$P\left(\frac{A}{E_1}\right)$ = Probability that an item produced by A is defective = 2%

$P\left(\frac{A}{E_2}\right)$ = Probability that an item produced by B is defective = 5%

(i) P (selected item is defective)

$$= P(A) = P(E_1A \text{ or } E_2A) = P(E_1A) + P(E_2A)$$

$$= P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)\left(\frac{A}{E_2}\right)$$

$$= (60\%)(2\%) + (40\%)(5\%)$$

$$= \frac{60}{100} \times \frac{2}{100} + \frac{40}{100} \times \frac{5}{100} = \frac{320}{10000} = 0.032$$

(ii) P (selected item is non-defective)

$$= P(\bar{A}) = P(E_1\bar{A} \text{ or } E_2\bar{A}) = P(E_1\bar{A}) + P(E_2\bar{A})$$

$$= P(E_1)P\left(\frac{\bar{A}}{E_1}\right) + P(E_2)P\left(\frac{\bar{A}}{E_2}\right)$$

$$= (60\%)(98\%) + (40\%)(95\%)$$

$$= \frac{60}{100} \times \frac{98}{100} + \frac{40}{100} \times \frac{95}{100} = \frac{9680}{1000} = 0.968$$

BAYE'S THEOREM:

If an event A can occur only with one of the n mutually exclusive and exhaustive events $B_1, B_2, \ldots B_n$ & if the conditional probabilities of the events.

$P\left(\frac{A}{B_1}\right), P\left(\frac{A}{B_2}\right) \ldots P\left(\frac{A}{B_n}\right)$ are known then, $P\left(\frac{B_i}{A}\right) = \frac{P(B_i)P\left(\frac{A}{B_i}\right)}{\sum_{i=1}^{n} P(B_i)P\left(\frac{A}{B_i}\right)}$

Proof:

The event A occurs with one of the 'n' mutually exclusive events $B_1, B_2, B_3, \ldots, B_n$

$$A = AB_1 + AB_2 + ... + AB_n$$

$$P(A) = P(AB_1) + P(AB_2) + ... + P(AB_n) = \sum_{i=1}^{n} P(AB_i)$$

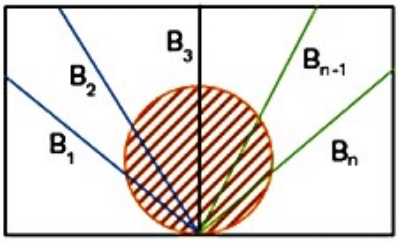
Note:

A = event what we have,

B_1 = event what we want,

$B_1, B_2, ..., B_n$ are alternative events.

Now,



$$P(AB_i) = P(A).P\left(\frac{B_i}{A}\right) = P(B_i).P\left(\frac{A}{B_i}\right)$$

$$P\left(\frac{B_i}{A}\right) = \frac{P(B_i)P\left(\frac{A}{B_i}\right)}{P(A)} = \frac{P(B_i).P\left(\frac{A}{B_i}\right)}{\sum_{i=1}^{n} P(AB_i)} = \frac{P(B_i).P\left(\frac{A}{B_i}\right)}{\sum_{i=1}^{n} P(B_i).P\left(\frac{A}{B_i}\right)}$$

Q. **Bag A contains 3 white and 2 black balls. Bag B contains 2 white and 2 black balls. One ball is drawn at random from A and transferred to B. One ball is selected at random from B and is found to be white. The probability that the transferred ball is white is**

(A) $\frac{8}{13}$ **(B)** $\frac{5}{13}$ **(C)** $\frac{4}{13}$ **(D)** $\frac{9}{13}$

Sol. Let E_1 and E_2 denote the events of the transferred ball being white and black, respectively. W denotes the drawn ball from B is white. By hypothesis,

$$P(E_1) = \frac{{}^3C_1}{{}^5C_1} = \frac{3}{5}, P(E_2) = \frac{{}^2C_1}{{}^5C_1} = \frac{2}{5}$$

$$P\left(\frac{W}{E_1}\right) = \frac{{}^3C_1}{{}^5C_1} = \frac{3}{5}, P\left(\frac{W}{E_2}\right) = \frac{{}^2C_1}{{}^5C_1} = \frac{2}{5}$$

By Bayes' theorem

$$P\left(\frac{E_1}{W}\right) = \frac{P(E_1)P\left(\frac{W}{E_1}\right)}{P(E_1)P\left(\frac{W}{E_1}\right) + P(E_2)P\left(\frac{W}{E_2}\right)} = \frac{\frac{3}{5}\times\frac{3}{5}}{\frac{3}{5}\times\frac{3}{5} + \frac{2}{5}\times\frac{2}{5}} = \frac{9}{13}$$

Q. **A letter is to come from either LONDON or CLIFTON. The postal mark on the letter legibly shows consecutive letters "ON". The probability that the letter has come from LONDON is**

(A) $\frac{12}{17}$ **(B)** $\frac{13}{17}$ **(C)** $\frac{5}{17}$ **(D)** $\frac{4}{17}$

Sol. Let the events be defined as

E_1: Letter coming from LONDON

E_2: Letter coming from CLIFTON

E_3: Two consecutive letters ON.

The word LONDON contains 5 types of consecutive letters (LO, ON, ND, DO, ON) of which there are two ON's. The word CLIFTON contains 6 types of consecutive letters (CL, LI, IF, FT, TO, ON) of which there is one "ON". Now

$$P(E_1) = \frac{1}{2} = P(E_2),$$

$$\Rightarrow P\left(\frac{E_3}{E_1}\right) = \frac{2}{5} \text{ and } P\left(\frac{E_3}{E_2}\right) = \frac{1}{6}$$

By Bayes' theorem

$$P\left(\frac{E_1}{E_3}\right) = \frac{\frac{1}{2}\times\frac{2}{5}}{\frac{1}{2}\times\frac{2}{5}+\frac{1}{2}\times\frac{1}{6}} = \frac{12}{17}$$

Q. **In a factory which manufactures bolts, machines A, B and C manufacture respectively 25%, 35% and 40 % of the bolts. Of their outputs, 5, 4 and 2 percent are respectively defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it is manufactured by the machine B?**

Sol. Let events B_1: the bolts is manufactured by machine A

B_2: the bolt is manufactured by machine B

B_3: the bolt is manufactured by machine C

Clearly, B_1, B_2, B_3 are mutually exclusive and exhaustive events and hence, they represent a partition of the sample space.

Let the event E be 'the bolt is defective'.

The event E occurs with B_1 or with B_2 or with B_3.

Given that, $P(B_1) = 25\% = 0.25, P(B_2) = 0.35$ and $P(B_3) = 0.40$

Again $P(E \mid B_1)$=Probability that the bolt drawn is defective given that it is manufactured by machine A = 5% = 0.05

Similarly, $P(E \mid B_2) = 0.04, P(E \mid B_3) = 0.02$

Hence, by Bayes' Theorem, we have

$$P(B_2 \mid E) = \frac{P(B_2)P(E \mid B_2)}{P(B_1)P(E \mid B_1) + P(B_2)P(E \mid B_2) + P(B_3)P(E \mid B_3)}$$

$$= \frac{0.35 \times 0.04}{0.25 \times 0.05 + 0.35 \times 0.04 + 0.40 \times 0.02} = \frac{0.0140}{0.0345} = \frac{28}{69}$$

Q. **In a test, an examinee either guesses or copies or know the answer for a multiple choice question having FOUR choice of which exactly one is correct. The probability that he makes a guess is 1/3 and the probability for copying is 1/6. The probability that is his answer is correct, given that he copied it is 1/8. The probability that he knew the answer, given that his answer is correct is**

(A) $\frac{5}{29}$ **(B)** $\frac{9}{29}$ **(C)** $\frac{24}{29}$ **(D)** $\frac{20}{29}$

Sol. Let the events be defined as

E_1 : Guessing

E_2 : Copying

E_3 : Knowing

E : Correct answer

By hypothesis

$$P(E_1) = \frac{1}{3}, P(E_2) = \frac{1}{6}, P(E_3) = 1 - \frac{1}{3} - \frac{1}{6} = \frac{1}{2}$$

$P\left(\frac{E}{E_1}\right) = \frac{1}{4}$ (out of four choice only one is correct)

$P\left(\frac{E}{E_2}\right) = \frac{1}{8}$

$P\left(\frac{E}{E_3}\right) = 1$

Therefore by Bayes' theorem

$$P\left(\frac{E_3}{E}\right) = \frac{P(E_3)P\left(\frac{E}{E_3}\right)}{P(E_1)P\left(\frac{E}{E_1}\right) + P(E_2)P\left(\frac{E}{E_2}\right) + P(E_3)P\left(\frac{E}{E_3}\right)} = \frac{\frac{1}{2} \times 1}{\frac{1}{3} \times \frac{1}{4} + \frac{1}{6} \times \frac{1}{8} + \frac{1}{2} \times 1} = \frac{24}{29}$$

Q. **A doctor is to visit a patient. From the past experience, it is known that the probabilities that he will come by train, bus, scooter or by other means of transport are respectively $\frac{3}{10}, \frac{1}{5}, \frac{1}{10}$ and $\frac{2}{5}$. The probabilities that he will be late are $\frac{1}{4}, \frac{1}{3}$ and $\frac{1}{12}$, if he comes by train, bus and scooter respectively, but he comes by other means of transport, then he will not be late. When he arrives, he is late. What is the probability that he comes by train?**

Sol. Let E be the event that the doctor visits the patient late and let T_1, T_2, T_3, T_4 be the events that the doctor comes by train, bus, scooter, and other means of transport respectively.

Then $P(T_1) = \frac{3}{10}, P(T_2) = \frac{1}{5}, P(T_3) = \frac{1}{10}$ and $P(T_4) = \frac{2}{5}$ (given)

$P(E \mid T_1)$ = Probability that the doctor arriving late comes by train $= \frac{1}{4}$

Similarly, $P(E \mid T_2) = \frac{1}{3}, P(E \mid T_3) = \frac{1}{12}$ and $P(E \mid T_4) = 0$, since he is not late if he comes by other means by other means of transport.

Therefore, by Bayes' Theorem, we have

$P(T_1 \mid E)$ = Probability that doctor arriving late comes by train

$$= \frac{P(T_1)P(E \mid T_1)}{P(T_1)P(E \mid T_1) + P(T_2)P(E \mid T_2) + P(T_3)P(E \mid T_3) + P(T_4)P(E \mid T_4)}$$

$$= \frac{\frac{3}{10} \times \frac{1}{4}}{\frac{3}{10} \times \frac{1}{4} + \frac{1}{5} \times \frac{1}{3} + \frac{1}{10} \times \frac{1}{12} + \frac{2}{5} \times 0} = \frac{3}{40} \times \frac{120}{18} = \frac{1}{2}$$

Hence, the required probability is $\frac{1}{2}$.

Q. **Suppose that the reliability of a HIV test is specified as follows : Of people having HIV, 90% of the test detect the disease but 10% go undetected. Of people free of HIV, 99% of the test are judged HIV-ive but 1% are diagnosed as showing HIV +ve. From a large population of which only 0.1% have HIV, one person is selected at random, given the HIV test, and the pathologist reports him/her as HIV +ve. What is the probability that the person actually has HIV?**

Sol. Let E denote the event that the person selected is actually having HIV and A be the event that the person's HIV test is diagnosed as +ve. We need to find $P(E \mid A)$

Also, E' denotes the event that the person selected is actually not having HIV. Clearly, {E,E'} is a partition of the sample space of all people in the population. We are given that

$$P(E) = 0.1\% = \frac{0.1}{100} = 0.001$$

$$P(E') = 1 - P(E) = 0.999$$

P(A|E) = P(Person tested as HIV +ve given that he/she is actually having HIV)

$$= 90\% = \frac{90}{100} = 0.9$$

and $P(A \mid E')$ = P (Person tested as HIV +ve given that he/she is actually not having HIV)

$$= 1\% = \frac{1}{100} = 0.01$$

Now, by Bayes' theorem

$$P(E \mid A) = \frac{P(E)P(A \mid E)}{P(E)P(A \mid E) + P(E')P(A \mid E')} = \frac{0.001 \times 0.9}{0.001 \times 0.9 + 0.999 \times 0.01} = \frac{90}{1089}$$

Thus, the probability that a person selected at random is actually having HIV given that he/she is tested HIV +ve is $\frac{90}{1089}$.

Q. **A bag contains 4 balls of unknown colours. Possibility of 1 white ball equals possibility of 2 or 3 or 4 white balls. A ball is drawn at random from it and is found to be white. The probability that all the balls in the bag are white is**

(A) $\frac{4}{5}$ **(B)** $\frac{1}{5}$ **(C)** $\frac{3}{5}$ **(D)** $\frac{2}{5}$

Sol. Let $W_j\ (j = 1,2,3,4)$ denote 1, 2, 3 and 4 white balls are in the bag. Let W be the ball drawn is white. Then $P(W_1) = P(W_2) = P(W_3) = P(W_4) = \frac{1}{4}$

$$P\left(\frac{W}{W_1}\right) = \frac{1}{4},\ P\left(\frac{W}{W_2}\right) = \frac{2}{4},\ P\left(\frac{W}{W_3}\right) = \frac{3}{4},\ P\left(\frac{W}{W_4}\right) = 1$$

Therefore by Bayes' theorem

$$P\left(\frac{W_4}{W}\right) = \frac{P(W_4)P\left(\frac{W}{W_4}\right)}{\sum_{j=1}^{4} P(W_j)P\left(\frac{W}{W_j}\right)} = \frac{\frac{1}{4} \times 1}{\frac{1}{4}\left(\frac{1}{4} + \frac{2}{4} + \frac{3}{4} + \frac{4}{4}\right)} = \frac{4}{10} = \frac{2}{5}$$

Q. **A lady has 10 coins in her purse, 8 of them are normal coins, one of them is a Double head (DH) and one is a Double tail (DT). She randomly draws a coin and tosses it for 5 times. The coin was found to fall headwise on all occasions. Find the probability that drawn coin was a DH coin.**

Sol. Event B_1 : Lady draws normal coin

Event B_2 : Lady draws DH coin.

Event B_3 : Lady draws DT coin.

Event A : Heads comes up on all 5 tosses.

Now, $P(B_1) = \frac{8}{10} = \frac{4}{5}$

$P(B_2) = \frac{1}{10}$

$P(B_3) = \frac{1}{10}$

$P\left(\frac{A}{B_1}\right) = \left(\frac{1}{2}\right)^5$

$P\left(\frac{A}{B_2}\right) = 1$

$P\left(\frac{A}{B_3}\right) = 0$

Using Baye's Theorem, required Probability

$$= P\left(\frac{B_2}{A}\right) = \frac{P(B_2)P\left(\frac{A}{B_2}\right)}{P(B_1)P\left(\frac{A}{B_1}\right) + P\left(\frac{A}{B_2}\right) + P(B_3)P\left(\frac{A}{B_3}\right)}$$

$$= \frac{\frac{1}{10} \times 1}{\frac{8}{10} \times \left(\frac{1}{2}\right)^5 + \frac{1}{10} \times 1 + 0}$$

$$= \frac{1}{\frac{1}{4} + 1} = \frac{4}{5}$$

Q. **There are two group of subjects one of which consists of 5 science subjects and 3 engineering subjects & other consists of 3 science and 5 engineering subjects. An unbiased die is cast. If the number 3 or 5 turns up, a subject is selected at random from first group, otherwise the subject is selected from 2nd group. Find the probability that an engineering subject is selected.**

Sol. Event B_1 : Subject is selected from 1st group.

Event B_2 : Subject is selected from 2nd group.

Event A : Engineering subject is selected.

$$\Rightarrow P(B_1) = \frac{2}{6} = \frac{1}{3}$$

$$\Rightarrow P(B_2) = \frac{4}{6} = \frac{2}{3}$$

$$\Rightarrow P(A) = P(A \cap B_1) + P(A \cap B_2)$$

(B_1 and B_2 are exhaustive)

$$= P(B_1).P\left(\frac{A}{B_1}\right) + P(B_2).P\left(\frac{A}{B_2}\right)$$

$$= \frac{1}{3} \times \left(\frac{3}{8}\right) + \frac{2}{3} \times \frac{5}{8}$$

$$= \frac{13}{24}$$

Q. **A bag contains 6 balls and it is not known of what colours they are. 3 balls are drawn from the bag and found to be all black. Find the probability that no black balls are left in the bag now. (Assume all number of black balls in the bag to be equally likely)**

Sol. There are 4 cases possible:

B_1 : There were 3 black balls in bag.

B_2 : There were 4 black balls in bag.

B_3 : There was 5 black balls in bag.

B_4 : All balls were black in colour.

Let event A be : 3 black balls are drawn.

Since it is given, $P(B_1) = P(B_2) = P(B_3) = P(B_4) = \frac{1}{4}$

(equally likely events).

$$P\left(\frac{A}{B_1}\right) = \frac{^3C_3}{^6C_3}$$

$$P\left(\frac{A}{B_2}\right) = \frac{^4C_3}{^6C_3}$$

$$P\left(\frac{A}{B_3}\right) = \frac{^5C_3}{^6C_3}$$

$$P\left(\frac{A}{B_4}\right) = \frac{^6C_3}{^6C_3}$$

Required probability $= P\left(\frac{B_1}{A}\right)$

$$\Rightarrow P\left(\frac{B_1}{A}\right) = \frac{P(B_1).P\left(\frac{A}{B_1}\right)}{\sum P(B_i)P\left(\frac{A}{B_i}\right)}$$

$$\Rightarrow P\left(\frac{B_1}{A}\right) = \frac{\frac{1}{4}\times\frac{^3C_3}{^6C_3}}{\frac{1}{4}\times\frac{^3C_3}{^6C_3}+\frac{1}{4}\times\frac{^4C_3}{^6C_3}+\frac{1}{4}\times\frac{^5C_3}{^6C_3}+\frac{1}{4}\times 1}$$

$$= \frac{^3C_3}{^3C_3 + ^4C_3 + ^5C_3 + ^6C_3}$$

$$= \frac{1}{1+4+10+20} = \frac{1}{35}$$

Q. **The contents of urn I and urn II are as following :**

urn	W	B
I	4	5
II	3	6

W → White balls

B→ Black balls

One urn is chosen at random and a ball is drawn and its colour noted and replaced back in same urn. Again a ball is drawn from the same urn and colour noted and replaced. The process is repeated 4 times and as a results 1 ball of white colour and 3 of black colour are noted. What is the probability that urn (I) is chosen?

Sol. B_1 : urn I is chosen

B_2 : urn II is chosen

A : 1W and 3 B balls are drawn.

$$P(B_1) = P(B_2) = \frac{1}{2} \quad P\left(\frac{A}{B_1}\right) = \frac{4}{9} \times \left(\frac{5}{9}\right)^3 \times {}^4C_1$$

$$P\left(\frac{A}{B_2}\right) = \frac{3}{9} \times \left(\frac{6}{9}\right)^3 \times {}^4C_1$$

Required Probability $= P\left(\frac{B_1}{A}\right)$

$$= \frac{P(B_1).P\left(\frac{A}{B_1}\right)}{\sum P(B_i).P\left(\frac{A}{B_i}\right)}$$

$$= \frac{\frac{1}{2} \times \frac{16 \times 5^3}{9^4}}{\frac{1}{2} \times \frac{16 \times 5^3}{9^4} + \frac{1}{2} \times \frac{12 \times 6^3}{9^4}}$$

$$= \frac{16 \times 5^3}{16 \times 5^3 + 12 \times 6^3} = \frac{4 \times 125}{4 \times 125 + 3 \times 2^3 \times 3^3}$$

$$= \frac{125}{125 + 162} = \frac{125}{287}$$

Q. **The probability that an archer hits the target when it is windy is 0.4; when it is not windy, her probability of hitting the target is 0.7. On any shot, the probability of gust of wind is 0.3. Find the probability that she**
(a) Hits the target on first shot.
(b) Hits the target exactly once in two shorts.

Sol. **(a)** **Case-I :** It is windy

$\Rightarrow$ Probability of hitting = (0.3) (0.4)

= 0.12

Case-II : It is not windy.

$\Rightarrow$ Probability of hitting the target $= (0.7)(0.7)$

$= 0.49$

Total Probability $= 0.12 + 0.49$

$= 0.61$

$\Rightarrow$ Probability of not hitting the target = 1 – 0.61
= 0.39

(b) Let event of hitting the target be A.

Then cases of hitting the target exactly one in two shots are : $(A\bar{A})$ & $(\bar{A}A)$

Now , P(A) = 0.61 ('a' part)

$\Rightarrow P(\bar{A}) = 0.39$

$\Rightarrow$ Required Probability $= P(A\bar{A}) + P(\bar{A}A)$
= (0.61)(0.39)(2)
= 0.4758

Q. **'A' writes a letter to his friend 'B' and gives it to his son to post it in a letter box, the reliability of his son being 3/4. Probability that a letter posted will get delivered is 8/9. At a later date, 'A' hears from 'B' that the letter has not reached him. Find the probability that the son did not post the letter at all.**

Sol. Event B_1 : Son post the letter.

Event B_2 : Son did not post the letter.

Event C : B did not receive the letter.

Required Probability $= P\left(\frac{B_2}{C}\right)$

$$= \frac{P(B_2)P\left(\frac{C}{B_2}\right)}{P(B_1)P\left(\frac{C}{B_1}\right) + P(B_2)P\left(\frac{C}{B_2}\right)}$$

$$= \frac{\left(\frac{1}{4}\right)(1)}{\left(\frac{3}{4}\right)\left(\frac{1}{9}\right) + \left(\frac{1}{4}\right)1} = \frac{3}{4}$$

Q. **A bag contains a number of cards with 30 % white on both sides, 50% white on one side and black on the other side and 20% black on both sides. The cards are mixed up, and a single card is drawn at random and placed on table, it's upper side shows up black. Probability that it's other side is also black is**

(A) $\frac{2}{6}$ **(B)** $\frac{4}{9}$ **(C)** $\frac{2}{3}$ **(D)** $\frac{2}{7}$

Sol. Event B_1 : Cards with white on both side is drawn.

Event B_2 : Card with white on one and black on other side is drawn.

Event B_3 : Card with black on both side is drawn.

Event A : card drawn has upper side black.

$\Rightarrow$ Required probability $= P\left(\frac{B_3}{A}\right)$

$$= \frac{P(B_3)P\left(\frac{A}{B_3}\right)}{\sum P(B_i)P\left(\frac{A}{B_i}\right)}$$

$$= \frac{(0.2)(1)}{(0.3)\times 0 + (0.5)\times(0.5) + (0.2)(1)}$$

$$= \frac{0.2}{0.45} = \frac{4}{9}$$

Q. **Suppose we have 10 coins such that if the ith coin is flipped, heads will appear with probability $\left(\frac{i}{10}\right)$. When 1 of the coins is randomly selected and flipped, it shows up head. The probability that it was the 5th coin is**

(A) $\frac{1}{9}$ **(B)** $\frac{1}{10}$ **(C)** $\frac{1}{11}$ **(D)** $\frac{1}{2}$

Sol. Event B_i : i^{th} coin was selected and flipped.

Event A : It shows up head.

$$P(B_i) = \frac{1}{10} \quad (i = 1, 2, 3, \ldots 10)$$

$$P\left(\frac{A}{B_i}\right) = \frac{i}{10} \quad (i = 1, 2, 3, \ldots 10)$$

Required Probability $= P\left(\frac{B_5}{A}\right)$

$$= \frac{P(B_5)P\left(\frac{A}{B_5}\right)}{\sum P(B_i)P\left(\frac{A}{B_i}\right)}$$

$$= \frac{\left(\frac{1}{10}\right)\left(\frac{5}{10}\right)}{\frac{1}{10}\left(\frac{1}{10}+\frac{2}{10}+\frac{3}{10}+\ldots+\frac{10}{10}\right)}$$

$$= \frac{5}{55} = \frac{1}{11}$$

Q. **A certain drug, manufactured by a company is tested chemically for its toxic nature. Let the event "The Drug is Toxic" be denoted by H & the event "The chemical Test Reveals that the Drug is Toxic" be denoted by S. Let $P(H) = a, P\left(\frac{S}{H}\right) = P\left(\frac{\bar{S}}{\bar{H}}\right) = 1 - a$. Then show that the probability that drug is not toxic given that the chemical test reveals that it is toxic, is free from 'a'**

Sol. Required Probability $= P\left(\frac{\bar{H}}{S}\right)$

$$= \frac{P(\bar{H})P\left(\frac{S}{\bar{H}}\right)}{P(\bar{H})P\left(\frac{S}{\bar{H}}\right) + P(H)P\left(\frac{S}{H}\right)}$$

$$= \frac{(1-a)(1-(1-a))}{(1-a)(1-(1-a)) + a(1-a)}$$

$$= \frac{a(1-a)}{(1-a)(a+a)} = \frac{1}{2} = \text{free from a}$$ **Hence proved.**

Extended Bay's Theorem

Q. **A bag contains 3 biased coins B_1, B_2, B_3 whose probability of falling head wise are $\frac{1}{3}, \frac{2}{3}, \frac{3}{4}$ respectively. A coin is drawn randomly and tossed, fell head wise. Find the probability that the same coin when tossed again will fall head wise.**

Sol. $P(\text{Head}) = P(H \cap B_1) + P(H \cap B_2) + P(H \cap B_3)$

$$= \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{3} + \frac{1}{3} \times \frac{3}{4}$$

P(getting head on 1st coin when head has occurred)

$$= \frac{\frac{1}{3}\times\frac{1}{3}}{\frac{1}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{3}+\frac{1}{3}\times\frac{3}{4}}$$

P(getting head again on 1st coin)

$$= \left(\frac{\frac{1}{3}\times\frac{1}{3}}{\frac{1}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{3}+\frac{1}{3}\times\frac{3}{4}}\right)\left(\frac{1}{3}\right) = \frac{4}{63}$$

Similarly, P(getting head again on 2nd win after getting head on it)

$$= \frac{\frac{1}{3}\times\frac{2}{3}}{\left(\frac{1}{3}\times\frac{1}{3}\right)+\left(\frac{1}{3}\times\frac{2}{3}\right)+\left(\frac{1}{3}\times\frac{3}{4}\right)}\times\frac{2}{3} = \frac{16}{63}$$

And P(getting head again on 3rd coin after getting a head on it)

$$= \frac{\frac{1}{3}\times\frac{3}{4}}{\left(\frac{1}{3}\times\frac{1}{3}\right)+\left(\frac{1}{3}\times\frac{2}{3}\right)+\left(\frac{1}{3}\times\frac{3}{4}\right)}\times\frac{3}{4} = \frac{9}{28}$$

So, probability for same coin falling head wise again is sum of above three

$$= \frac{4}{63}+\frac{16}{63}+\frac{9}{28} = \frac{16+64+81}{252}$$

$$= \frac{161}{252} = \frac{23}{36}$$

Q. **A bag contain 6R, 4W balls. 5 balls are drawn one by one without replacement and were found to be atleast two white. Find the probability that the next draw of a ball from this bag will give a white ball.**

Sol. Three cases possible :

B_1 : 4 balls are white and 1 ball is red

B_2 : 3 white balls and 2 red balls are drawn

B_3 : 2 white and 3 red balls are drawn

$$P(B_1) = \frac{{}^4C_4 \times {}^6C_1}{{}^{10}C_5} = \frac{6}{{}^{10}C_5}$$

$$P(B_2) = \frac{{}^4C_3 \times {}^6C_2}{{}^{10}C_5} = \frac{60}{{}^{10}C_5}$$

$$P(B_3) = \frac{{}^4C_2 \times {}^6C_3}{{}^{10}C_5} = \frac{120}{{}^{10}C_5}$$

$$P\left(\frac{W}{B_1}\right) = 0;\ P\left(\frac{W}{B_2}\right) = \frac{1}{5}$$

$$P\left(\frac{W}{B_3}\right) = \frac{2}{5}$$

$P(B_1$ given that atleast 2 balls are white)

$$= \frac{P(B_1)}{P(B_1) + P(B_2) + P(B_3)} = \frac{6}{186}$$

$P(B_2$ given that atleast 2 balls are white)

$$= \frac{P(B_2)}{P(B_1)P(B_2) + P(B_3)} = \frac{60}{186}$$

Similarly, $P(B_3$ given that atleast 2 balls are white) $= \frac{120}{186}$

$$P(\text{getting a white ball after } B_1) = \frac{6}{186} \times 0 = 0$$

$$P(\text{getting a white ball after } B_2) = \frac{60}{186} \times \frac{1}{5}$$

$$P(\text{getting a white ball after } B_3) = \frac{120}{186} \times \frac{2}{5}$$

$$\Rightarrow \text{Total required probability} = 0 + \frac{60}{5 \times 186} + \frac{240}{5 \times 186}$$

$$= \frac{60}{186} = \frac{10}{31}$$

Q. **A purse contains 4 coins, each coin is either a rupee or a 50 paise coin. 2 coins are drawn successfully without replacement and were found to be both rupee coins. If both these coins are replaced in the bag, what is the probability that next drawing will give a 50 paise coin?**

Sol. There are 3 cases possible:

B_1 : All coins are rupee.

B_2 : There are 3 rupee coins and one 50 paise coin

B_3 : There are two rupee and two 50 paise coins.

A : 2 coins are drawn and found to be both rupee.

$$P\left(\frac{A}{B_1}\right) = 1$$

$$P\left(\frac{A}{B_2}\right) = \frac{{}^3C_2}{{}^4C_2} = \frac{3}{6}$$

$$P\left(\frac{A}{B_3}\right) = \frac{{}^2C_2}{{}^4C_2} = \frac{1}{6}$$

$$\Rightarrow P\left(\frac{B_1}{A}\right) = \frac{1}{1+\frac{3}{6}+\frac{1}{6}} = \frac{6}{10}$$

$$\Rightarrow P\left(\frac{B_2}{A}\right) = \frac{\frac{3}{6}}{1+\frac{3}{6}+\frac{1}{6}} = \frac{3}{10}$$

$$\Rightarrow P\left(\frac{B_3}{A}\right) = \frac{\frac{1}{6}}{1+\frac{3}{6}+\frac{1}{6}} = \frac{1}{10}$$

In case of B_1, after replacement, probability of getting a 50 paise coin $= 0 = p_1$

Similarly, in case of B_2, $P(50p \text{ coin}) = p_2 = \frac{1}{4}$

And P(50 p coin in case of B_3) $= p_3 = \frac{2}{4}$

$$\Rightarrow \text{Total probability} = P\left(\frac{B_1}{A}\right)p_1 + P\left(\frac{B_2}{A}\right)p_2 + P\left(\frac{B_3}{A}\right)p_3$$

$$= \frac{6}{10} \times 0 + \frac{3}{10} \times \frac{1}{4} + \frac{1}{10} \times \frac{2}{4} = \frac{1}{8}$$

PROBABILITIES THROUGH STATISTICAL (STOCHASTIC) TREE DIAGRAM :

Q. **A box contains three coins A, B and C**
A : Normal coin; B: Double Headed (DH) coin; C: a weighted coin so that

$$P(H) = \frac{1}{3}$$

A coin is randomly selected & tossed.
(A) Find the probability that head appears.
(B) If head appears find the probability that it is a normal coin P(A/H).
(C) Find the probability that tail appears.
(D) If tail appears, find the probability that it is a weighted coin P(C/T).

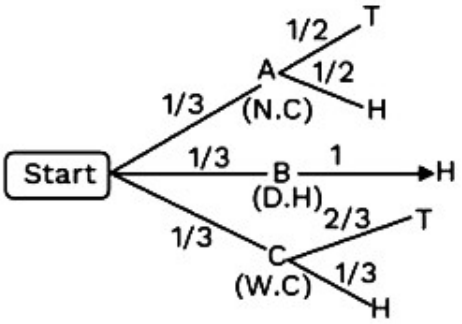
Sol. **(A)** $P(H) = \frac{1}{3}\cdot\frac{1}{2} + \frac{1}{3}.1 + \frac{1}{3}\cdot\frac{1}{3} = \frac{11}{18}$

(B) $P\left(\frac{A}{H}\right) = \frac{P(A \cap H)}{P(H)} = \frac{\frac{1}{3}\cdot\frac{1}{2}}{\frac{11}{18}} = \frac{3}{11}$

(C) $P(T) = \frac{1}{3}\cdot\frac{1}{2} + \frac{1}{3}.0 + \frac{1}{3}\cdot\frac{2}{3} = \frac{7}{18}$

Or $1 - P(H) = 1 - \frac{11}{18} = \frac{7}{18}$

(D) $P\left(\frac{C}{T}\right) = \frac{P(C \cap T)}{P(T)} = \frac{\frac{1}{3}\cdot\frac{2}{3}}{\frac{7}{18}} = \frac{4}{7}$



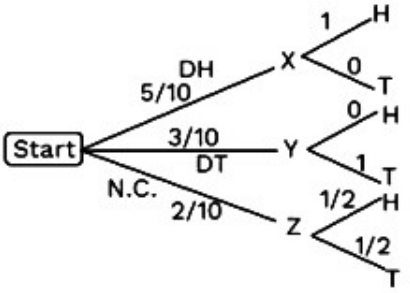
Q. A box contains 10 coins
- 5 coins DH denoted by say X
- 3 coins DT denoted by say Y
- 2 coins normal denoted by Z

A coin is drawn at random from the box and tossed, fell headwise. Find the probability that it was a normal coin.

Sol. $P\left(\frac{Z}{H}\right) = \frac{P(H \cap Z)}{P(H)}$

$P(H) = \frac{5}{10}\cdot 1 + \frac{2}{10}\cdot\frac{1}{2} = \frac{6}{10}$

$\Rightarrow P\left(\frac{Z}{H}\right) = \frac{\frac{2}{10}\cdot\frac{1}{2}}{\frac{6}{10}} = \frac{1}{6}$

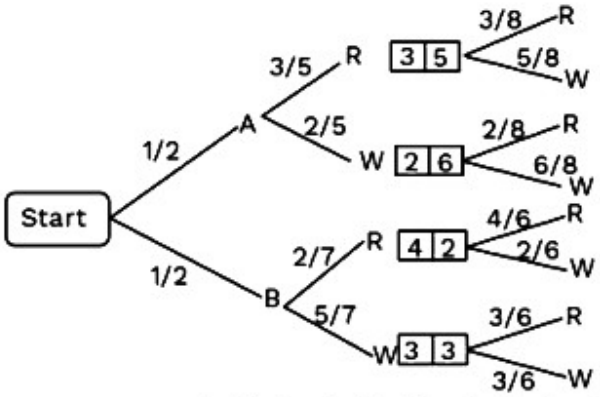


Q. **Let the contents of the boxes A and B with respect to number of R and W marbles is as given below:**

Bag	R	W
A	3	2
B	2	5

A bag is selected at random; a marble is drawn and put into the other box; then a marble is drawn from the second box. Find the probability that both marbles drawn are of same colour.

Sol.



$$P(RR \cup WW) = \frac{1}{2}\cdot\frac{3}{5}\cdot\frac{3}{8} + \frac{1}{2}\cdot\frac{2}{5}\cdot\frac{6}{8} + \frac{1}{2}\cdot\frac{2}{7}\cdot\frac{4}{6} + \frac{1}{2}\cdot\frac{5}{7}\cdot\frac{3}{6} = \frac{901}{1680}$$

Q. **A bag contains 1 red and 2 blue balls. A trial consists of selecting a ball at random noting its colour and replacing it together with an additional ball of same colour. Given that 3 trials are made, find the probability that**
(i) Atleast 1 blue ball is drawn
(ii) Exactly 1 blue ball is drawn
(iii) Given that all 3 balls drawn are of same colour, find the probability that they are all red

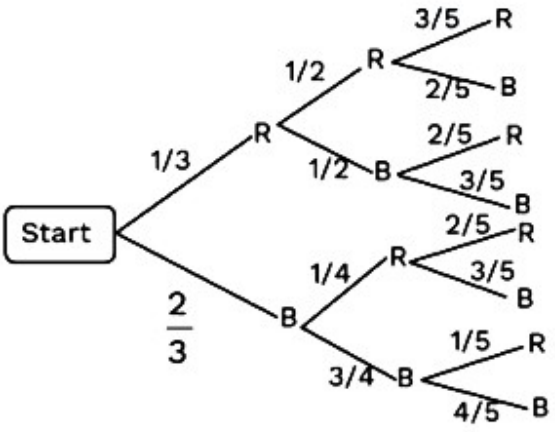
Sol. **(i)** P(atleast 1 Blue ball is drawn)

$= 1 - P(RRR)$

$= 1 - P(\text{no blue ball is drawn})$

$$= 1 - \frac{1}{3}\times\frac{1}{2}\times\frac{3}{5}$$

$$= 1 - \frac{1}{10} = \frac{9}{10}$$



(ii) P(exactly 1 Blue ball is drawn)

$$= P(RRB) + P(RBR) + P(BRR)$$

$$= \frac{1}{3}\times\frac{1}{2}\times\frac{2}{5} + \frac{1}{3}\times\frac{1}{2}\times\frac{2}{5} + \frac{2}{3}\times\frac{1}{4}\times\frac{2}{5}$$

$$= \frac{1}{15} + \frac{1}{15} + \frac{1}{15} = \frac{1}{5}$$

(iii) A : ball drawn are of same colour

B : All balls are red.

$$\text{Required Probability} = P\left(\frac{B}{A}\right)$$

$$= \frac{P(B \cap A)}{P(A)}$$

$$= \frac{P(RRR)}{P(RRR) + P(BBB)}$$

$$= \frac{\frac{1}{3} \times \frac{1}{2} \times \frac{3}{5}}{\frac{1}{3} \times \frac{1}{2} \times \frac{3}{5} + \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5}}$$

$$= \frac{\frac{1}{10}}{\frac{1}{10} + \frac{2}{5}} = \frac{1}{5}$$

Q. **Box A contains nine cards numbered 1 through 9, and box B contains five numbered 1 through 5. A box is chosen at random and a card is drawn and not replaced; if the card shows an even number, another card is drawn from the same box. If the card shows an odd number, a card is drawn from the other box;**
(i) What is the probability that both cards show even numbers?
(ii) If both cards show even numbers, what is the probability that they come from box A?
(iii) What is the probability that both cards show odd numbers?

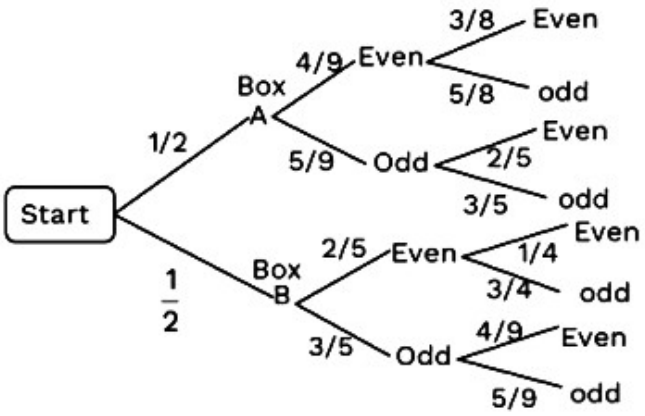
Sol. **(i)** P(Both cards show even number)

$$= P(B_A EE) + P(B_B EE)$$

$\{B_1 \rightarrow \text{Box}_1 \text{ and } B_2 \rightarrow \text{Box}_2\}$

$$= \frac{1}{2} \times \frac{4}{9} \times \frac{3}{8} + \frac{1}{2} \times \frac{2}{5} \times \frac{1}{4}$$

$$= \frac{1}{12} + \frac{1}{20} = \frac{2}{15}$$



(ii) Required probability

$$= \frac{P(B_A EE)}{P(B_A EE) + P(B_B EE)}$$

$$= \frac{\frac{1}{12}}{\frac{1}{12} + \frac{1}{20}} = \frac{5}{8}$$

(iii) P(Both cards show odd Numbers)

$$= P(B_A 00) + P(B_B 00)$$

$$= \frac{1}{2} \times \frac{5}{9} \times \frac{3}{5} + \frac{1}{2} \times \frac{3}{5} \times \frac{5}{9}$$

$$= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

MATHEMATICAL EXPECTATION
(PRACTICAL USE OF PROBABILITY IN DAY-TO -DAY LIFE) :

It is worthwhile indicating that if 'P' represents a person's chance of success in any venture and 'M' the sum of money which he will receive in case of success, then the sum of money denoted by **'P·M'** is called his expectation.

Q. **Two players of equal skill A and B are playing a game. They leave off playing (due to some force majeure conditions) when A wants 3 points and B wants 2 to win. If the prize money is Rs. 16000/-. How can the referee divide the money in a fair way.**

Sol. A wins if he scores 3 points before B scores 2.

Probability of A's scoring a point = Probability of B's scoring at point $= \frac{1}{2}$

Hence, required probability that A succeeds

$$= P(AAA) + P(BAAA) + P(ABAA) + P(AABA)$$

$$= \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{5}{16}$$

Probability that B Succeeds $= 1 - \frac{5}{16} = \frac{11}{16}$

$\therefore$ A's expectation $= \frac{5}{16} \times 16000 = 5000$

B's expectation $= \frac{11}{16} \times 16000 = 11000$

Q. **Two hunters A and B shot a bear simultaneously. The bear was shot dead with one hole in its body. Probability of A shooting the bear is 0.8 and that of B shooting the bear is 0.4 The hide was sold for Rs.280/- divide the money in a fair way.**

Sol. There are 4 cases:

$A\bar{B}, \bar{A}B$ and $\bar{A}\bar{B}$

(Where A & B represents successful hit by A & B respectively.)

Since there was only one hole on the body, possible cases are $A\bar{B}$ and $\bar{A}B$.

$$P(\text{A hits the target}) = \frac{P(A\bar{B})}{P(A\bar{B}) + P(\bar{A}B)}$$

$$= \frac{0.8 \times 0.6}{0.8 \times 0.6 + 0.2 \times 0.4} = \frac{0.48}{0.56}$$

Similarly, P(B hits the target) $= \frac{0.08}{0.56}$

$\Rightarrow$ A's share $= P(A) \times 280 = \frac{0.48}{0.56} \times 280 = \text{Rs.}240$

$\Rightarrow$ B's share $= P(B) \times 280 = \frac{0.08}{0.56} \times 280 = \text{Rs}\,40$

Q. **An unbiased coin is tossed five times. Suppose that a variable X is assigned the value k when k consecutive heads are obtained for k=3, 4, 5, otherwise X takes the value –1. Then the expected value of X, is : [JEE Main 2020]**

(A) $\frac{1}{8}$ **(B)** $\frac{3}{16}$ **(C)** $-\frac{1}{8}$ **(D)** $-\frac{3}{16}$

Sol. k=3$\rightarrow$ cases are HHHTT, HHHTH, THHHT, TTHHH, HTHHH

$\Rightarrow$ Total 5 cases favourable

$\Rightarrow P(X = 3) = \frac{5}{2^5} = \frac{5}{32}$

$\Rightarrow$ for k=4, cases are HHHHT, THHHH

$\Rightarrow$ total 2 favourable cases.

$\Rightarrow P(X = 4) = \frac{2}{32}$

$\Rightarrow$ for k=5$\Rightarrow$ 1 favourable case$\rightarrow$HHHHH

$\Rightarrow P(X = 5) = \frac{1}{32}$

$\therefore P(X = -1) = 1 - \frac{5}{32} - \frac{2}{32} - \frac{1}{32} = \frac{24}{32}$

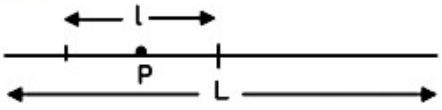
Expected Value of $X = (-1)P(X = -1) + 3P(X = 3) + 4P(X = 4) + 5P(X = 5)$

$$= \frac{-24}{32} + \frac{15}{32} + \frac{8}{32} + \frac{5}{32}$$

$$= \frac{4}{32} = \frac{1}{8}$$

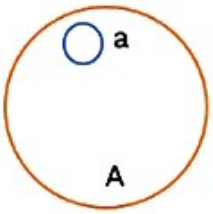
GEMETRICAL PROBABILITY (CONTINUOUS SAMPLE SPACE):

(1) One-dimensional Probability:



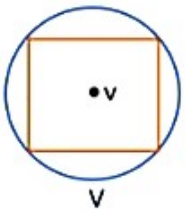
$$P = \frac{\text{favourable length}}{\text{total length}}$$

(2) Two-dimensional Probability:



$$P = \frac{\text{favourable area}}{\text{total area}}$$

(3) Three-dimensional Probability:



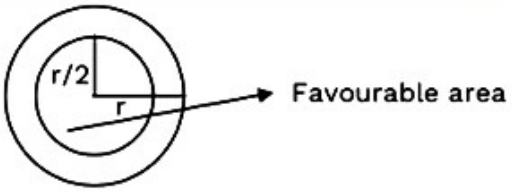
$$P = \frac{\text{favourable volume}}{\text{Total volume}}$$

Q. **A point is taken inside a circle of radius r, find the probability that the point is closer to center than the circumference.**

Sol. $n(S) = \pi r^2$

$$n(A) = \pi\left(\frac{r}{2}\right)^2$$

$$P = \frac{\pi\left(\frac{r}{2}\right)^2}{\pi r^2} = \frac{1}{4}$$

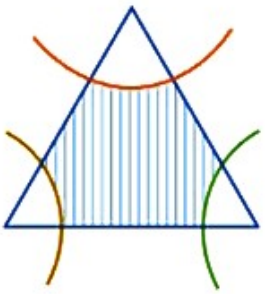


Q. **A point is selected at random inside a equilateral triangle whose length of side is 3. Find the probability that its distance from any corner is greater than 1.**

Sol. Area of sector $= \frac{r^2\theta}{2}$

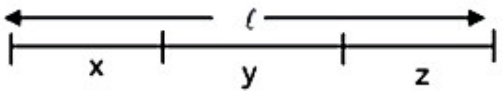
$$n(S) = \frac{\sqrt{3}}{4}.9$$

$$n(A) = \frac{\sqrt{3}}{4}.9 - 3\frac{1.1}{2}.\frac{\pi}{3} \qquad \therefore P(A) = \frac{\frac{\sqrt{3}}{4}.9 - \frac{\pi}{2}}{\frac{\sqrt{3}}{4}.9} = 1 - \frac{2\pi}{9\sqrt{3}}$$



Q. **A stick of length l is broker into three parts, find the probability that these three parts form a triangle.**

Sol. $x > 0\ y > 0, \ell - (x + y) > 0,\ z = \ell - (x + y)$

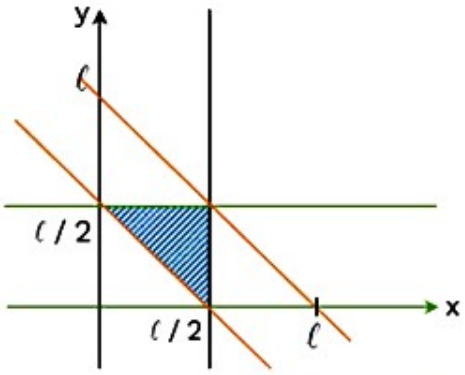


For a triangle sum of two sides should be greater than third side

$x + y > \ell - (x + y) \Rightarrow x + y > \frac{\ell}{2}$ ...(i)

$x + \ell - (x + y) > y \Rightarrow \ell > 2y \Rightarrow y < \frac{\ell}{2}$ ...(ii)

Similarly $x < \frac{\ell}{2}$ ...(iii)



So required probability $= \frac{\frac{1}{2} \times \frac{\ell}{2} \times \frac{\ell}{2}}{\frac{1}{2} \times \ell \times \ell} = \frac{1}{4}$

Q. **A circle of radius 'r' is inscribed in a square of side 2r. Find the probability that a point chosen at random is inside the square but outside the circle.**

Sol. $a = 2r$

Area of circle $= \pi r^2$

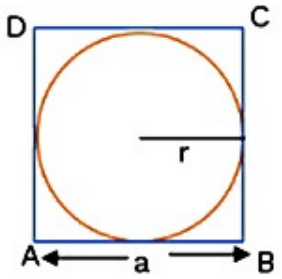
Area of square $= a^2$

$= 4r^2$

$\Rightarrow$ Area inside square but outside the circle $= 4r^2 - \pi r^2$

$\Rightarrow$ Required Probability $= \frac{4r^2 - \pi r^2}{4r^2}$

$= 1 - \frac{\pi}{4}$



Q. **Equation of an ellipse is given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Find the probability that a point chosen at random is outside the ellipse and inside the curve $(x^2 - a^2)(y^2 - b^2) = 0$.**

Sol. Area of ellipse $= \pi ab$

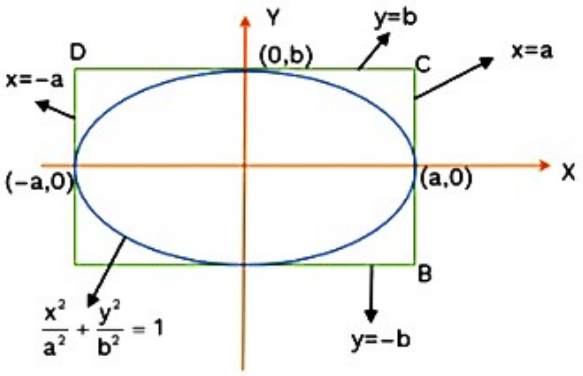
Area of rectangle

$ABCD = (2a)(2b) = 4ab$

Area outside the ellipse but inside rectangle $= (4-\pi)ab$

$\Rightarrow$ Required Probability $= \dfrac{(4-\pi)ab}{4ab}$

$= 1 - \dfrac{\pi}{4}$



Q. **Two points are picked at random on the unit circle $x^2 + y^2 = 1$. The probability that the chord joining the two points has length at least 1, is**

(A) $\frac{4}{9}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{1}{6}$ **(D)** $\frac{2}{3}$

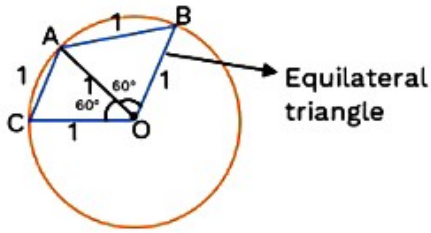
Sol. If we start from point A and draw a chord of length 1 on either side (AB or AC) total angle subtended $= 60° + 60° = 120°$.

$\Rightarrow$ Beyond point B or C, length of chord exceeds unit 1.

$\Rightarrow$ Required points cover $240°$ angle.

$(360° - 120° = 240°)$

$\Rightarrow$ Probability $= \dfrac{240°}{360°} = \dfrac{2}{3}$.



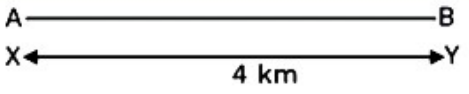
Q. **A starts from a town 'X' any time between 1 PM to 4 PM and walks towards the town 'Y' on a straight road with a speed of 4km/hr and B starts from 'Y' at any time between 1 PM to 4PM and walks towards 'X' at 4 km/hr. Assuming all times of starting all equally likely, find the odds in favour of their meeting on the way.**

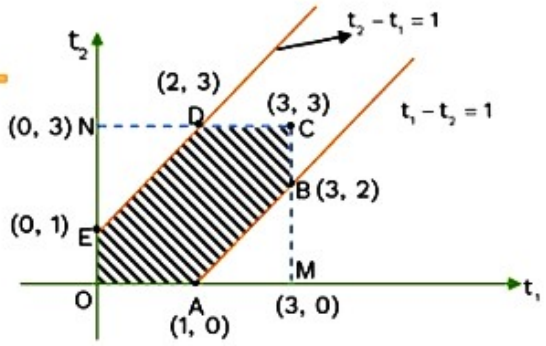
Sol. Let A starts 't_1' hr before 4 pm

Let B starts 't_2' hr before 4 pm

$\Rightarrow 0 \le t_1 \le 3$ and $0 < t_2 \le 3$

Also, A and B have to meet on the way. So, one must not reach other end before other party





starts. Since time taken to reach other end is $\frac{4}{4} = 1$ hr,

$\Rightarrow |t_1 - t_2| \leq 1$

Total Area = $Ar(\square OMCN) = 3 \times 3 = 9$

Favourable Area $= 9 - (\Delta DEN) - ar(\Delta BAM)$

$$= 9 - 2\left(\frac{1}{2} \times 2 \times 2\right)$$

$$= 5$$

$\Rightarrow$ Probability that A and B meets on the way = $p = \frac{5}{9}$

$\Rightarrow$ Odds in favour of there meeting on the way $= \frac{p}{1-p} = \frac{\frac{5}{9}}{\frac{4}{9}} = 5 : 4$

CONCIDENCE TESTIMONY

If p_1 and p_2 are the probabilities of speaking the truth of two independent witnesses A and B who give the same statement

P (their combined statement is true)

$$= P(H_1 / H_1 \cup H_2) = \frac{p_1 p_2}{p_1 p_2 + (1-p_1)(1-p_2)}$$

Where H_i means both speaks the truth and H_2 means both speaks false.

In this case it has been assumed that we have no knowledge of the event except the statement made by A and B.

- However if p is the probability of the happening of the event before their statement then

P (their combined statement is true) $= \frac{pp_1p_2}{pp_1p_2 + (1-p)(1-p_1)(1-p_2)}$

Here it has been assumed that the statement given by all the independent witness can be given in two ways only, so that if all the witnesses tell falsehoods, they agree telling the same falsehood.

- If this is not case and c is the chance of their coincidence testimony then the

Probability that the statement is true = pp_1p_2

Probability that the statement is false $= (1-p).c\,(1-p_1)(1-p_2)$

However, chance of coincidence testimony is taken only if the joint statement is not contradicted by any witness.

Q. **A speaks the truth 3 out of 4 times, and B 5 out of 6 times. What is the probability that they will contradict each other in stating the same fact?**

Sol. $P(A) = \frac{3}{4}; P(B) = \frac{5}{6}$

$$P\text{ (contradict)} = \frac{3}{4}.\frac{1}{6} + \frac{5}{6}.\frac{1}{4} = \frac{8}{24} = \frac{1}{3}$$

Q. **A speaks truth 3 times out of 4, and B 7 times out of 10. They both assert that a white ball has been drawn from a bag containing 6 balls of different colour; find the probability of the truth of their assertion. P(A) = 3/4; P(B) = 7/10**

Sol. There are 2 hypothesis

(i) their coincidence testimony is true

(ii) it is false

H_1 : while ball is actually drawn & both speaks the truth

$$P(H_1) = \frac{1}{6}.\frac{3}{4}.\frac{7}{10}$$

H_2 : (white has not been withdrawn) and (their statement coincides) and they both speak false

$$P(H_2) = \frac{5}{6} \times \frac{1}{5} \times \frac{1}{5} \times \frac{1}{4} \times \frac{3}{10}$$

Let E : their assertion is true

$$\therefore P(E) = P\left(\frac{H_1}{H_1 \cap H_2}\right) = \frac{\frac{1}{6}.\frac{3}{4}.\frac{7}{10}}{\frac{1}{6}.\frac{3}{4}.\frac{7}{10} + \frac{5}{6}\left(\frac{1}{25}\right).\frac{1}{4}.\frac{3}{10}} = \frac{35}{36}$$

RANDOM VARIABLE

Random variables are of two types:

(i) Discrete random variable.

(ii) Continuous random variables.

(i) A random variable is called a discrete random variable if it can take only finitely many values. For example, in the experiment of drawing three cards from a pack of playing cards, the random variable "number of kings drawn" is a discrete random variable taking either 0 or 1 or 2 and 3.

(ii) A random variable is called a continuous random variable if it can take any value between certain limits. For example, height, weight of students in a class are continuous random variables.

Probability Distribution of a discrete random variable :

Let x be discrete random variable assuming values $x_1, x_2, x_3, \ldots x_n$ corresponding to the various outcomes of random experiment. If the probability of occurrence of $x = x_i$ is $P(x_i) = p_i, 1 \le i \le n$ such the $p_1 + p_2 + p_3 + \ldots + p_n = 1$, then the function, $P(x_i) = p_i, 1 \le i \le n$ is called the probability function of the random variable x and the set $\{P(x_1), P(x_2), P(x_3), \ldots, P(x_n)\}$ is called the probability distribution of x.

Q. **Three balls are drawn one by one without replacement from a bag containing 5 white and 4 red balls. Find the probability distribution of the number of red balls drawn.**

Sol. Let x denote the discreate random variable "number of red balls".

5 White
4 Red

$\therefore$ The possible values of x are 0, 1, 2, 3.

Let R_i be the event of drawing a red ball from the bag in the ith draw, i = 1, 2, 3.

$$P(x=0) = P(\bar{R}_1\bar{R}_2\bar{R}_3) = P(\bar{R}_1)\left(\frac{\bar{R}_2}{\bar{R}_1}\right)P\left(\frac{\bar{R}_3}{\bar{R}_1\bar{R}_2}\right) = \frac{5}{9}\times\frac{4}{8}\times\frac{3}{7} = \frac{60}{504} = \frac{5}{42}$$

$$P(x=1) = P(R_1\bar{R}_2\bar{R}_3 \quad \text{or} \quad \bar{R}_1R_2\bar{R}_3 \quad \text{or} \quad \bar{R}_1\bar{R}_2R_3)$$

$$= P(R_1)P\left(\frac{\bar{R}_2}{R_1}\right)P\left(\frac{\bar{R}_3}{R_1\bar{R}_2}\right) + P(\bar{R}_1)\times P\left(\frac{R_2}{\bar{R}_1}\right)P\left(\frac{\bar{R}_3}{\bar{R}_1R_2}\right) + P(\bar{R}_1)P\left(\frac{\bar{R}_2}{\bar{R}_1}\right)P\left(\frac{R_3}{\bar{R}_1\bar{R}_2}\right)$$

$$= \frac{4}{9}\times\frac{5}{8}\times\frac{4}{7} + \frac{5}{9}\times\frac{4}{8}\times\frac{4}{7} + \frac{5}{9}\times\frac{4}{8}\times\frac{4}{7} = \frac{240}{504} = \frac{10}{21}$$

$$P(x=2) = P(R_1R_2\bar{R}_3 \quad \text{or} \quad R_1\bar{R}_2R_3 \quad \text{or} \quad \bar{R}_1R_2R_3)$$

$$= P(R_1)P\left(\frac{R_2}{R_1}\right)P\left(\frac{\bar{R}_3}{R_1R_2}\right) + P(R_1)P\left(\frac{\bar{R}_2}{R_1}\right)P\left(\frac{R_3}{R_1\bar{R}_2}\right) + P(\bar{R}_1)P\left(\frac{R_2}{\bar{R}_1}\right)P\left(\frac{R_3}{\bar{R}_1R_2}\right)$$

$$= \frac{4}{9}\times\frac{3}{8}\times\frac{5}{7} + \frac{4}{9}\times\frac{5}{8}\times\frac{3}{7} + \frac{5}{9}\times\frac{4}{8}\times\frac{3}{7} = \frac{180}{504} = \frac{5}{14}$$

$$P(x=3) = P(R_1R_2R_3) = P(R_1)P\left(\frac{R_2}{R_1}\right)P\left(\frac{R_3}{R_1R_2}\right) = \frac{4}{9}\times\frac{3}{8}\times\frac{2}{7} = \frac{24}{504} = \frac{1}{21}$$

The required Probability distribution is

x	0	1	2	3
P(x)	$\frac{5}{42}$	$\frac{10}{21}$	$\frac{5}{14}$	$\frac{1}{21}$

Previous Year's Question

Two cards are drawn successively with replacement from a well shuffled deck of 52 cards. Let X denote the random variable of number of aces obtained in the two drawn cards. Then P(x = 1) + P(X = 2) equals : **[JEE (Main) 2019]**

(A) $\frac{49}{169}$ (B) $\frac{52}{169}$ (C) $\frac{24}{169}$ (D) $\frac{25}{169}$

Sol. **(D)**

Total number of aces = 4

$\Rightarrow$ Probability of getting an ace $= \frac{4}{52} = \frac{1}{13}$

$$\Rightarrow P(X=1) = P\left(\text{Ace}, \overline{\text{Ace}}\right) + P\left(\overline{\text{Ace}}, \text{Ace}\right)$$

$$= \frac{1}{13} \times \left(1 - \frac{1}{13}\right) + \left(1 - \frac{1}{13}\right)\frac{1}{13}$$

$$= \frac{24}{13^2}$$

Similarly, $P(X=2) = P(\text{Ace}, \text{Ace})$

$$= \frac{1}{13} \times \frac{1}{13} = \frac{1}{13^2}$$

$$\Rightarrow P(X=1) + P(X=2) = \frac{24}{169} + \frac{1}{169} = \frac{25}{169}$$

Q. **A random variable x has the following probability distribution values of x.**

x	0	1	2	3	4	5	6	7
P(x)	0	K	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Find (i) k (ii) $P(x < 6)$ (iii) $P(x \geq 6)$ (iv) $P(0 < x < 5)$

Sol. (i) $\sum_{i=0}^{7} P(x_i) = 1$

$\Rightarrow 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$\Rightarrow 10k^2 + 9k - 1 = 0$

$\Rightarrow (10k - 1)(k + 1) = 0$

$\Rightarrow k = \frac{1}{10}$ or -1

$\Rightarrow k = \frac{1}{10}$ ($\because$ k=–1 not possible)

(ii) $p(x < 6) = P(0) + P(1) + P(2) + P(3) + P(4) + P(5)$

$= 8k + k^2$

$= \frac{8}{10} + \frac{1}{100} = 0.81$

(iii) $P(x \geq 6) = 1 - P(x < 6)$

$= 1 - 0.81 = 0.19$

(iv) $P(0 < x < 5) = P(x < 6) - P(0) - P(5)$

$= 0.81 - 0 - 0.01$

$= 0.8$

MEAN AND VARIANCE OF A PROBABILITY DISTRIBUTION

(1) Mean :

If a random variable X assumes the values $x_1, x_2, ..., x_n$ with probabilities $p_1, p_2, ...p_n$ respectively then the mean of X is defined by

Random variable (x_i)	Probability (p_i)	$p_i x_i$
x_1	p_1	$p_1 x_1$
x_2	p_2	$p_2 x_2$
⋮	⋮	⋮
⋮	⋮	⋮
x_n	p_n	$p_n x_n$

Then mean $(\mu) = \frac{\sum_{i=1}^{n} p_i x_i}{\sum_{i=1}^{n} p_i} = \sum_{i=1}^{n} x_i p_i$ $\left\{\sum_{i=1}^{n} p_i = 1\right\}$

(2) Variance :

$$\sigma^2 = \sum_{i=1}^{n} p_i (x_i - \mu)^2 = \sum_{i=1}^{n} p_i \left(x_i^2 - 2\mu x_i + \mu^2\right) = \sum_{i=1}^{n} \left(p_i x_i^2 - 2\mu p_i x_i + p_i \mu^2\right)$$

$$= \sum_{i=1}^{n} p_i x_i^2 - 2\mu \sum_{i=1}^{n} p_i x_i + \mu^2 \sum_{i=1}^{n} p_i = \sum_{i=1}^{n} p_i x_i^2 - 2\mu^2 + \mu^2$$

$$= \sum_{i=1}^{n} p_i x_i^2 - \mu^2$$

(3) Standard Deviation :

$$SD = +\sqrt{\sigma^2}$$

Q. **Two bad eggs are accidently mixed with 10 good eggs. 3 eggs are drawn simultaneously from the basket. Find the mean and variance of the number of bad eggs drawn.**

Sol. $x = 0$

$$P(0) = \frac{{}^{10}C_3}{{}^{12}C_3} = \frac{10.9.8}{12.11.10} = \frac{6}{11}$$

$x = 1$

$$P(1) = \frac{{}^{2}C_1 \times {}^{10}C_2}{{}^{12}C_3} = \frac{9}{22}$$

$x = 2$

$$P(2) = \frac{{}^{2}C_2 \cdot {}^{10}C_1}{{}^{12}C_3} = \frac{10 \cdot 6}{12 \cdot 11 \cdot 10} = \frac{1}{22}$$

x_i	p_i	$p_i x_i$
0	$\frac{6}{11}$	0
1	$\frac{9}{22}$	$\frac{9}{22}$
2	$\frac{1}{22}$	$\frac{2}{22}$

$$\mu = \sum p_i x_i = \frac{11}{22} = \frac{1}{2}$$

$$\therefore \sum p_i x_i^2 = 0 + \frac{9}{22} + \frac{4}{22}$$

$$\sigma^2 = \sum p_i x_i^2 - \mu^2 = \frac{13}{22} - \frac{1}{4} = \frac{15}{44}$$

BINOMIAL PROBABILITY DISTRIBUTION

Let an experiment has n independent trials and each of the trial has two possible outcomes i.e. success or failure.

If number of success in the experiment is a random variable then

x_i	p_i	$p_i x_i$	$p_i x_i^2$
0	${}^nC_0 p^0 q^n$	$0 \times {}^nC_0 p^0 q^n$	$0^2 \cdot {}^nC_0 p^0 q^n$
1	${}^nC_1 p^1 q^{n-1}$	$1 \times {}^nC_1 p^1 q^{n-1}$	$1^2 \cdot {}^nC_1 p^1 q^{n-1}$
2	${}^nC_2 p^2 q^{n-2}$	⋮	⋮
3	${}^nC_3 p^3 q^{n-3}$	⋮	⋮
⋮	⋮	⋮	⋮
r	${}^nC_r p^r q^{n-r}$	$r \times {}^nC_r p^r \cdot q^{n-r}$	$r^2 \cdot {}^nC_r p^r q^{n-r}$

$$P(x = r) = {}^n C_r p^r . q^{n-r}$$

where p= probability of getting success
and q= probability of getting failure
Mean of BPD of a random variable

$$\mu = \sum p_i x_i = \sum_{r=0}^{n} r \cdot {}^n C_r \cdot p^r \cdot q^{n-r} = \sum_{r=0}^{n} r \cdot \frac{n}{r} \cdot {}^{n-1} C_{r-1} \cdot p^r . q^{n-r} = p \cdot n \sum_{r=1}^{n} {}^{n-1}C_{r-1} \cdot p^{r-1} q^{n-r}$$

$$= np\left[{}^{n-1}C_0 \cdot p^0 q^{n-1} + {}^{n-1} C_1 \cdot p^1 q^{n-2} + \ldots + {}^{n-1} C_{n-1} p^{n-1} q^0\right]$$

$$\Rightarrow \mu = np(p+q)^{n-1} = np$$

Variance of BPD of a random variable

$$\sum p_i x_1^2 = \sum_{n=0}^{n} r^2 \cdot {}^n C_r \cdot p^r q^{n-r} = \sum_{r=0}^{n} r^2 \cdot \frac{n}{r} \cdot {}^{n-1} C_{r-1} \cdot p^r \cdot q^{n-r}$$

$$= n \sum_{r=0}^{n} (r - 1 + 1) \cdot {}^{n-1} C_{r-1} \cdot p^r \cdot q^{n-r}$$

$$= n\left[\sum_{r=0}^{n} (r-1) \cdot {}^{n-1} C_{r-1} \cdot p^r \cdot q^{n-r} + \sum_{r=0}^{n} {}^{n-1}C_{r-1} \cdot p^r \cdot q^{n-r}\right]$$

$$= p^2 \cdot n(n-1)(p+q)^{n-2} + pn \cdot (p+q)^{n-1}$$

$$= p^2 \cdot n(n-1) + pn$$

$$\therefore \sigma^2 = p^2 n^2 - p^2 n + pn - n^2 p^2 \quad \{\because \mu = np\}$$

$$\sigma^2 = pn(1-p) = npq \ \textbf{(VARIANCE)}$$

Standard deviation of BPD of a random variable :

Positive value of square root of variance is called standard derivation.

$$SD = +\sqrt{\sigma^2} = \sqrt{npq}$$

Q. **A pair of dice is thrown 5 times. If getting a doublet is considered as a success then find the mean & variance of the successes.**

Sol. Here n = 5 and favourable sample space are (1, 1), (2, 2), (3, 3),(4, 4), (5, 5), (6, 6)

$$\therefore p = \frac{1}{6} \text{ and } q = \frac{5}{6}$$

$$\text{Mean} = \mu = np = \frac{5}{6}$$

$$\text{Variable} = \sigma^2 = npq = \frac{25}{36}$$

Q. **If difference between mean & variance of a BPD is 1 and difference between squares is 11 then find the probability of getting exactly 3 successes.**

Sol. Given

$$np - npq = 1 \quad \ldots(i)$$

$$n^2p^2 - n^2p^2q^2 = 11 \quad \ldots(ii)$$

$$\therefore np + npq = 11$$

$$npq = 5 \qquad q = \frac{5}{6} \qquad p = \frac{1}{6} \qquad n = 36$$

$$\therefore \text{Required probability} = {}^{36}C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{33}$$

Q. **If X follows a binomial distribution with mean 3 and variance $\left(\frac{3}{2}\right)$, find**

(i) $P(X \geq 1)$ **(ii)** $P(X \leq 5)$

Sol. We know that mean = np and variance = npq

$$\therefore np = 3 \text{ and } npq = \frac{3}{2} \Rightarrow 3q = \frac{3}{2} \Rightarrow q = \frac{1}{2}$$

$$\therefore p = (1 - q) = \left(1 - \frac{1}{2}\right) = \frac{1}{2}$$

Now, $np = 3$ and $p = \frac{1}{2} \Rightarrow n \times \frac{1}{2} = 3 \Rightarrow n = 6$

So, the binomial distribution is given by

$$P(X=r) = {}^nC_r . p^r . q^{(n-r)} = {}^6C_r \times \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{(6-r)} = {}^6C_r \left(\frac{1}{2}\right)^6$$

(i) $P(X \geq 1) = 1 - P(X = 0)$

$$= 1 - {}^6C_0 \cdot \left(\frac{1}{2}\right)^6 = \left(1 - \frac{1}{64}\right) = \frac{63}{64}$$

(ii) $P(X \leq 5) = 1 - P(X = 6)$

$$= 1 - {}^6C_6 \left(\frac{1}{2}\right)^6 = \left(1 - \frac{1}{64}\right) = \frac{63}{64}$$

Q. **If the sum of the mean and variance of a binomial distribution for 5 trials is 1.8, find the distribution.**

Sol. Mean = np and variance = npq

It is given that n = 5 and mean + variance = 1.8

$\therefore np + npq = 1.8$, where $n = 5$

$\Leftrightarrow 5p + 5pq = 1.8$

$\Leftrightarrow p + p(1-p) = 0.36 \quad [\because q = (1-p)]$

$\Leftrightarrow p^2 - 2p + 0.36 = 0$

$\Leftrightarrow 100p^2 - 200p + 36 = 0$

$\Leftrightarrow 25p^2 - 50p + 9 = 0$

$\Leftrightarrow 25p^2 - 45p - 5p + 9 = 0$

$\Leftrightarrow 5p(5p-9) - (5p-9) = 0$

$\Leftrightarrow (5p-9)(5p-1) = 0$

$\Leftrightarrow p = \frac{1}{5} = 0.2 \quad [\because p \text{ cannot exceed } 1]$

Thus, $n = 5, p = 0.2$, and $q = (1-p) = (1-0.2) = 0.8$

Let X denote the binomial variate. Then, the required distribution is

$P(X = r) = {}^n C_r \cdot p^r \cdot q^{(n-r)} = {}^5 C_r \cdot (0.2)^r \cdot (0.8)^{(5-r)}$ where r = 0, 1, 2, 3, 4, 5

Q. **The mean and variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 success is**

(A) $\frac{28}{256}$ **(B)** $\frac{37}{256}$ **(C)** $\frac{128}{256}$ **(D)** $\frac{219}{256}$

Sol. In a Binomial distribution,

mean = np=4

Variance = npq = 2

$\Rightarrow 4q = 2$

$\Rightarrow q = \frac{1}{2}$

Since $p + q = 1, p = 1 - \frac{1}{2} = \frac{1}{2} \Rightarrow n = 8$

Now, for 2 success, $P(n = 8, r = 2) = {}^{8}C_{2}\left(\frac{1}{2}\right)^{2}\left(\frac{1}{2}\right)^{6} = \frac{28}{256}$

MISCELLANEAOUS QUESTION

Q. **A lot contains 20 articles. The probability that the lot contains exactly 2 defective articles is 0.4 and the probability that the lot contains exactly 3 defective articles is 0.6. Articles are drawn from the lot at random one by one without replacement and are tested till all defective articles are found . Find the probability that the testing procedure ends at the 12th testing.**

Sol. **Case-I :** The lot contains 2 defective articles. Testing Process ends at 12th testing.

$\Rightarrow 2^{nd}$ defective article discovered at 12th testing.

$\Rightarrow$ One defective article discovered in first 11 testing's.

$\Rightarrow$ In first 11 testing's, 10 good and 1 defective article drawn.

$\Rightarrow$ Probability $= P_1 = \left(\frac{{}^{18}C_{10} \times {}^{2}C_{1}}{{}^{20}C_{11}}\right) \times \left(\frac{1}{9}\right)$

Where $\frac{1}{9}$ is probability of picking defective article on 12th testing.

Case-II : The lot contains 3 defective articles.

$\Rightarrow 3^{rd}$ defective article drawn on 12th testing

$\Rightarrow$ In first 11 testing, 2 defective and 9 good articles drawn out of 17 good articles and 3 defective artices.

$\Rightarrow$ Probability $= P_2$

$= \left(\frac{{}^{17}C_{9} \times {}^{3}C_{2}}{{}^{20}C_{11}}\right) \times \left(\frac{1}{9}\right)$

Total Probability $= P(\text{Case} - I) \times P_1 + P(\text{case} - II) \times P_2$

$$= \left[(0.4)\left(\frac{{}^{18}C_{10} \times {}^{2}C_{1}}{{}^{20}C_{11}}\right) + 0.6\left(\frac{{}^{17}C_{9} \times {}^{3}C_{2}}{{}^{20}C_{11}}\right)\right]\frac{1}{9}$$

$$= \left[\frac{0.4 \times 9 \times 11 \times 2}{19 \times 20} + \frac{0.6 \times 9 \times 10 \times 11 \times 3}{18 \times 19 \times 20}\right]\frac{1}{9}$$

$$= \frac{4.4 + 5.5}{190}$$

$$= \frac{99}{1900}$$

Q. **A box contains 5 tubes, 2 of them defective and 3 good one. Tubes are tested one by till the 2 defective tubes are discovered. What is the probability that the testing procedure comes to an end of**
(i) IInd testing **(ii) IIIrd testing**

Sol. **(i)** Testing procedure ends on IInd testing if both defective tubes are drawn first.

Probability$= \frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$

(ii) Ton end procedure on IIIrd testing, 3 cases are possible :
(G-Good, D-Defective tube)

$\Rightarrow$ GDD $\rightarrow$ probability $= p_1 = \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{1}{10}$

$\Rightarrow$ DGD $\rightarrow$ Probability $= p_2 = \frac{2}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{1}{10}$

$\Rightarrow$ GGG $\Rightarrow$ probability $= \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{1}{10}$

In this case, since all good tubes are known, remaining ones are defective.

$\Rightarrow$ Total probability$= p_1 + p_2 + p_3 = \frac{3}{10}$

Q. **n whole numbers are taken at random are multiplied together. Show that the chance of digit at unit place of their product is**

(A) 1, 3, 7 or 9 is $\left(\frac{2}{5}\right)^n$ **(B) 2, 4, 6 or 8 is** $\frac{4^n - 2^n}{5^n}$

(C) 5 is $\frac{5^n - 4^n}{10^n}$ **(D) 0 is** $\frac{10^n - 8^n - 5^n + 4^n}{10^n}$

Sol. **(A)** Unit place will contain 1, 3, 7 or 9 only when none of the number is even or odd multiple of 5.

So, only four choice out of 10 for every number: 1, 3, 7, 9.

$\Rightarrow$ Probability $= \left(\frac{4}{10}\right)\left(\frac{4}{10}\right)\left(\frac{4}{10}\right)$...n times

$= \left(\frac{2}{5}\right)^n$ **Hence proved.**

(B) Unit place contain 2, 4, 6 or 8 only when none of the numbers contains 5 or 0 in its unit place and atleast one of them contain number 2 or 4 or 6 or 8 in their unit place.

P(numbers does not include multiple of 5) $= \left(\frac{8}{10}\right)^n = \left(\frac{4}{5}\right)^n$

P(number only include 1, 3, 7 or 9) $= \left(\frac{2}{5}\right)^n$.

$\Rightarrow$ Required Probability $= \left(\frac{4}{5}\right)^n - \left(\frac{2}{5}\right)^n$

$= \frac{4^n - 2^n}{5^n}$ **Hence proved.**

(C) Unit place will have 5 if none of the number are even and atleast one have 5 in its unit place.

$\Rightarrow$ Required probability $= \left(\frac{5}{10}\right)^n - P(\text{none of them have 5})$

$= \left(\frac{5}{10}\right)^n - \left(\frac{4}{10}\right)^n$

$= \frac{5^n - 4^n}{10^n}$ **Hence proved.**

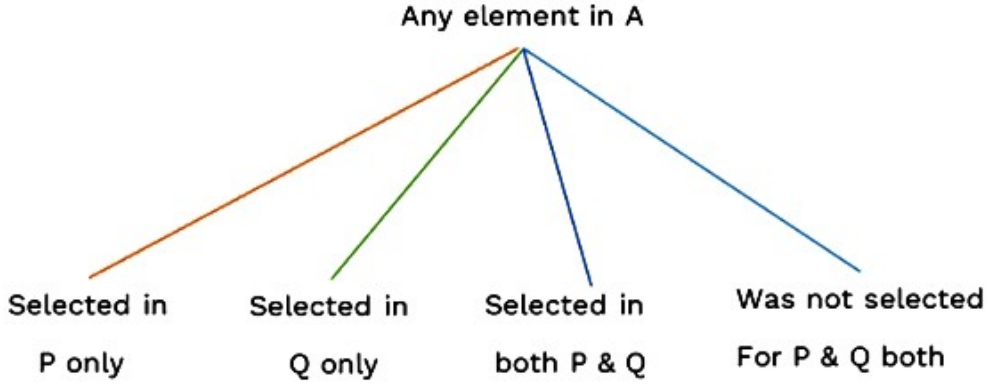
(D) For 0 to be on unit place, two cases possible.
(i) There is multiple of 10 among the numbers
(ii) Else, there is atlest an even number with unit place 2, 4, 6 or 8 and odd multiple of 5.
Number of such possible ways.
= Total-cases when none of them is multiple of 5-cases where unit place of product is 5

$= 10^n - 8^n - \left(5^n - 4^n\right)$

$\Rightarrow$ Probability $= \frac{10^n - 8^n - 5^n + 4^n}{10^n}$ **Hence proved.**

Q. **'A' is a set containing n distinct elements. A subset 'P' of 'A' is chosen at random. The set 'A' is reconstructed by replacing the elements of 'P'. A subset 'Q' of 'A' is again chosen at random. Find the probability that 'P' & 'Q' have no common elements.**

Sol.



From above figure, we can see that every element has 4 options. So , total number of ways of establishing P and Q subset is 4^n.

Also number of ways such that P & Q does not have common elements is 3^n.

(Every element can only go in P or go in Q or neither of them. So, three choices available)

$\Rightarrow$ P(P and Q have no common elements) $= \frac{3^n}{4^n}$.

Q. **Let S = {1, 2, 3, 4, 5, 6, 7, 8}, 2 subsets P & Q of S are selected randomly then find the probability that**

(i) $\mathbf{P \cap Q = \phi}$

(ii) An onto mapping exists from P to Q

(iii) $\mathbf{n(P \cup Q) = 8}$ **&** $\mathbf{n(P \cap Q) = 0}$.

(iv) $\mathbf{n(P \cup Q) = 6}$ **&** $\mathbf{n(P \cap Q) = 2}$.

Sol. **(i)** Probability that $P \cap Q = \phi = \left(\frac{3}{4}\right)^n$

Where n= number of element in S ($\because$ Q.5)

$\Rightarrow$ Probability$= \left(\frac{3}{4}\right)^8$

(ii) Number of ways of selecting number of elements for P and Q is $9 \times 9 = 81$

An onto mapping can exists from P to Q only when $n(P) \geq n(Q)$

Number of ways of selecting different number of elements for P and Q is ${}^{8}C_{2}\,(n(P) \geq n(Q))$

Number of ways of selecting equal number of element for P & Q is 8 (1, ...,8)

$\Rightarrow$ Required Probability $= \dfrac{{}^{8}C_{2} + 8}{81} = \dfrac{36}{81} = \dfrac{4}{9}$

(iii) $n(P \cup Q) = 8 \quad \& \; n(P \cap Q) = 0$

$\Rightarrow$ Every element in S is either selected in P or Q.

$\Rightarrow$ 2 choices for every element

$\Rightarrow$ Number of favourable ways $= 2^{8}$

$\Rightarrow$ Total number of ways $= 4^{8}$.

$\Rightarrow$ Probability $= \dfrac{2^{8}}{4^{8}} = \left(\dfrac{1}{2}\right)^{8}$.

(iv) $n(P \cup Q) = 6$ & $n(P \cap Q) = 2$.

Following cases are possible.

n(P)	n(Q)	Number of ways
2	6	${}^{8}C_{2} \times {}^{6}C_{0} \times {}^{6}C_{4}$
3	5	${}^{8}C_{2} \times {}^{6}C_{1} \times {}^{5}C_{3}$
4	4	${}^{8}C_{2} \times {}^{6}C_{2} \times {}^{4}C_{2}$
5	3	${}^{8}C_{2} \times {}^{6}C_{3} \times {}^{3}C_{1}$
6	2	${}^{8}C_{2} \times {}^{6}C_{4} \times {}^{2}C_{0}$

Here ${}^{8}C_{2} \rightarrow$ Selecting 2 common element.

Total number of favourable ways $= {}^{8}C_{2}\,[15 + 60 + 90 + 60 + 15]$

$= 28[240]$

$= 6720$

Total number of ways $= 2^{8} \times 2^{8} = 2^{16}$

$\Rightarrow$ Probability $= \dfrac{6720}{2^{16}} = \dfrac{105}{1024}$

Q. **Two numbers a and b are selected from the set of natural number then the probability that a^2+b^2 is divisible by 5 is**

(A) $\frac{9}{25}$ **(B)** $\frac{7}{18}$ **(C)** $\frac{11}{36}$ **(D)** $\frac{17}{81}$

Sol. Each number have 10 choices (0 to 9) for its unit place. Unit place decides if a^2+b^2 is multiple of 5.

Favourable pairs are (1, 2); (2, 1); (0, 0); (1, 3); (3, 1); (1, 7); (7, 1); (1, 8); (8, 1); (2, 9); (9, 2); (2, 4); (4,2)(2, 6)(6, 2); (3, 4); (4, 3); (3, 6); (6, 3); (3, 9); (9, 3); (4, 7); (7, 4); (4, 8); (8, 4); (5, 0); (0, 5); (5, 5); (6, 7); (7, 6); (6, 8); (8, 6); (7, 9); (9, 7);(8, 9) and (9, 8).

$\Rightarrow$ Total 36 pairs possible for unit places.

$\Rightarrow$ Total number of cases = $10 \times 10 = 100$

$\Rightarrow$ Probability$=\frac{36}{100}=\frac{9}{25}$.

Q. **Integers a, b, c and d not necessarily distinct, are chosen independently and at random from the set $S=\{0,1,2,3,...2006,2007\}$. If the probability that $|ad-bc|$ is even, is $\frac{p}{q}$ where p and q are relatively prime then find the value of $(p+q)$.**

Sol. Total number of ways of selecting a, b, c, d is 2008^4.

ad-bc is even only when either both ad and bc are even or both are odd.

Case-I: ad and bc are both odd.

$\Rightarrow$ a, b, c and d are odd.

Number of ways $=(1004)^4$

Case-II: ad and bc are both even.

Number of cases where ad is even $=(2008)^2-\underbrace{(1004)^2}_{\text{where a \& d are both odd}}$

Similarly, number of cases where bc is even $=(2008)^2-(1004)^2$

$\Rightarrow$ Total possible number of ways in case II $=\left(2008^2-1004^2\right)^2$

$=(1004)^4\times 9$

$\Rightarrow$ Total favourable number of ways $=(1004)^4+9\times(1004)^4$

$=10\times(1004)^4$

$$\Rightarrow \text{Required probability} = \frac{10 \times (1004)^4}{(2008)^4}$$

$$= \frac{10}{16} = \frac{5}{8} = \frac{p}{q}$$

$$\Rightarrow p + q = 13$$

Q. **Two distinct number x & y are chosen at random from the set {1, 2, 3, 4,...3n}. Find the probability that $x^2 - y^2$ is divisible by 3.**

Sol. $x^2 + y^2 = (x - y)(x + y)$ divisible by 3.

A : Cases where x + y is divisible by 3

B : Cases where x – y is divisible by 3

Number of favourable cases = $A \cup B$

A :→ For every $x = 3r + 1$ or $3r + 2$, there are n possible y which are of the form 3m+2 or 3m+1 respectively such that x+ y=3r + 3m+3 which is divisible by 3.

Number of pairs possible=$2n^2$

→ Also, for every $x = 3r$, there are $(n - 1)$ other multiple of 3 available for y so that

x + y is divisible by 3. Number of such pairs possible $= n(n-1) = n^2 - n$.

B : $x - y$ will be divisible by 3 only if both are of the form 3r + 1 or both of the form 3r + 2 or both are of the form 3r.

There are 'n' numbers each of the form 3r. 3r + 1 and 3r+ 2.

Choosing 2 number x and y where they are of the form 3r+1 or 3r+2 has number of ways $= 2 \times n \times (n - 1)$

$$= 2n^2 - 2n$$

$\Rightarrow$ Number of favourable cases $= n(A \cup B)$

$$= (2n^2) + (n^2 - n) + (2n^2 - 2n)$$

$$= 5n^2 - 3n$$

$\Rightarrow$ Total possible cases $= 3n(3n - 1)$

$$\text{Required Probability} = \frac{5n^2 - 3n}{3n(3n-1)} = \frac{5n - 3}{9n - 3}$$

Q. **If two whole numbers x and y are randomly selected from the set of natural numbers, then find the probability that $x^3 + y^3$ is divisible by 8.**

Sol. **Case-I :** If x and y both are even, then $x^3 + y^3$ will be divisible by 8.

P(x and y both are even) $= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Case-II : If one is even and other is odd, $x^3 + y^3$ will be odd, so not possible.

Case-III : Both x and y are odd.

$x^3 + y^3 = (x + y)(x^2 + y^2 - xy)$

$\Rightarrow x^2 + y^2 - xy$ will be odd number.

So, x+ y must be divisible by 8.

$\rightarrow$ Let x & y $8r + 1$ or $8r + 3$ or $8r + 5$ or $8r + 7$.

For every possible x, there is only 1 out of 4 choices for y so that x + y is divisible by 8.

$$\text{Probability} = \underbrace{\frac{1}{2}}_{x \text{ is odd}} \times \underbrace{\frac{1}{2}}_{y \text{ is odd}} \times \underbrace{\frac{1}{4}}_{\substack{x+y \\ \text{divisible} \\ \text{by 8}}} = \frac{1}{16}$$

Total probability $= \frac{1}{4} + 0 + \frac{1}{16} = \frac{5}{16}$

Q. **If the integers m and n are chosen at random from 1 to 100, then the probability that a number of the form $7^m + 7^n$ is divisible by 5 equals**

(A) $\frac{1}{4}$ **(B)** $\frac{1}{7}$ **(C)** $\frac{1}{8}$ **(D)** $\frac{1}{49}$

Sol. All numbers in 1 to 100 are of the form $4r, 4r + 1, 4r + 2, 4r + 3$.

There are 25 numbers each

Unit place for $7^{4r} = 1$

Unit place for $7^{4r+1} = 7$

Unit place for $7^{4r+2} = 9$

Unit place for $7^{4r+3} = 3$

Possible pairs such that $7^m + 7^n$ is divisible by 5 are

$(m,n) \equiv (4r, 4r + 2); (4r + 2, 4r); (4r + 1, 4r + 3); (4r + 3, 4r + 1)$

$\Rightarrow$ Total number of cases $= 4 \times 4 = 16$

$\Rightarrow$ Probability $= \frac{4}{16} = \frac{1}{4}$

Q. **The probability that an item produced by a factory is defective is 'p'. From a certain lot, a sample of 'n' items is drawn with replacement. If it contains no defective items, the lot is accepted, while if it has more than two defective items, the lot is rejected. If the sample has one or two defective items, an independent sample of 'm' items is drawn with replacement from the lot and combined with previous sample. If the combined sample does not contain more than two defective items, the batch is accepted. Find the probability that the batch is accepted.**

(A) $(1-p)^n + np(1-p)^{n+m}$

(B) $(1-p)^n + np(1-p)^{n+m-1}$

(C) $(1-p)^n + np(1-p)^{n+m-2}(1+(m-1)p) + {}^nC_2(1-p)^{m+n-2}p^2$

(D) $(1-p)^{n-2}\left((1-p)^2 + \frac{n(n-1)p^2}{2}\right) + np(1-p)^{n+m-2}(1+(m-1)p)$

Sol. **(C)**

There are 3 cases possible:

A: None of the items drawn were defective.

B: One of the items drawn was defective.

C: Two of the items drawn were defective.

Probability that an item drawn was not defective is 1 – p.

So, Required probability = P(A) + P(B) + P(C)

$$= (1-p)^n + {}^nC_1(1-p)^{n-1}p\left[(1-p)^m + {}^mC_1(1-p)^{m-1}p\right] + {}^nC_2(1-p)^{n-2}p^2(1-p)^m$$

↓	↓	↓	↓	↓
None defective	1 defective	none defective in sample of next 'm' items	1 defective in next 'm' items	2 defective

$$= (1-p)^n + np(1-p)^{n+m-2}(1+(m-1)p) + {}^nC_2(1-p)^{m+n-2}p^2$$

Q. **Each packet of blades sold contains a coupon which is equally likely to bear the letters A, B or C. If 'm' packets are purchased, what is the probability that the coupons cannot be used to spell BAC.**

(A) $\left(\frac{2}{3}\right)^m$ **(B)** $\frac{3.2^m - 2}{3^m}$ **(C)** $\frac{2^m}{3^{m-1}}$ **(D)** $\frac{2^m - 1}{3^{m-1}}$

Sol. **(D)**

A → number of ways such that there is no A

B → number of ways such that there is no B

C → number of ways such that there is no C

$n(A \cup B \cup C) = S_1 - S_2 + S_3 = {}^3C_1\, 2^m - {}^3C_2 1^m + 0$

where $S_1 \rightarrow A + B + C$

$S_2 \rightarrow A \cap B + B \cap C + C \cap A$

$S_3 \rightarrow A \cap B \cap C$

Required Probability = $P = \dfrac{{}^3C_1 2^m - {}^3C_2 (1)^m}{3^m} = \dfrac{2^m - 1}{3^{m-1}}$

Q. **Suppose m, n are real numbers randomly chosen in [0, 1]. Determine the probability that the distance between the roots of the equation $x^2 + mx + n = 0$ is not greater than 1.**

(A) $\dfrac{2}{3}$ **(B)** $\dfrac{1}{6}$ **(C)** $\dfrac{1}{3}$ **(D)** $\dfrac{1}{2}$

Sol. **(C)**

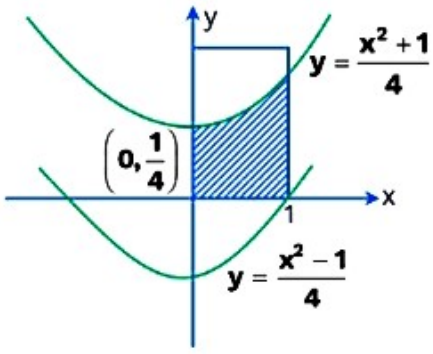
$|x_1 - x_2| = \sqrt{(x_1 + x_2)^2 - 4x_1x_2} = \sqrt{m^2 - 4n} \le 1$

$|m^2 - 4n| \le 1$

$\Rightarrow -1 \le x^2 - 4y \le 1$

$A = \int_0^1 \dfrac{x^2 + 1}{4} dx = \dfrac{1}{3}$

$P = \dfrac{\text{Favourable area}}{\text{Total area}} = \dfrac{1/3}{1} = \dfrac{1}{3}$



Q. **A natural number 'x' is chosen at random from the first 1000 natural numbers. If [·] denotes the greatest integer function, then the probability that**

$\left[\dfrac{x}{2}\right] + \left[\dfrac{x}{3}\right] + \left[\dfrac{x}{5}\right] = \dfrac{31x}{30}$ **is**

(A) $\dfrac{31}{1000}$ **(B)** $\dfrac{4}{125}$ **(C)** $\dfrac{33}{1000}$ **(D)** $\dfrac{65}{1000}$

Sol. **(C)**

$$\frac{x}{2}-\left\{\frac{x}{2}\right\}+\frac{x}{3}-\left\{\frac{x}{3}\right\}+\frac{x}{5}-\left\{\frac{x}{5}\right\}=\frac{31x}{30}$$

$$\Rightarrow \quad \left\{\frac{x}{2}\right\}+\left\{\frac{x}{3}\right\}+\left\{\frac{x}{5}\right\}=0$$

$$\Rightarrow \quad \frac{x}{2}\in I,\frac{x}{3}\in I \,\&\, \frac{x}{5}\in I$$

$$\Rightarrow \quad x = 30\ k,\ k \in I$$

There are 33 such integers in first 1000 natural numbers.

$$\Rightarrow \quad P=\frac{33}{1000}$$

Q. **If a and b are chosen randomly from the set consisting of numbers 1, 2, 3, 4, 5, 6 with replacement. Then the probability that $\lim\limits_{x\to 0}\left(\frac{a^x+b^x}{2}\right)^{2/x}=6$ is**

(A) $\frac{1}{18}$ **(B)** $\frac{5}{36}$ **(C)** $\frac{1}{9}$ **(D)** $\frac{1}{6}$

Sol. **(C)**

$$\lim_{x\to 0}\left(\frac{a^x+b^x}{2}\right)^{2/x}=e^{\lim\limits_{x\to 0}\left(\frac{a^x+b^x}{2}-1\right)\frac{2}{x}}=e^{\ln ab}=ab$$

$$\Rightarrow \quad ab = 6$$

(a, b) = (1, 6), (6, 1), (2, 3), (3, 2)

$$P=\frac{4}{36}=\frac{1}{9}$$

Q. **A seven-digit number of $a_1a_2a_3a_4a_5a_6a_7$ (all digits distinct) is formed randomly. The probability that number formed satisfy $a_1 > a_2 > a_3 < a_4 < a_5 < a_6 < a_7$ is**

(A) $\frac{11}{1572}$ **(B)** $\frac{1}{168}$ **(C)** $\frac{7}{1512}$ **(D)** $\frac{5}{1512}$

Sol. **(D)**

7 digits are distinct and a_3 is the smallest

$\therefore$ Number of favourable ways = $^{10}C_7$ × 1 × $^{6}C_2$

selecting 7 digit | selecting a_3 | selecting a_1, a_2

$$P = \frac{\left(\frac{10\times9\times8\times6\times5}{3\times2\times2}\right)}{9\times9\times8\times7\times6\times5\times4}$$

$$= \frac{5}{1512}$$

ONE OR MORE THAN ONE CORRECT QUESTIONS

Q. **Of three independent events E_1, E_2, E_3, the chance that only E_1 occurs is a, that only E_2 occurs is b, and only E_3 occurs is c and probability of none of E_1, E_2, E_3 occuring is x. Then**

(A) $P(E_2) = \frac{x}{b+x}$ **(B)** $P(E_2) = \frac{b}{b+x}$

(C) $x^2 = (a + x)(b + x)(c + x)$ **(D)** $P(E_3) = \frac{X}{c+x}$

Sol. **(BC)**

$$P\left(\bar{E}_1 \cap \bar{E}, \cap \bar{E}_3\right) = x = P(\bar{E}_1)P\left(\bar{E}_2\right)P(\bar{E}_3)$$

$$P\left(E_1 \cap \bar{E}_2 \cap \bar{E}_3\right) = a = P(E_1)P\left(\bar{E}_2\right)P(\bar{E}_3)$$

$$P\left(\bar{E}_1 \cap E_2 \cap \bar{E}_3\right) = b = P(\bar{E}_1)P(E_2)P\left(\bar{E}_3\right)$$

$$P\left(\bar{E}_1 \cap \bar{E}, \cap E_3\right) = x = P\left(\bar{E}_1\right)P\left(\bar{E}_2\right)P(E_3)$$

$$\frac{a}{x} = \frac{P(E_1)}{P\left(\bar{E}_1\right)} = \frac{P(E_1)}{1-P(E_1)} \Rightarrow P(E_1) = \frac{a}{a+x}$$

Similarly, $P(E_2) = \frac{b}{b+x}$ and $P(E_3) = \frac{c}{c+x}$

$$P(\bar{E}_1)P\left(\bar{E}_2\right)P\left(\bar{E}_2\right) = x = \frac{x}{a+x}\frac{x}{b+x}\frac{x}{c+x}$$

$\Rightarrow (a + x)(b + x)(c + x) = x^2$

Q. **In a city, a person owns independently a sedan car with probability $\frac{3}{10}$ and a SUV with probability $\frac{4}{10}$. If he has sedan only, then he keeps a driver with probability $\frac{6}{10}$, whereas if he owns SUV only, then he keeps a driver with probability $\frac{7}{10}$, whereas if he keeps both type of cars then his probability of keeping a driver $\frac{9}{10}$ is. Then**

(A) Probability that person keeps a driver is $\frac{412}{1000}$

(B) Probability that person keeps a driver is $\frac{71}{125}$

(C) Given that person keeps driver, then probability that he owns Sedan is $\frac{54}{103}$

(D) Given that person keeps driver, then probability that he owns Sedan is $\frac{76}{103}$

Sol. **(AC)**

D → event that person keeps driver

E_1 → event that person own sedan

E_2 → event that person own SUV

$$P(D) = P(E_1 \cap \bar{E}_2)P(D/E_1 \cap \bar{E}_2) + P(\bar{E}_1 \cap E_2)P(D/\bar{E}_1 \cap E_2) + P(E_1 \cap E_2)P(D/E_1 \cap E_2)$$

$$= \frac{3}{10}\times\frac{6}{10}\times\frac{6}{10} + \frac{7}{10}\times\frac{4}{10}\times\frac{7}{10} + \frac{3}{10}\times\frac{4}{10}\times\frac{9}{10} = \frac{103}{250}$$

$$P(E_1/D) = \frac{P(E_1 \cap \bar{E}_2)P(D/E_1 \cap \bar{E}_2) + P(E_1 \cap E_2)P(D/E_1 \cap E_2)}{P(D)} = \frac{\left(\frac{54}{250}\right)}{\left(\frac{103}{250}\right)} = \frac{54}{103}$$

Comprehension-1 for Q.1 to Q.2:

The values of a and b are equally possible in the square $|a| \le 1$, $|b| \le 1$. Consider the events

A = {The roots of quadratic expression $x^2 + 2ax + b$ are real}

B = {The roots of $x^2 + 2ax + b$ are positive}.

1. **P(A) =**

(A) $\frac{2}{3}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{1}{6}$ **(D)** $\frac{3}{8}$

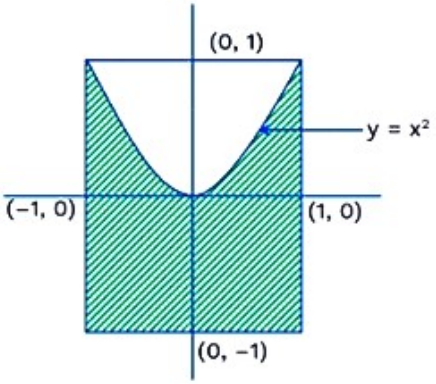
Sol. **(A)**

$4(a^2 - b) \geq 0$

$b \leq a^2$

Area of the curve $y=x^2$ with lines $x = 1$, $x = -1$ and $y = -1$ is equal to 8/3.

Total area available is equal to $2 \times 2 = 4$



So, required probability = $P(A) = \frac{2\left(\frac{1}{3}\right) + 2(1)}{2(2)} = \frac{8}{12} = \frac{2}{3}$

2. **P(B) =**

(A) $\frac{3}{8}$ **(B)** $\frac{2}{9}$ **(C)** $\frac{1}{6}$ **(D)** $\frac{1}{12}$

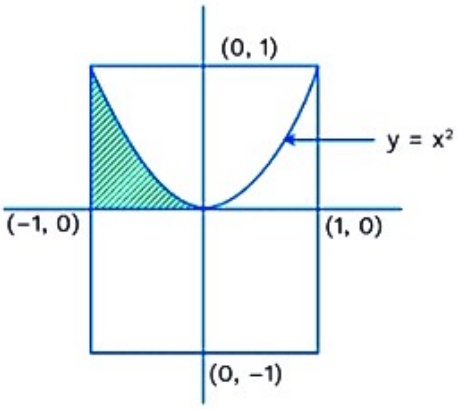
Sol. **(D)**

$D \geq 0 \Rightarrow b \leq a^2$

$-a > 0 \Rightarrow a < 0$

$\& f(0) > 0 \Rightarrow b > 0$

$P(B) = \frac{\frac{1}{3}}{2(2)} = \frac{1}{12}$



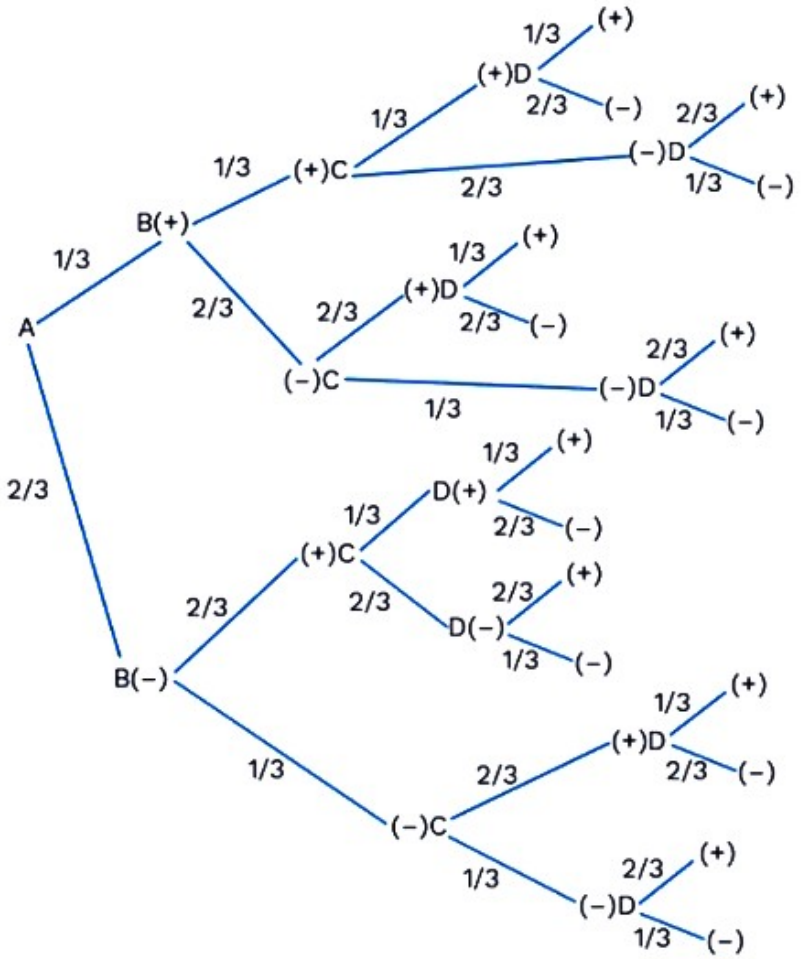
Comprehension-2 for Q.3 to Q.4:

A slip of paper is given to A, who marks it with either a plus sign or a minus sign, the probability of his writing a plus sign is $\frac{1}{3}$. He then passes the slip to B, who may leave it unchanged or change the sign before passing it to C. Next C passes the slip to D after perhaps changing the sign. Finally D passes it to an honest judge after perhaps changing the sign. It is known that B, C, D each change the sign with probability $\frac{2}{3}$.

3. **The probability that judge see a plus sign on the slip is equal to**

(A) $\frac{4}{9}$ **(B)** $\frac{38}{81}$ **(C)** $\frac{40}{81}$ **(D)** $\frac{41}{81}$

Sol. **(D)**



$$P=\left(\frac{1}{3}\times\frac{1}{3}\times\frac{1}{3}\times\frac{1}{3}\right)+\frac{1\times1\times2\times2}{3^4}+\frac{1\times2\times(2\times1+1\times2)}{3^4}+\frac{2\times2\times(1\times1+2\times2)}{3^4}$$

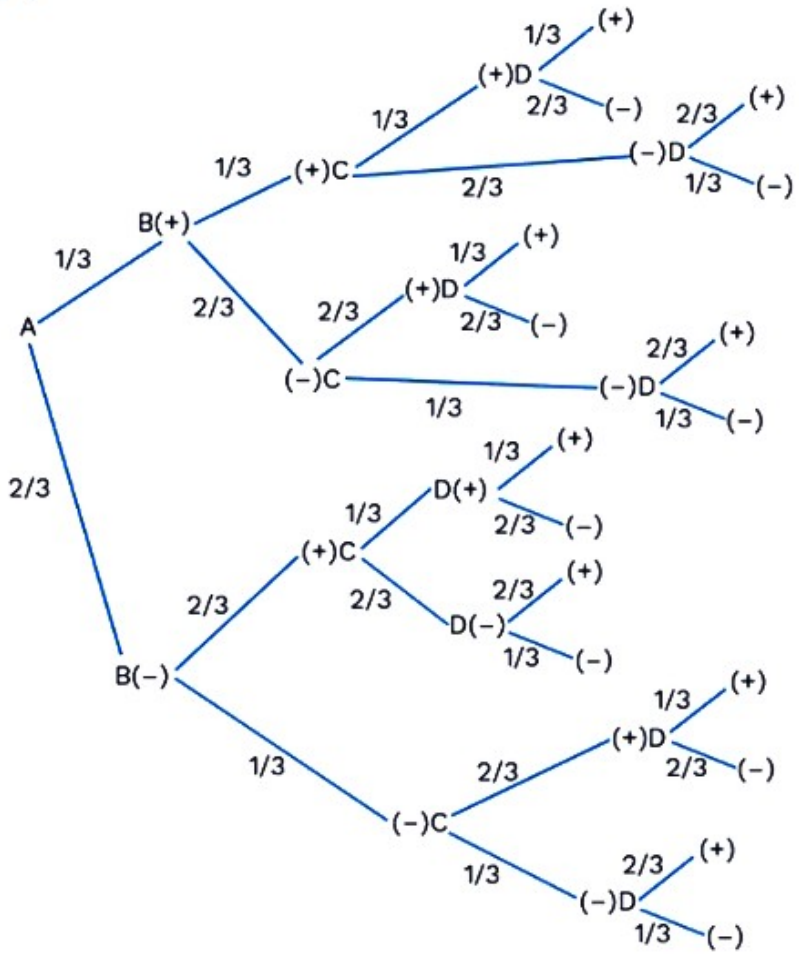
$$+\frac{2\times1\times(2\times1+1\times2)}{3^4}$$

$$P=\frac{5+8+20+8}{81}=\frac{41}{81}$$

4. **If the judge see a plus sign on the slip, then the probability that A originally wrote a plus sign is equal to**

(A) $\frac{13}{41}$ **(B)** $\frac{14}{41}$ **(C)** $\frac{13}{40}$ **(D)** $\frac{11}{40}$

Sol. **(A)**



$$P = \frac{\frac{1}{3}\times\frac{1}{3}\times\left(\frac{1}{3}\times\frac{1}{3}+\frac{2}{3}\times\frac{2}{3}\right)+\frac{1}{3}\times\frac{2}{3}\times\left(\frac{2}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{3}\right)}{\frac{41}{81}} = \frac{5+2(4)}{41} = \frac{13}{41}$$

Q. **If a, b, c are numbers obtained by throwing a dice thrice, such that $a^2 + b^2 + c^2 \le ab + bc + ca$, then the probability that a point (a, b, c) lies inside the tetrahedron formed by coordinate planes and $x + y + z = 12$ is P. Then 16P is equal to**

Sol. **(8)**

$a^2 + b^2 + c^2 \le ab + bc + ca$

$((a - b)^2 + (b - c)^2 + (c - a)^2) \le 0$

$\Rightarrow a = b = c$

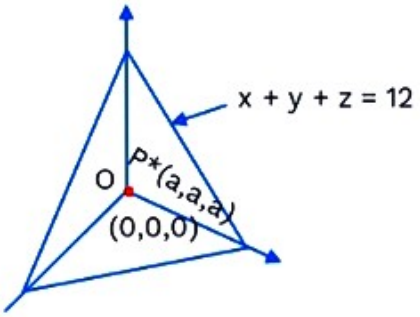
So, total six possibilities: (1,1,1), (2,2,2), (3,3,3), (4,4,4), (5,5,5) and (6,6,6)

$\because$ Points O (0, 0, 0) and P(a, a, a) lie on name side of plane x + y + z = 12

$\Rightarrow -12 (a + a + a - 12) > 0$

$\Rightarrow 0 < a < 4$

$\Rightarrow a = \{1, 2, 3\}$



Required Probability = $P = \frac{3}{6} = \frac{1}{2}$

So, 16P = 8.

Q. **A 7 digit number with all digits distinct is randomly formed using digits {1, 2, 3, ... 7}. The probability that formed number is such that product of any four consecutive digits is divisible by 10 is P, then the value of [10P] is (where [·] denote greatest integer function)**

Sol. First four digits must include 5 but last four digits also have the same condition.

So, it is only possible if 5 lies in the middle, i.e., 4th place.

_____5_____

Now any 4 consecutive digits will have 5 among them.

If their product is to be divisible by 10, they must also have an even number among them.

But there is a case where 1,3,5 is on one side and 2,4,6 are on other. In such a case, side with all odd digits does not fulfil required conditions.

So, Required Probability = P = 1 – (all 2, 4, 6 in one side of 5)

$= 1 - \frac{3!3! \times 2}{7!}$

$= \frac{69}{70}$

Q. **A pack of playing cards was found to contain only 51 cards. If the first 13 cards which are examined are all black. If the probability that the missed one is red is equal to $\frac{p}{q}$ where p, q are relatively prime natural numbers, then p + q is equal to**

Sol. **(5)**

Let E_1 be the event that missing card is black.

Let E_2 be the event that missing card is Red.

Let A be the event that first 13 cards drawn are all black.

$$P(A/E_1) = \frac{{}^{25}C_{13}}{{}^{51}C_{13}}$$

$$P(A/E_2) = \frac{{}^{26}C_{13}}{{}^{51}C_{13}}$$

$$P(E_2/A) = \frac{P(E_2)\,P(A/E_2)}{P(E_1)\,P(A/E_1)+P(E_2)\,P(A/E_2)}$$

$$= \frac{\frac{1}{2}\cdot\frac{{}^{26}C_{13}}{{}^{51}C_{13}}}{\frac{1}{2}\cdot\frac{{}^{25}C_{13}}{{}^{51}C_{13}}+\frac{1}{2}\cdot\frac{{}^{26}C_{13}}{{}^{51}C_{13}}} = \frac{{}^{26}C_{13}}{{}^{25}C_{13}+{}^{26}C_{13}}$$

$$= \frac{1}{\frac{13}{26}+1} = \frac{26}{39} = \frac{2}{3}$$

So, p = 2, q = 3

p + q = 5